Learner Guide

Faculty of Information Technology

Networks 511

Year 1 *Semester 1*

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FACULTY OF INFORMATION TECHNOLOGY

LEARNER GUIDE
MODULE: NETWORKS 511
(1ST SEMESTER)

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CHAPTER 1: NETWORK TOPOLOGIES AND OSI REFERENCE MODEL

1.1. TYPES OF TOPOLOGIES AND THE OSI MODEL

LEARNING OUTCOMES

After reading this Section of the guide, the learner should be able to:

- **Understand the Network Topologies and the Network infrastructure.**
- **Know the Protocols and how they complement each other**
- **Describe the seven layers of the OSI model**
- **Master and analyse scenarios and refer the matching OSI layer**
- **Identify types of applications and protocols used on a network**
- **Distinguish between the client-server and peer-to-peer models used to control access to a network**
- **Describe various networking hardware devices and the most common physical topologies**
- **HOW Encapsulation/De-capsulation works**
- **Explore best practices for safety when working with networks and computers**
- **Describe the seven-step troubleshooting model for solving a networking problem**

1.1.1 What is a network, or computer network?

It is a group or system of interconnected people or things or a collection of computers and other hardware interconnected by communication channels that allow sharing of resources and information. Where at least one process in one device is able to send/receive data to/from at least one process residing in a remote device, then the two devices are said to be in a network. Simply, more than one computer interconnected through a communication medium for information interchange is called a computer network.

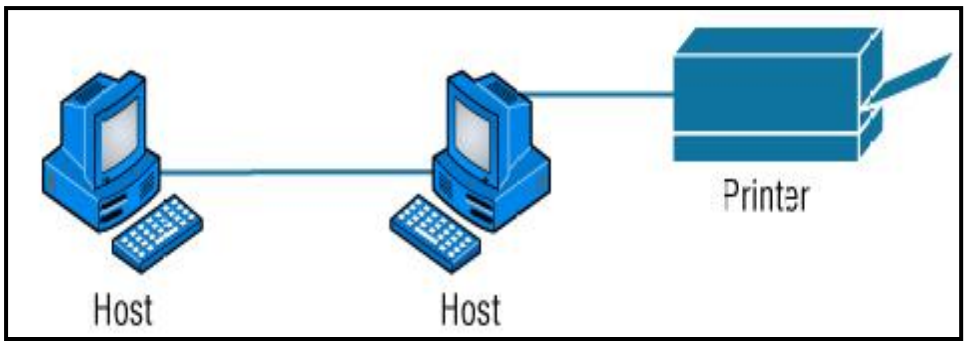


Figure 1: Basic Network

For reference purpose: pp.44 Networks + Study Guide 3rd Edition by Todd Lammle.

Using networks offers advantages relative to using a stand-alone computer—that is, a computer that is not connected to other computers and that uses software applications and data stored on its local disks. Most importantly, networks enable multiple users to share devices (e.g., printers, scanners, cameras etc.) and data (such as spread sheet and other types of files), which are collectively known as the network's resources. Sharing devices saves money. For example, rather than buying 20 printers for 20 staff members, a company can buy two printers, divide the 20 staff members into two groups and the two groups will share the two printers [one printer for each group of ten] over a network. Sharing devices also saves time.

For example, it's faster for co-workers to share data over a network than to copy data to a removable storage device and physically transport the storage device from one computer to another— as in an outdated file sharing method commonly referred to as a sneaker net (presumably because people wore sneakers when walking from computer to computer). Before networks, transferring data via floppy disks was the only possible way to share data.

Networks also allow you to manage, or administer, resources on multiple computers from a central location.

Imagine you work in the Information Technology (IT) department of a multinational bank and must verify that each of 5000 employees around the globe uses the same version of a database program. Without a network, you would have to visit every employee's machine to check and install the proper software. With a network, however, you could check the software installed on computers around the world from the computer on your desk. Because they allow you to share devices and administer computers centrally,

networks increase productivity. It's not surprising, then, that virtually all organizations depend on their networks to stay competitive.

Computers talk to each other using binary code which comprises of 0s and 1s in a sequence explaining what they want to do.

Discussion: Research on the History of Network and Internet and present it in class.

1.1.2 MANs and WANs

- **Metropolitan area network (MAN)** - a group of connected LANs in the same geographical area Also known as a **Campus Area Network (CAN)**
- **WAN (wide area network):** a group of LANs that spread over a wide geographical area – Internet is the largest and most varied WAN
- **MANs and WANs** often use different transmission methods and media than LANs
- **PAN (personal area network):** smallest network – A network of personal devices

MANs and WANs diagram below,

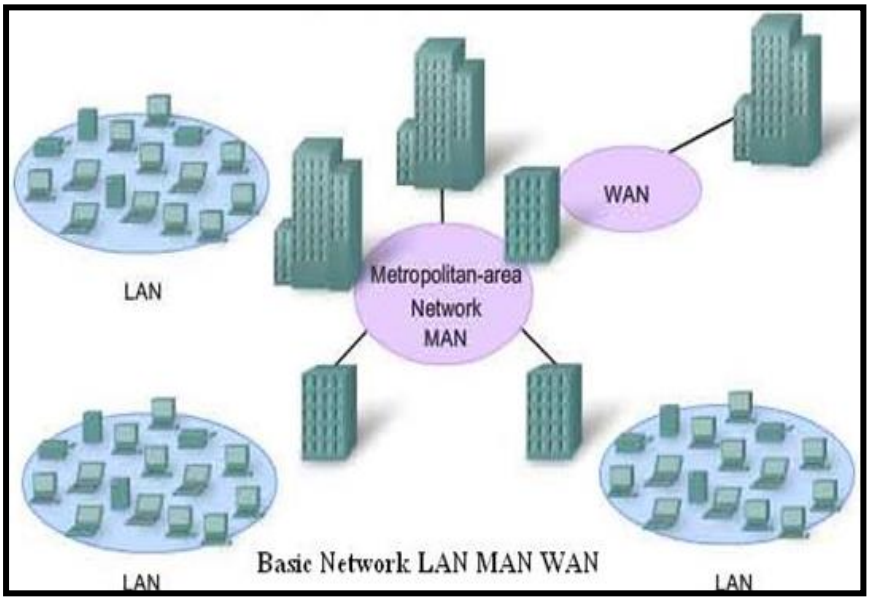


Figure 2: MANs & WANs

1.2 Selection of Network Architecture

1.2.1 Peer-to-Peer

A computer can be configured on a network in different ways relative to other computers.

Computers can have different levels of control over shared resources. They can also be made to

communicate and share resources according to different hierarchical schemes. The following sections describe two fundamental network models: peer-to-peer and client/server. The simplest form of a network is a peer-to-peer network. In a peer-to-peer network, every computer can communicate directly with every other computer. By default, no computer on a peer-to-peer network has more authority than another. However, each computer can be configured to share only some of its resources and prevent access to other resources. Traditional peer-to-peer networks typically consist of two or more general-purpose personal computers, with modest processing capabilities. Every computer is capable of sending and receiving information to and from every other computer, as shown below;

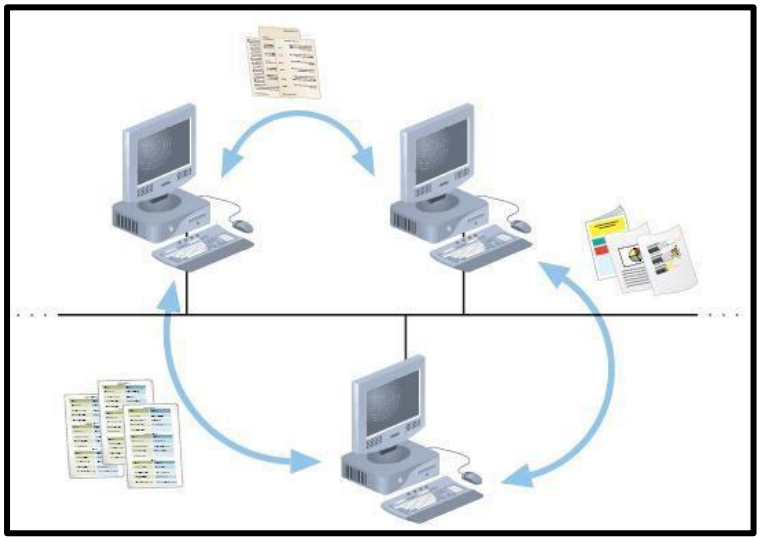


Figure 3: Resource sharing on a simple peer-to-peer network

The following are advantages of using traditional peer-to-peer networks:

- They are simple to configure. For this reason, they may be used in environments in which time or technical expertise is scarce.
- They are often less expensive to set up and maintain than other types of networks. This fact makes them suitable for environments in which saving money is critical.
- The following are disadvantages of using traditional peer-to-peer networks:
- They are not very flexible. As a peer-to-peer network grows larger, adding or changing significant elements of the network may be difficult.
- They are also not necessarily secure—meaning that in simple installations, data and other resources shared by network users can be easily discovered and used by unauthorized people.
- They are not practical for connecting more than a handful of computers, because they do not always centralize resources.

For example, if your computer is part of a peer-to-peer network that includes five other computers, and each computer user stores her spread sheets and word-processing files on her own hard disk, whenever your colleagues want to edit your files, they must access your machine on the network. If one colleague saves a changed version of one of your spread sheets on her hard disk, you'll find it difficult to keep track of which version is the most current. As you can imagine, the more computers you add to a peer-to-peer network, the more difficult it becomes to find and manage resources.

A common way to share resources on a peer-to-peer network is by modifying the file-sharing controls via the computer's operating system. For example, you could choose to create a directory on your computer's hard disk called "SharedDocs" and then configure the directory to allow all networked computers to read its files. On a peer-to-peer network, each user is responsible for configuring her computer to allow access to certain resources and prevent access to others. In other words, resource sharing is not controlled by a central computer or authority. Because access depends on many different users, it might not be uniform or secure.

Although traditional peer-to-peer networks are typically small and contained within a home or office, examples of very large peer-to-peer networks have emerged to take advantage of the Internet. These newer types of peer-to-peer networks (commonly abbreviated P2P networks) link computers from around the world to share files between each other's hard disks. Unlike the older style of peer-to-peer networking, they require specialized software (besides the computer's operating system) to allow resource sharing. Examples of these networks include Gnutella, Freenet, and the original Napster. In 2001, Napster, which allowed users around the globe to share music files, was forced to cease operation due to charges of copyright infringement from musicians and music producers. Later, the service was redesigned to provide legitimate music file-sharing services. More recently, a company called BitTorrent has made a unique high-speed data transfer technology (also called BitTorrent) the foundation of its business. The company specializes in allowing companies and individuals to share video, audio, software, and games over the Internet.

1.2.2 Client/Server Networks

Another way of designing a network is to use a central computer, known as a server, to facilitate communication and resource sharing between other computers known as clients on the network. Clients usually take the form of personal computers, also known as workstations. A network that uses a server to enable clients to share data, data storage space, and devices is known as a client/server network. (The term client/server architecture is sometimes used to refer to the design of a network in which clients rely on servers for resource sharing and processing.) In terms of resource sharing and control, you can compare the client/server network to a public library. Just as a

librarian manages the use of books and other media by patrons, a server manages the use of shared resources by clients. For example, if a patron does not have the credentials to check out books, the librarian prevents the patron from doing so. Similarly, a server allows only authorized clients to access its resources.

Every computer on a client/server network acts as a client or a server. (It is possible, but uncommon, for some computers to act as both.) Clients on a network can still run applications from and save data to their local hard disk. But by connecting to a server, they also have the option of using shared applications, data, and devices. Clients on a client/server network do not share their resources directly with each other, but rather use the server as an intermediary. Clients and servers communicate through connectivity devices such as switches and/or routers.

The below figure illustrates how resources are shared on a client/server network.

To function as a server, a computer must be running a Network Operating System [NOS].

A(n) NOS is a special type of software designed to do the following:

- Manage data and other resources for a number of clients.
- Ensure that only authorized users access the network.
- Control which type of files a user can open and read.
- Restrict when and from where users can access the network.
- Dictate which rules computers will use to communicate.
- Supply applications to clients.

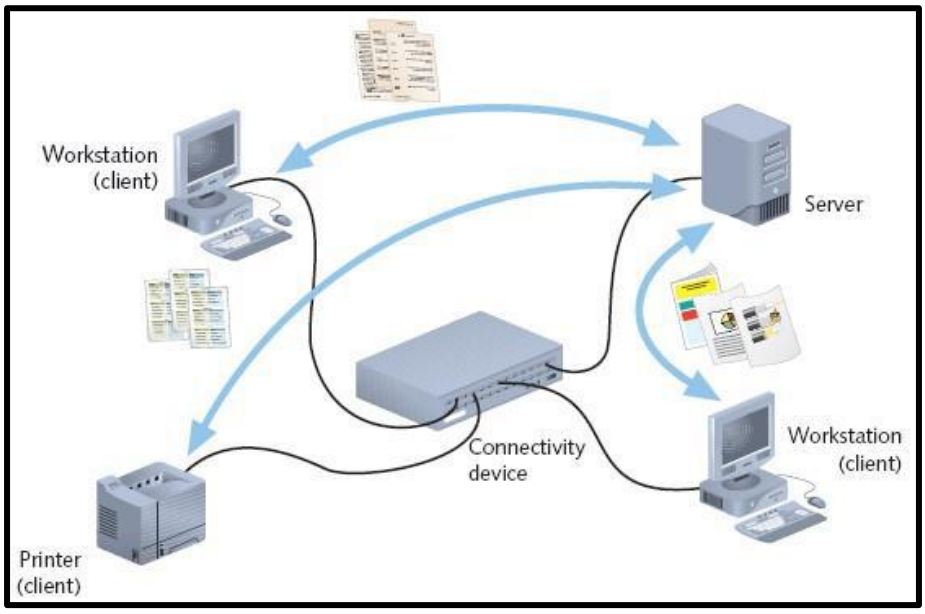


Figure 4: Resource sharing on a client/server network

Examples of popular network operating systems include various forms of UNIX and Linux, Microsoft Windows Server 2003 or Server 2008, and Mac OS X Server. (By contrast, a stand-alone computer, or

a client computer, uses an operating system, such as Windows XP or Windows Vista, which has more limited resource management capabilities.)

Usually, servers have more memory, processing power, and storage capacity than clients. They may even be equipped with special hardware designed to provide network management functions beyond that provided by the network operating system. For example, a server might contain an extra hard disk and specialized software so that if the primary hard disk fails, the secondary hard disk automatically takes its place.

Although client/server networks are typically more complex in their design, architecture and maintenance than peer-to-peer networks, they offer many advantages over peer-to-peer networks, such as:

- User logon accounts and passwords for anyone on a server-based network Access to multiple shared resources (such as data files or printers) can be centrally granted to a single user or groups of users.
- Problems on the network can be monitored, diagnosed, and often fixed from one location.
- Servers are optimized to handle heavy processing loads and dedicated to handling requests from clients, enabling faster response times.
- Because of their efficient processing and larger disk storage, servers can connect more than a handful of computers on a network.

Together, these advantages make client/server networks easier to manage, more secure, and exceedingly more powerful than peer-to-peer networks. They are also more scalable—that is, they can be more easily added onto and extended—than peer-to-peer networks.

1.2.3 NETWORK INFRASTRUCTURE IMPLEMENTATION

1.2.4. Types of Network Topology

Network Topology is the schematic description of a network arrangement, connecting various nodes (sender and receiver) through lines of connection.

i. BUS Topology

Bus topology is a network type in which every computer and network device is connected to single cable. When it has exactly two endpoints, then it is called Linear Bus topology.

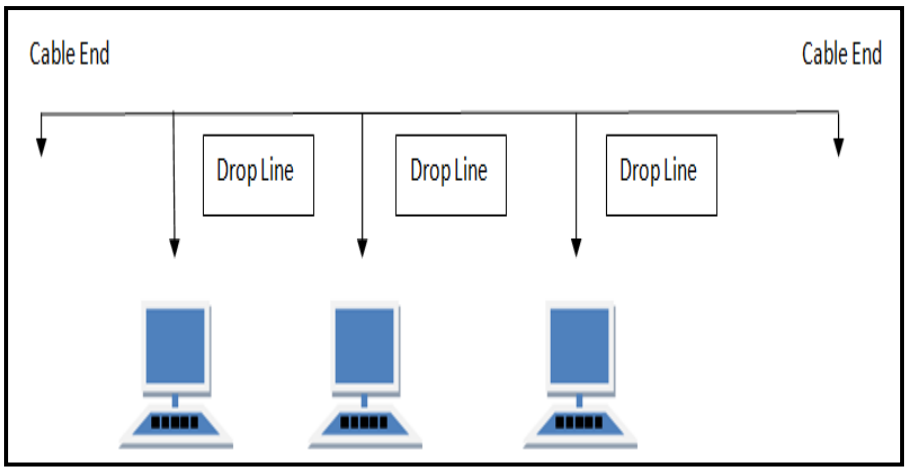


Figure 5: Bus Topology

Structures of Bus Topology

- It uses one direction when transmitting data.
- A single cable is used to connect every device.

Advantages of Bus Topology

- It is cost effective.
- Less Cabling compared to other network topology.
- Normally it is used in small networks.
- It is easy to understand.
- Expansion is easy, join only two cables together.

Disadvantages of Bus Topology

- Cable failure then the whole network fails.
- If network traffic is heavy or nodes are more the performance of the network decreases.
- Cable has a limited length.
- It is slower than the ring topology.

ii. RING Topology

It is called ring topology because it forms a ring as each computer is connected to another computer, with the last one connected to the first. There are exactly two neighbors for each device.

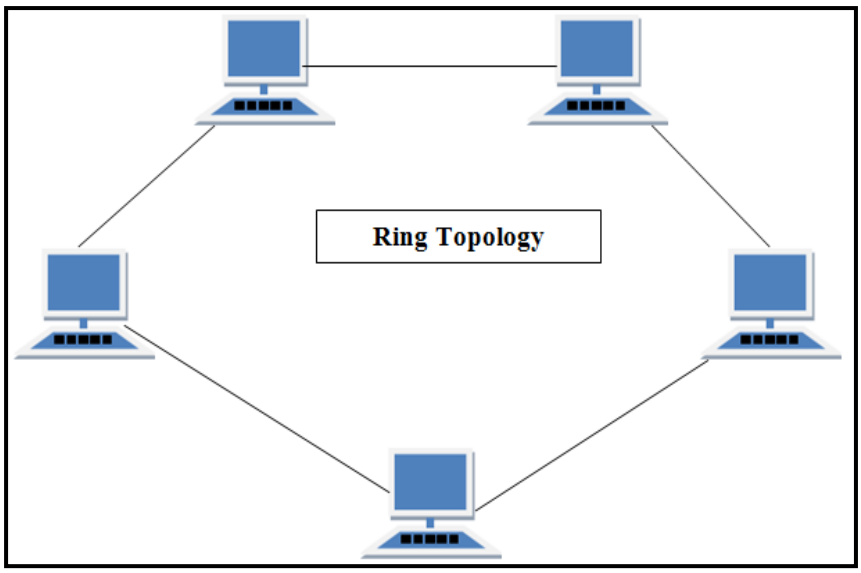


Figure 6: Ring Topology

Structures of Ring Topology

- A number of repeaters are used for Ring topology with large number of nodes, because if someone wants to send some data to the last node in the ring topology with 100 nodes, then the data will have to pass through 99 nodes to reach the 100th node. Hence to prevent data loss repeaters are used in the network.
- The transmission is unidirectional, but it can be made bidirectional by having 2 connections between each Network Node, it is called Dual Ring Topology.
- In Dual Ring Topology, two ring networks are formed, and data flow is in opposite direction in them. Also, if one ring fails, the second ring can act as a backup, to keep the network up.
- Data is transferred in a sequential manner that is bit by bit. Data transmitted, has to pass through each node of the network, till the destination node.

Advantages of Ring Topology

- Transmitting network is not affected by high traffic or by adding more nodes, as only the nodes having tokens can transmit data.
- Cheap to install and expand

Disadvantages of Ring Topology

- Troubleshooting is difficult in ring topology.
- Adding or deleting the computers disturbs the network activity.
- Failure of one computer disturbs the whole network.

iii. STAR Topology

In this type of topology all the computers are connected to a single hub through a cable. This hub is the central node and all others nodes are connected to the central node.

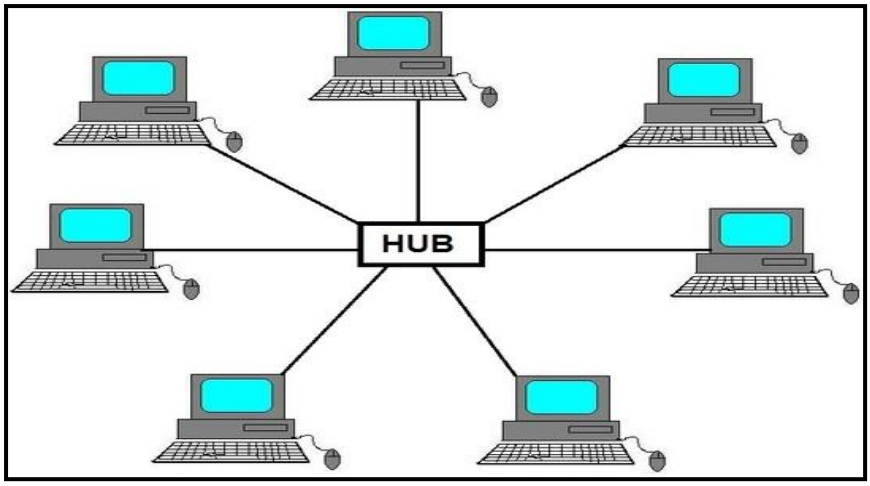


Figure 7: Star Topology

Structures of Star Topology

- Every node has its own dedicated connection to the hub.
- Hub acts as a repeater for data flow.
- Can be used with twisted pair, Optical Fiber or coaxial cable.

Advantages of Star Topology

- Fast performance with few nodes and low network traffic.
- Hub can be upgraded easily.
- Easy to troubleshoot.
- Easy to setup and modify.
- Only that node is affected which has failed, rest of the nodes can work smoothly.

Disadvantages of Star Topology

- Cost of installation is high.
- Expensive to use.
- If the hub fails then the whole network is stopped because all the nodes depend on the hub.
- Performance is based on the hub that is it depends on its capacity

iv. Mesh Topology

It is a point-to-point connection to other nodes or devices. All the network nodes are connected to each other. Mesh has n (n-2)/2 physical channels to link n devices.

There are two techniques to transmit data over the Mesh topology, they are:

1. **Routing**
2. **Flooding**

1. Routing

In routing, the nodes have a routing logic, as per the network requirements. Like routing logic to direct the data to reach the destination using the shortest distance. Or, routing logic

which has information about the broken links, and it avoids those nodes etc. We can even have routing logic, to re-configure the failed nodes.

2. Flooding

In flooding, the same data is transmitted to all the network nodes; hence no routing logic is required. The network is robust, and it's very unlikely to lose the data. But it leads to unwanted load over the network.

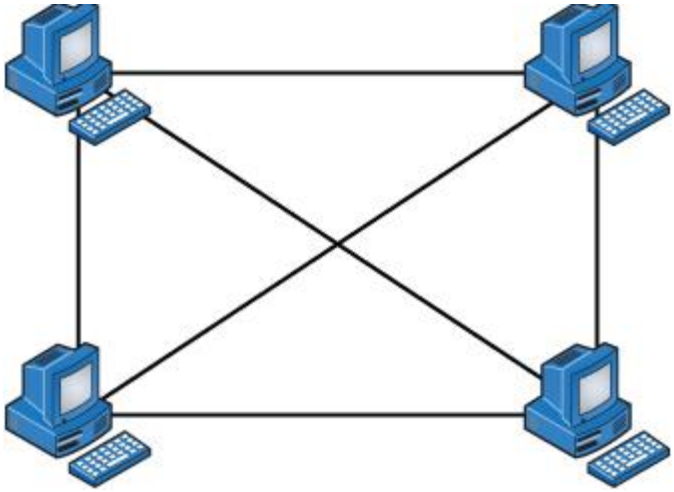


Figure 8: Mesh Topology

Types of Mesh Topology

1. Partial Mesh Topology: In this topology some of the systems are connected in the same fashion as mesh topology but some devices are only connected to two or three devices.
2. Full Mesh Topology: Each and every nodes or devices are connected to each other.

Structures of Mesh Topology

- Fully connected.
- Robust.
- Not flexible.

Advantages of Mesh Topology

- Each connection can carry its own data load.
- It is robust.
- Fault is diagnosed easily.
- Provides security and privacy.

Disadvantages of Mesh Topology

- Installation and configuration is difficult.
- Cabling cost is more.
- Bulk wiring is required.

v. TREE Topology

It has a root node and all other nodes are connected to it forming a hierarchy. It is also called hierarchical topology. It should at least have three levels to the hierarchy.

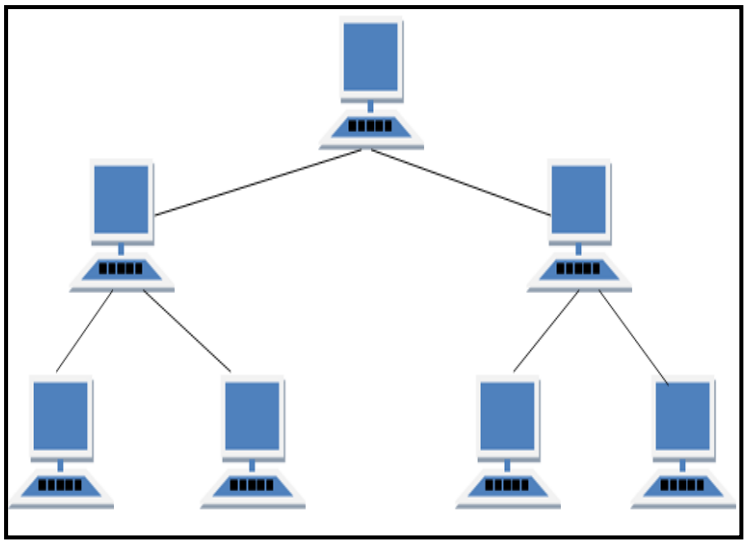


Figure 9: Tree Topology

Structures of Tree Topology

- Ideal if workstations are located in groups.
- Used in Wide Area Network.

Advantages of Tree Topology

- Extension of bus and star topologies.
- Expansion of nodes is possible and easy.
- Easily managed and maintained.
- Error detection is easily done.

Disadvantages of Tree Topology

- Heavily cabled.
- Costly.
- If more nodes are added maintenance is difficult.
- Central hub fails, network fails.

vi. Hybrid Topology

It is two different types of topologies which is a mixture of two or more topologies. For example if in an office in one department ring topology is used and in another star topology is used, connecting these topologies will result in Hybrid Topology (ring topology and star topology).

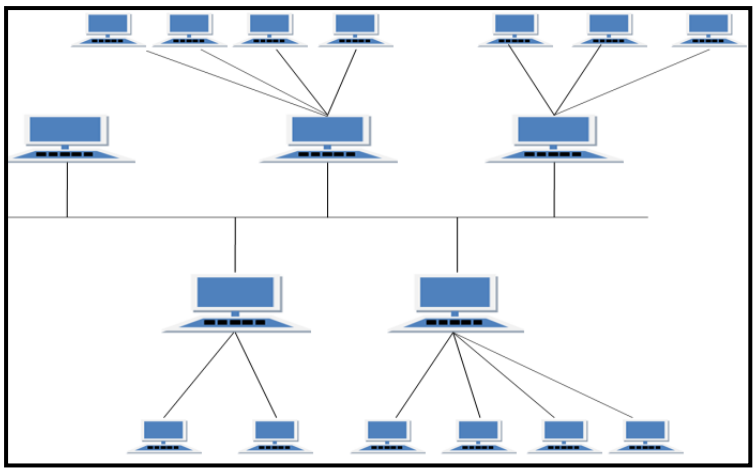


Figure 10: Hybrid Topology

Structures of Hybrid Topology

- It is a combination of two or topologies
- Inherits the advantages and disadvantages of the topologies included

Advantages of Hybrid Topology

- Reliable as Error detecting and troubleshooting is easy.
- Effective.
- Scalable as size can be increased easily.
- Flexible.

Disadvantages of Hybrid Topology

- Complex in design.
- Costly.

vii. Point- to- Point Topology

Point-To-Point: A data transmission that involves one transmitter and one receiver.

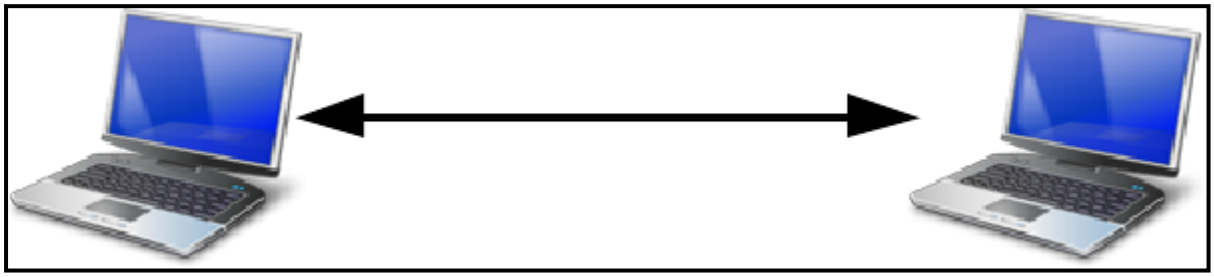


Figure 11: Point to Point

viii. **Point-To-Multipoint:** A communications arrangement in which one transmitter issues signals to multiple receivers. The receivers may be undefined, as in a broadcast transmission, or defined, as in a non-broadcast transmission. **RTT (Round Trip Time):** The length of time it

takes for a packet to go from sender to receiver, then back from receiver to sender. RTT is usually measured in milliseconds. **Simplex:** A type of transmission in which signals may travel in only one direction over a medium.

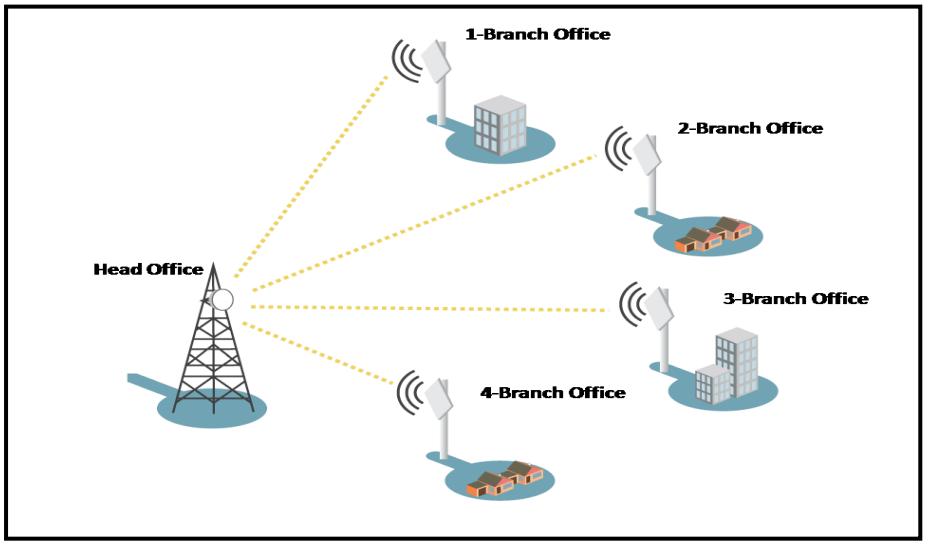


Figure 12: Point to Multipoint

1.3 Topology Selection, Backbones, and Segments

The introduction to networks has introduced you to all the types of network topologies and you can now select the right type of network. But all you have to know is about backbones and segments. Quality is the guide to your selection for network type also they do have their pros and cons to consider

There some questions to consider;

What is the budget for the job?

- How much fault tolerance and security do you need?
- Is the network going to expand? And how often is it going to be configured?
- Scalability of the network is needed because if it's going grow and change over the years.

Selection of Topology

To design a nice and cost effective solution for few computers in a room getting a network cards and use wireless that will be the best choice it's easy to setup and no cabling.

Got the right topology for the right network;

- Cost
- Ease of installation
- Ease of maintenance
- Fault tolerance requirement
- Security requirement

Network Backbone

To have a standard way of communicating with each other intelligibly and the type of network we are referring to. This is why we divide in different parts called **Backbones** and **Segments.**

Let's show the backbone and the segments on a network;

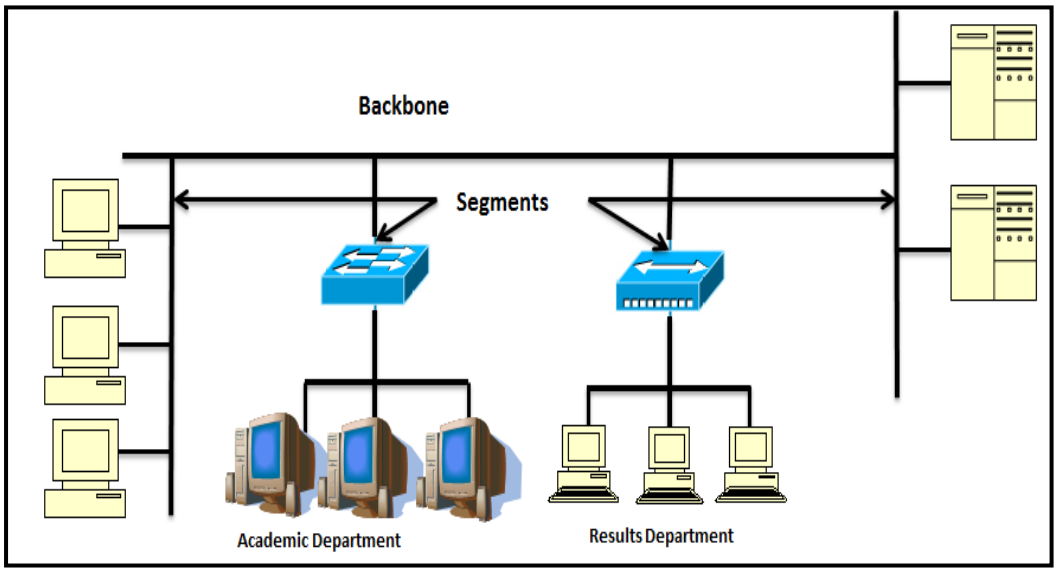


Figure 13:Backbone and Segments

Further reading on point to point and multipoint topology selecting Network topology:
Network Study Guide 3rd Edition, Todd Lammle, (pp.16-21).

Summary

There was a strong foundation built for you to understand networking with some components mentioned in building one such as router, Switches and hubs. Good understanding of various types of network connection methods (architecture) like Peer-to-Peer and Client server. Added to what you learnt were the logical and physical network topologies with their structures and Pros and Cons. This will equip you with the needed knowledge on selecting the right network topology for your network.

1.4 Open Systems Interconnection (OSI) reference model – Is a seven-layer model developed by ISO in the 1980s to categorize the layers of communication.

The OSI model is the seven layer architecture. It defines seven layers or levels in a complete communication system.

Please refer to the below diagram starting from the sender down through to the first layer and gets to the receiving end;

1.4.1 OSI MODEL FEATURES:

1. Big picture of communication over network is understandable through this OSI model.
2. We see how hardware and software work together.
3. We can understand new technologies as they are developed.
4. Troubleshooting is easier by separate networks.
5. Can be used to compare basic functional relationships on different networks.

I would like you to start by understanding the seven layers and their work before explaining the layers protocols.

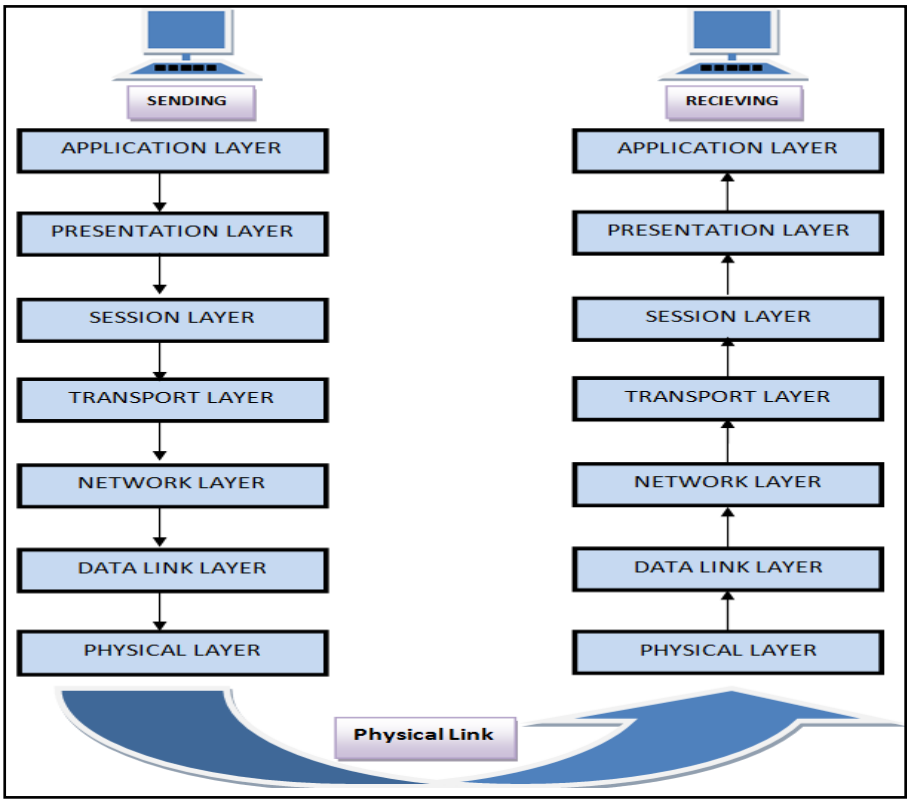


Figure 14: OSI Model

Layer 7: Application Layer

- Describes the interface between two applications, on separate computers
- Application layer protocols are used by programs that fall into two categories:
  - Provide services to a user, such as a browser and Web server
  - Utility programs that provide services to the system, such as SNMP that monitor and gather information about network traffic. Payload - data that is passed between applications or utility programs and the OS (Operating Systems)

Layer 6: Presentation Layer

- This layer helps to understand data representation in one form on a host to the other host and data is reformatted, compressed, and/or encrypted in a way that the receiving application can read.

Layer 5: Session layer

- Manages and synchronize the conversation between two different applications.
- Transfer of data from source to destination session layer streams of data are marked and are resynchronized properly, so that the ends of the messages are not cut prematurely and data loss is avoided.

Layer 5: Transport Layer

- Responsible for transporting Application layer payloads from one application to another.
- Functions such as Multiplexing, Segmenting or Splitting on the data are done by this layer
- It receives messages from the Session layer above it, converts the message into smaller units and passes it on to the Network layer.
- Transport layer can be very complex, depending upon the network requirements.

  Transport layer breaks the message (data) into small units so that they are handled more efficiently by the network layer.

Layer 4: Transport Layer

- This layer provides end to end data delivery among hosts.
- This layer takes data from the above layer and breaks it into smaller units called Segments and then gives it to the Network layer for transmission.

Layer 3: Network Layer

- This layer helps to move messages from one node to another and defines the path which the packets will follow or be routed to reach the destination.

Layer 2: Data Link Layer

- This layer takes the raw transmission data (signal, pulses etc.) from the Physical Layer and makes Data Frames, and sends that to the upper layer and vice versa.
- This layer also checks any transmission errors and sorts it out accordingly.

Layer 1: Physical Layer

- This layer deals with hardware technology and actual communication mechanism such as signaling, voltage, and cable type and wave length.
- It converts the digital/analog bits into electrical signal or optical signals.

- Data encoding is also done in this layer.

1.4.2 OSI LAYERS PROTOCOLS

There are some protocols that operate with these OSI model layers that we are going to discuss.

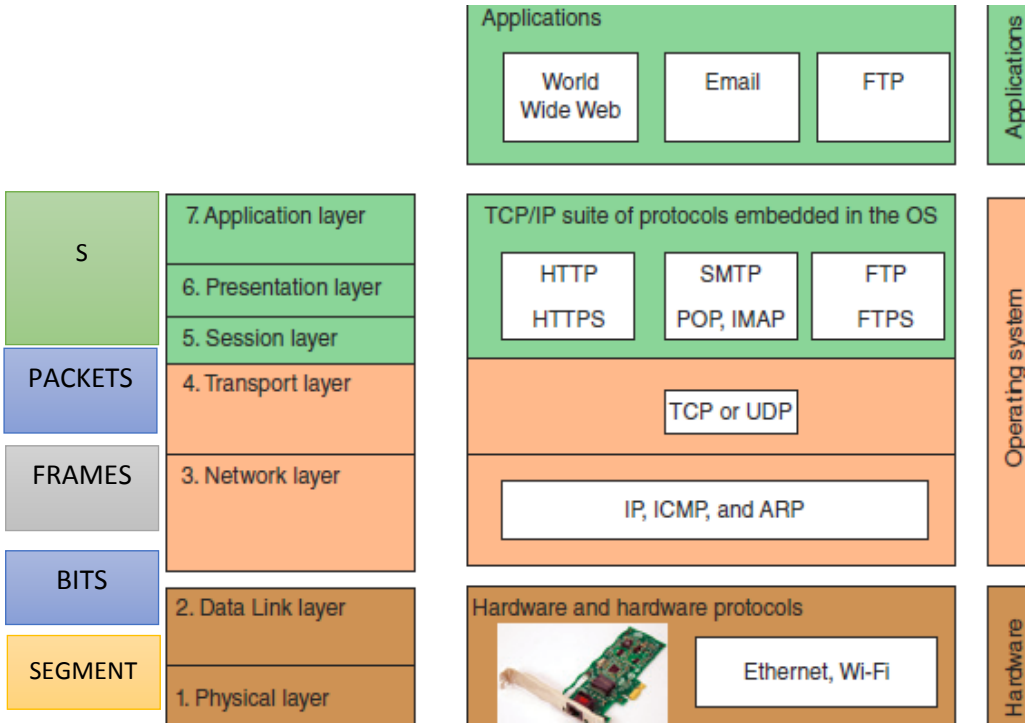


Figure 15: OSI Protocol structure

Advantages of OSI reference model:

- OSI model distinguishes well between the services, interfaces and protocols.
- Protocols of OSI model are very well hidden.
- Protocols can be replaced by new protocols as technology changes.
- Supports connection oriented services as well as connectionless service.

Drawbacks of OSI reference model:

- Model was devised before the invention of protocols.
- Fitting of protocols is tedious task.
- It is just used as a reference model.

1.5. ENCAPSULATION/DE-CAPSULATION

Encapsulation is a process to hide or protect a process from the possibility of outside interference or misuse of the system while simplifying the use of the system itself, also makes one type of network data packets to other data types. Encapsulation occurs when a protocol that is on the lower layer receives data from the protocol that is at a higher layer and put the data into a data format that is understood by the protocol. Access to the internal system so arranged through a set of interfaces.

Let's use a letter example encapsulation process in the process of mail delivery, if a letter would be sent but without the envelope, address and postage. The letter should have an identity in order to get to the destination; if it does not have an identity then the letter will not be able to get to the destination. Envelopes with address and stamp the same as the data encapsulation.

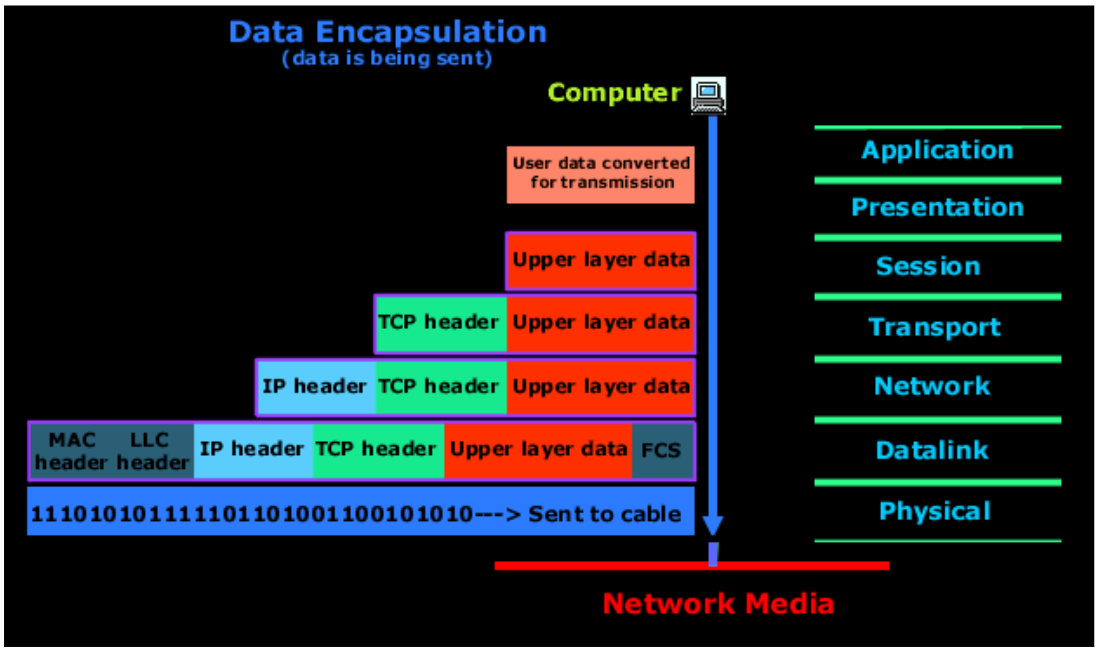


Figure 16: Encapsulation/ Decapsulation

1.5.1 Encapsulation process varies in each layer as follows:

1. Initially the data was made, when starting the process of sending, the data fell through the application layer (layer 7) responsible for the exchange of information from the computer to the network, and basically this is an interface layer between network applications used by users. This layer serves to define the request of the user. Then the data passed to the Presentation layer (layer 6), which layer is responsible for determining whether he needs to do the encryption on this request or to other forms of translational data.

Then forwarded to the session layer (layer 5) in which this layer will check whether an application requesting the information and verify that the services requested on the server. Any information will

be passed down an added header of each layer. However, the processing layer 5, 6 and 7 sometimes it is not necessary to the header.

2. Data arrived at the Transport layer (layer 4), ensuring that he already possessed a proper connection with the server and start the process by converting the information into the form of segments. Error checking and merging data coming from the same application done in this transport layer and the wholeness data is guaranteed here as well. L4PDU formed from this process.

3. The next segment is forwarded to the Network layer (layer 3), here the segments received earlier and added the network address for the station to request and network address for the requested server. Segments will be converted into a packet-packet, then network layer makes

Network header, in which there are also addresses the network layer, and placed L4PDU behind it, and formed L3PDU.

4. Then the packet-packet was passed to the Data Link layer (layer 2) and the packages were arranged and then be wrapped again into the individual frames, one example of this process is to provide the MAC address of destination and source MAC address which is then used that information to make a trailer.

Because a packet can be delivered via many devices and routers, this is where the role of MAC addresses in packets sent between a router and other routers. Then be transmitted to the media. All the information added by each layer before (as an actual file requests) must fit into a size of 46-1500 bytes of data fields in the Ethernet frame. The data link layer is responsible for sending the frame according to the topology used. L2PDU formed in this process.

5. Finally, the data arrived at the Physical layer (layer 1), information will be taken from the source to the destination. Since the physical layer frame is not known, he would pass that information into the form of bits. The addition of the header does not occur at this layer. The Physical layer is associated with the hardware. Finally, the bits

They'll be synchronized and then converted into electrical signals in the form of high and low voltage and then transmitted through the media. Example of the cable to the destination, it is in accordance with the characteristics that determine the physical layer of a layer sequence of events in which the bit stream moving through a physical medium.

At each layer there is LxPDU (Layer N Protocol Data Unit), which is a form of bytes in the header-trailer on the data. PDU is the processes on each layer of the OSI model. At each layer is also formed a new formation, the layer 2 PDU including the header and trailer is called the frame formation. In the third layer is called packet (packet). While in layer 4 are called segments (segments).

After the encapsulation process, and then sent to the server and the server will perform this process in reverse, i.e. from the physical layer to application layer, this process is called de-capsulation. If the encapsulation performed packaging, then the de-capsulation will perform the opening of the sheaths was through his layers.

1.5.2 Decapsulation Process

Decapsulation is the inverse of the encapsulation process. Encapsulation is the process of wrapping the data while the de-capsulation process is a process of opening packs. The process was reversed from the encapsulation process. Encapsulation the process starts from the uppermost layer (Application Layer) to the lowest layer (Physical layer) whiles the de-capsulation process starts from the lowest layer (Physical Layer) to the uppermost layer (Application Layer).

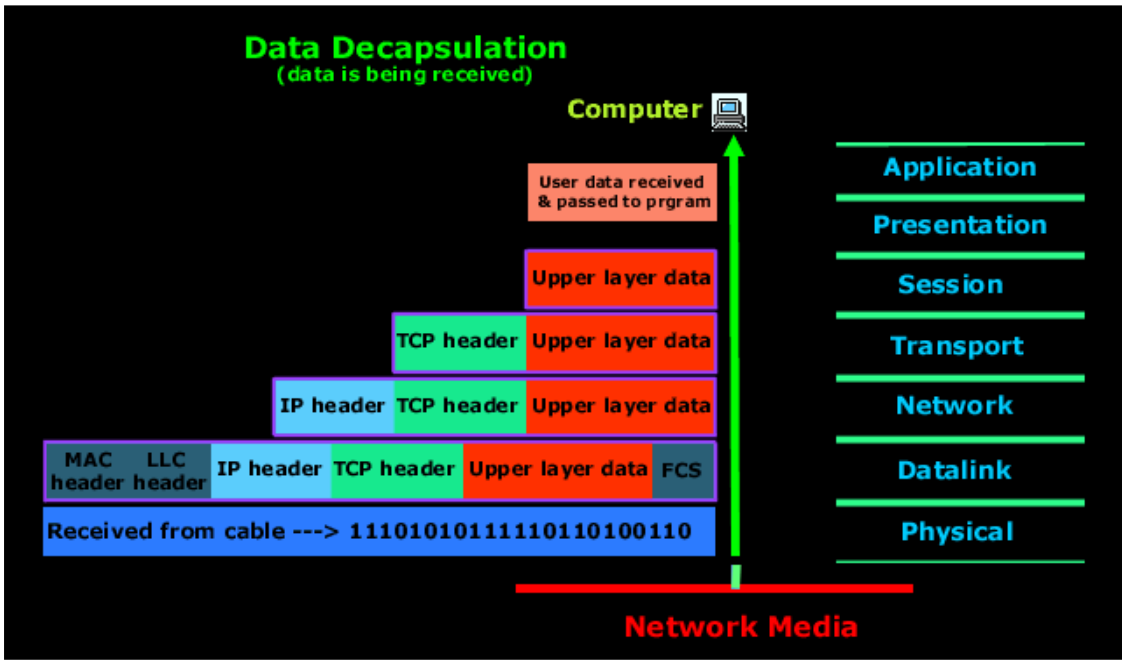


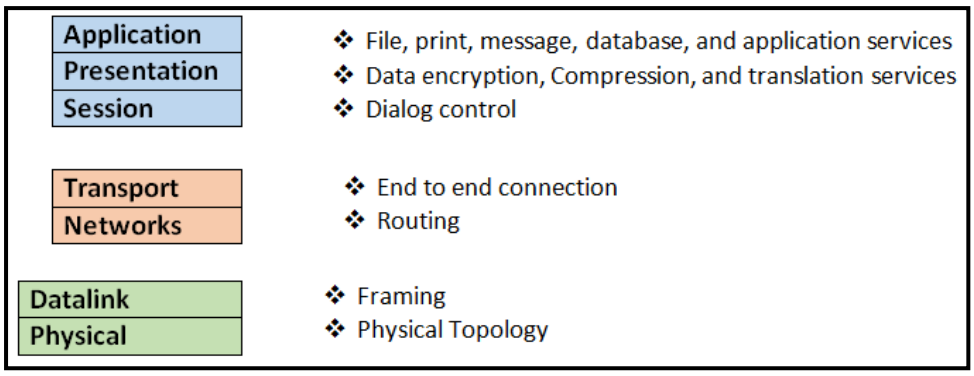
Figure 17: encapsulation / Decapsulation showing steps.

Further reading web link: http://www.firewall.cx/networking-topics/the-osi-model/179-osi-data-encapsulation.html

Key Terms

Topology, Hybrid, Mesh, Star, Bus, and Tree, OSI, Framing Routing, compression, Encryption and end to end connection.

Summary



Assessment Questions

1. What are the major roles of Data Link Layer (DLL) in an OSI model?

   A. Detection & Correction of transmission errors
   B. IP routing
   C. Gateway Allocator
   D. Making sure frames are not generated.

2. Which OSI layer allows the transmission and reception of data segments to a session layer in addition to the provision of flow control, sequence numbering and message acknowledgment?

   a. Network Layer
   b. Session Layer
   c. Transport Layer
   d. Application Layer

3. Which network component/s is/are involved in the Presentation layer of an OSI model?

   a. Gateway
   b. Multiplexer & Oscilloscope
   c. Amplifier & Hub
   d. All of the above

4. Which is the only layer of OSI layer that prevents itself from adding its own header to the data during the data transmission process?

   a. Application layer
   b. Network layer
   c. Physical layer
   d. None of the above

5. Which protocol/s is used in data link layer of an OSI model?

   a. SMB

b. NCP
c. LLC
d. ARFC

TCP is a ..... protocol.

a. stream-oriented

b. message-oriented

c. block-oriented

d. packet-oriented.

7. Which of the following is not the layer of TCP/IP protocol?

A. Physical layer

B. link layer

C. network layer

D. transport layer.

8. TCP groups a number of bytes together into a packet called a ....

A. user datagram

B. segment

C. datagram

D. packet

9. The ..... of TCP/IP protocol is responsible for figuring out how to get data to its destination.

A. application layer
B. link layer
C. network layer

D. transport layer.

10. TCP is a(n) ..... transport protocol.
A. protocol delivery
B. reliable
C. best-effort delivery
D. effortless delivery

11. ..... is the protocol that hides the underlying physical network by creating a virtual network view.

A. Internet Protocol (IP)

B. Internet Control Message Protocol(ICMP)

C. Address Resolution Protocol(ARP)

D. Bootstrap Protocol(BOOTP)

12. To use the services of UDP, we need ..... socket addresses.

A. Four

B. two

C. three

D. four

13. Which of the following is not the name of Regional Internet Registries (RIR) to administer the network number portion of IP address?

A. American Registry for Internet Numbers (ARIN)

B. Roseaux IP Europeans (RIPE)

C. Europeans Registry for Internet Numbers(ERIN)

D. Asia Pacific Network Information Center(APNIC)

14. UDP packets are called .....

A. user datagrams

B. segments

C. frames

D. packets

15. ..... addresses use 21 bits for the and 8 bits for the portion of the IP address for TCP/IP network.

A. Class A

B. Class B

C. Class C

D. Class D

16. UDP packets have fixed-size header of ..... bytes.

A. 16

B. 8

C. 32

D. 64

17. ..... messages are never sent in response to datagrams with a broadcast or a multicast destination address.

A. ICMP

B. ARP

C. IP

D. BOOTP

18. TCP assigns a sequence number to each segment that is being sent. The sequence number for each segment is number of the ..... byte carried in that segment.

A. First

B. last

C. middle

D. zero

19. ..... is responsible for converting the higher level protocol address (IP addresses) to physical network addresses.

A. Internet Protocol (IP)

B. Internet Control Message Protocol (ICMP)

C. Address Resolution Protocol (ARP)

D. Bootstrap Protocol (BOOTP)

20. UDP and TCP are both ..... layer protocols.

A. data link

B. network

C. transport

D. interface

21. ..... is a process-to-process protocol that adds only port addresses, checksum error control, and length information to the data from upper layer.

A. TCP

B. UDP

C. IP

D. ARP

22. Which of the following functions does UDP perform?

A. Process-to-process communication

B. Host-to-host communication

C. End-to-end reliable data delivery

D. Interface-to-interface communication.

23. A port address in TCP/IP is .....bits long.

A. 32

B. 48

C. 16

D. 64

24. When the IP layer of a receiving host receives a datagram, ...

A. delivery is complete

B. a transport layer protocol takes over

C. a header is added

D. a session layer protocol takes over

25. TCP/IP is a ..... hierarchical protocol suite developed before the OSI model.

A. seven-layer

B. five-layer

C. six-layer

D. four-layer

26. In the client-server model, what is the primary protocol used for communication between a browser and Web server?

A. FTP

B. TCP

C. HTTP

D. SSL

Answer: C. HTTP

27. Which two encryption protocols might be used to provide secure transmissions for browser and Web server communications?

A. HTTP and HTTPS

B. SSL and TLS

C. SSL and HTTP

D. TCP and UDP

Answer: B. SSL and TLS

28. Apache is a popular example of what type of networking software?

A. Web server

B. Browser

C. Email server

D. Email client

Answer: A. Web server

29. Which email protocol allows an email client to download email messages to the local computer?

A. IMAP4

B. SMTP

C. TCP

D. POP3

Answer: D. POP3

30. Which email protocol allows an email client to read mail stored on the mail server?

A. IMAP4

B. SMTP

C. TCP

D. POP3

Answer: A. IMAP4

31. Which client-server application allows an administrator to control a remote computer, but does not encrypt or secure the communication between client and server?

A. Telnet

B. Remote Desktop

C. FTP

D. SSH

Answer: A. Telnet

32. Which application embedded in Windows operating systems allows remote control of a computer and uses the RDP secure protocol for transmissions?

A. Telnet

B. Remote Desktop

C. FTP

D. SSH

Answer: B. Remote Desktop

33. What service provided by Windows Server 2012 R2 allows a computer to serve up applications to other computers on the network?

A. Remote Desktop Services

B. Windows 8.1

C. File Transfer Protocol

D. Active Directory

Answer: A. Remote Desktop Services

34. List three types of services a network might support that are considered part of unified communications or convergence.

A. File transfers, print services, and conversational voice

B. User authentication, streaming live audio and video, and print services

C. Web services, email services, and file services

D. Conversational voice, streaming live audio and voice, and streaming stored audio and voice

Answer: D. Conversational voice, streaming live audio and video, and streaming stored audio and video

35. Which Session layer protocol is a streaming live video teleconference likely to use on the network?

A. UDP

B. SMTP

C. RTP

D. TCP

Answer: C. RTP

36. A network consists of 10 computers, all running Windows 7 Professional. One computer acts as a file server and serves up data to other computers on the network. Which networking model does the network use?

Answer: Peer-to-peer model

37. In Question 36, suppose one computer is upgraded from Windows 7 Professional to Windows Server 2012 R2. Which networking model can the network now support that it could not support without the upgrade?

Answer: Client-server model

38. What is the name of the domain controller database that Windows Server 2012 R2 uses to store data about user access and resources on the network?

Answer: Active Directory

39. A network consists of seven computers and a network printer all connected directly to one switch. Which network topology does this network use?

Answer: Star topology

40. In Question 39 suppose a new switch is connected to the first switch by way of a network cable and three computers are connected to the new switch. Which network topology is now used?

Answer: Star-bus topology

41. What is the fundamental distinction between a Layer 2 switch and a router?

Answer: A Layer 2 switch belongs only to its local network, and a router belongs to two or more networks.

42. What is the fundamental distinction between a node and a host?

Answer: A host is a computer that hosts a resource on the network, and a node is any computer or device that can be addressed on the network.

43. What is the fundamental distinction between a MAN and a WAN?

Answer: A MAN covers a small geographical area, and a WAN covers a large geographical area.

44. What is a message called that is delivered by TCP? What is a message called that is delivered by UDP? At which layer do the two protocols work?

Answer: Segment

Answer: Datagram

Answer: Transport layer

45. Which type of address is used at the Transport layer to identify the receiving application?

Answer: Port number

46. Is TCP or UDP normally used when streaming live video? Why?

Answer: UDP, because guaranteed delivery is not as important as fast transmissions.

47. At the Network layer, what is a message called?

Answer: A packet

48. What is the primary protocol used at the Network layer?

Answer: IP (Internet Protocol)

49. At the Network layer, what type of address is used to identify the receiving host?

Answer: IP address

50. What is a PDU called at the Link layer?

Answer: A frame

51. At the Link layer, which type of network address is used to identify the receiving node?

Answer: Physical address, MAC address, hardware address, or Data Link layer address

52. Why is it important to wear an ESD strap when installing a server in a rack?

Answer: To protect the server against static electricity, or ESD

53. A computer is unable to access the network. When you check the LED lights near the computer's network port, you discover the lights are not lit. Which layer of the OSI model are you using to troubleshoot this problem? At which two layers does the network adapter work?

Answer: Physical layer

Answer: Link layer and Physical layer

54. A user complains that he cannot access a particular Web site, although he is able to access other Web sites. At which layer of the OSI model should you begin troubleshooting the problem?

Answer: Application layer

55. A user complains that Skype drops her videoconference calls and she must reconnect. At which layer of the OSI model should you begin troubleshooting? Which OSI layer is responsible for not dropping the Skype connection?

Answer: Application layer, Session layer

52. Which of the following best describes a network's physical topology?

a. The method by which multiple nodes transmit signals over a shared communications channel

b. The physical layout of a network

c. The distance spanned by a network's cable and wireless infrastructure

d. The software used to ensure reliable connections between nodes on a network

53. Which of the following is the most popular type of modern network architecture for business?

a. Client-server

b. Terminal/mainframe

c. Peer-to-peer

d. Mainframe/dial-up

54. Which of the following elements is *not* required for a client to connect to a server on a client-server LAN?

a. Protocols

b. Media

c. Email account

d. Client software

55. Which of the following are capable of acting as a network server? (Choose all that apply.)

a. Windows XP

b. Windows Server 2012 R2

c. Windows 7

d. Linux/UNIX

56. Network protocols are used to do which of the following? (Choose all that apply.)

a. To ensure reliable delivery of data

b. To determine the nearest printer for a print job

c. To interpret keyboard commands

d. To indicate the source and destination addresses for data packets

57. True or False? On a client-server network, clients may have only one protocol installed at any

time.

58 A significant difference between the peer-to-peer and client-server network types is that a peer-to-peer network:

a. Is more difficult to set up

b. Does not allow for resource sharing between workstations

c. Does not usually provide centralized management for shared resources

d. Is more secure

59 Why is it necessary for each client on a client-server network to have a unique address?

Answer: Otherwise, the device on the network won't know where to send packets.

60. Which of the following is *not* a network type option for virtual computers in VMware Workstation 10?

a. Bridged

b. NAT

c. Structured

d. Host-only

61. Which virtual disk types are available in VMware Workstation 10? (Choose all that apply.)

a. MFM

b. IDE

c. SCSI

d. SATA

62. One disk file is created for every _____ of virtual disk capacity.

a. 2 MB

b. 2 GB

c. 20 MB

d. 20 GB

63. What is the primary difference between bridged and NAT as a network type?

Answer: Bridged network gives the virtual computer direct access to the external network, whereas NAT grants access through the host's IP address.

64. True or False? Three 2-GB virtual machines could all run simultaneously on a physical machine with 4 GB of RAM.

65. What physical topology would you use to create your peer-to-peer network where all the workstations are connected to a single switch?

a. Bus

b. Tree

c. Star

d. Cube

66. Which of the following operating systems will allow you to create a peer-to-peer network from a group of workstations? (Choose all that apply.)

a. MS-DOS

b. Windows 8.1

c. Linux

d. Windows 7

67. Which of the following components are *not* necessary to create a peer-to-peer network from a group of workstations? (Choose all that apply.)

a. Client operating system

b. NIC or motherboard with onboard NIC

c. Web browser

d. Network operating system

68. Multipoint topology is

A. Bus

B. Star

C. Mesh

D. Ring

68. In mesh topology, every device has a dedicated topology of

A. Multipoint linking

B. Point to point linking

C. None of Above

D. Both a and b

69. Bus, ring and star topologies are mostly used in the

A. LAN

B. MAN

C. WAN

D. Internetwork

70. Combination of two or more topologies are called

A. Star

B. Bus

C. Ring

D. Hybrid

71. During Encapsulation/De-encapsulation, at which layer of the hybrid TCP/IP-OSI model a trailer is usually added to the message?

a. Physical

b. Transport

c. Data Link

d. Application

e. Internet

72. What is called the message created at the Internet layer?

a. frame

b. IP packet

c. TCP segment

d. datagram

73. What is called the message created at the Data-Link layer?

a. a frame

b. an IP packet

c. a TCP segment

d. a datagram

74. Which of the following statement(s) is/are true regarding Encapsulation/De-encapsulation?

a. No message is created at the Physical layer

b. As soon as a message is created at a layer, it is passed down to the next lower-layer.

c. During De-encapsulation process, protocol-related information are removed from the received message and passed down to the next lower-layer.

d. None of the above.

75. In which of these Network configurations the File Server system can be implemented:

(Select all correct answers)

a. Microcomputer-to-Mainframe

b. Terminal-to-Mainframe

c. Microcomputer-to-LAN

d. All of the above

76. Client/Server systems are found in: (Circle all correct answers)

a. Microcomputer-to-LAN configurations

b. Microcomputer-to-Mainframe configurations

c. Terminal-to-Mainframe configurations

d. Terminal-to-Microcomputer configurations

In Client/Server system, the Client could be a terminal and the

server could be a Mainframe. T F

77. Inquiry/Response applications are typical applications used with

Terminal-to-Mainframe configuration T F

78. Which of the following are true regarding Client/Server System?

a) It is platform independent

b) Heavier processing is done by the client computer

c) The client and the server should have some processing capabilities

d) No processing done by the server

e) All the processing is done by user PC

79. Which of the following are true regarding File Server System

a) It is platform independent

b) The client and the server should have some processing capabilities

c) No processing done by the server

d) All the processing is done by user PC

80. Which of the following configuration is used for the Internet web service, i.e. for access websites?

a) File Server System access

b) Client/server system

c) Internet access system

d) All of the above

Review Questions

1. What are the three basic LAN topologies?
2. What common WAN topology often results in multiple connections to a single site (leading to a high degree of fault tolerance) and has one-to-many connections?
3. What is the term for a device that shares its resources with other network devices?
4. What network model draws a clear distinction between devices that share their resources and devices that do not?
5. Which network topology or connection type can be implemented with only two endpoints?
6. What device is generally implemented as a star topology?
7. What does MPLS stand for?
8. What does WAN stand for?
9. Will a computer that shares no resources most likely be connected to the backbone or to a segment?
10. Which LAN topology is characterized by all devices being daisy-chained together with the devices at each end being connected to only one other device?
11. What is the primary difference between peer-to-peer and client-server architectures?

On a peer-to-peer network consisting of four Windows 7 or 8.1 workstations, each user can individually control which of her local data files she wants to share with other users.

a. True

b. False

Discussion Question

Activity 1: Case Study Problem

RGI is an independent, business for distance, and contact learning consisting of an Owner, a CAO, an administrator, and five Deans and they are deciding to implement a network. The company occupies half of a small building in an office park. The volume of business had been stable for the past three years, but recently it has been increasing. To handle the increased business volume, two new agents will be hired.

Illustration of the current business network arrangement

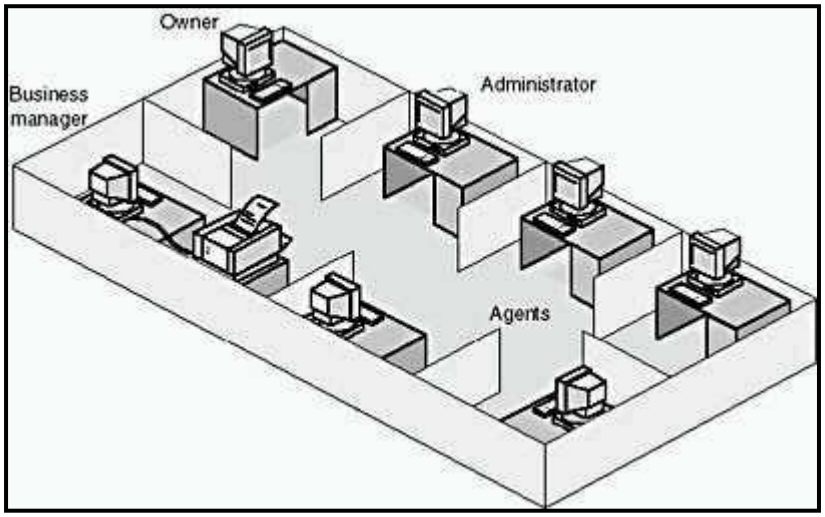


Figure 18: Case study model

Everyone in RGI Company has a computer, but the business manager has the only printer. These computers are not connected by any form of networking. When agents need to print a document, they must first copy the file to a floppy disk, and then carry it to the business manager's computer, where they are finally able to print it. Similarly, when staff members want to share data, the only means available is to copy the data on one computer to a floppy disk and insert the disk in another computer.

Recently, problems have arisen. The business manager is spending too much time printing other people's documents; and it is frequently unclear which copy of a given document is the current and authoritative version.

Your task is to design a network for this company.

To clarify the task of choosing a solution, you ask some questions.

Circle the most appropriate answers to the following questions:

1. Which type of network would you suggest for this company?
   - ❖ Peer-to-peer
   - ❖ Server-based
2. Which network topology would be most appropriate in this situation?
   - ❖ Bus
   - ❖ Ring
   - ❖ Star
   - ❖ Mesh
   - ❖ Star bus
   - ❖ Star ring

3. In groups of four or five discuss names and describe network topologies? Be aware of the difference between physical networks (what humans see) and logical networks (what the equipment can "see").
4. To show your understanding of LAN and WAN. Explain where to apply each and give example of the real world.
5. Which LAN topology is characterized by all devices being daisy-chained together with the devices at each end being connected to only one other device?

Reference:

Recommended books

i. CompTIA Network + Study guide by Todd Lammle (pages 2-21 and 27- 53)
ii. CompTIA Network + Guide to Networks by Tamara Dean (pages 1-25)
iii. Networks+ Study Guide 3rd Edition, Chapter 2 by Todd Lammle pp. 29-48

WEB:

i. https://www.tutorialspoint.com//ipv4/ipv4_osi_model.htm
ii. http://www.learn44.com/introduction-to-the-osi-reference-model-layered-approach/
iii. https://www.tutorialspoint.com/ipv4/ipv4_osi_model.htm
iv. http://www.gocomputertraining.com/network-fundamentals.html
v. http://www.studytonight.com/computer-networks/network-topology-types
vi. http://www.webopedia.com/quick_ref/OSI_Layers.asp

vii. http://www.omnisecu.com/tcpip/osi-model.php

Facebook:

i. https://www.facebook.com/search/top/?q=network%20topologies
ii. https://www.facebook.com/pages/Networktopology/109488015737624?fref=ts&rf=115047478510671
iii. https://www.facebook.com/search/top/?q=osi%20model

CHAPTER 2 ETHERNET

LEARNING OUTCOME

- ***Understanding Ethernet Standards***
- ***Know the type of cables available for networks***
- ***Be able to configure and understand addressing schema***
- ***After reading this section, learners will be able to demonstrate the ability to*** distinguish ***between different forms of signalling.***
- ***Understand types of layers and their work with Ethernet***
- ***Understand Ethernet frames***
- ***Good knowledge on how Half and Full Duplex work***
- ***Know how to work with Binaries and MAC addresses***

2.1 CURRENT ETHERNET STANDARDS

Introduction

Ethernet is a type of network cabling and signaling specifications developed by Xerox in the late 1970.

The 10BASE-T Ethernet standard in 1990, a lot has changed. Twisted-pair cabling went from 10BASE-T with a data rate of 10 Mb/s over category 3 cabling in 1990, to 100BASE-TX at 100 Mb/s over category 5 in 1995, to 1000BASE-T at 1 Gb/s over category 5e or higher in 1999, to 10GBASE-T at 10 Gb/s over category 6A in 2006. The future will be from 40GBASE-T at 40 GB/s over future category 8 cabling. Fiber is another story—moving from 1 GB/s over multimode and single mode fiber in 1998 to 10 GB/s in 2003 to 100 GB/s in 2010. That's speed increased by a factor of 100 in just 12 years!

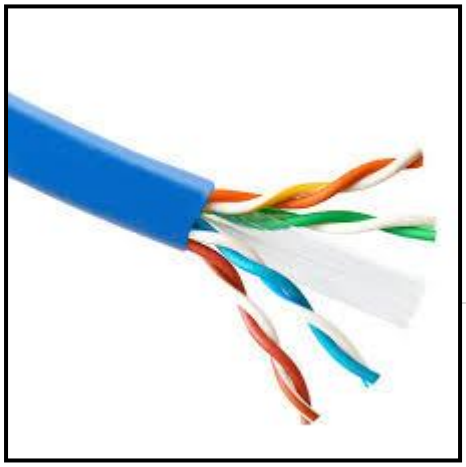


Figure 19: Types of UTP and STP

The cabling standards were developed in 1989 by TIA-568 standard the same time that the IEEE 10BASE-T standard was published. The TIA-568 established the framework for structured cabling still exists today, including the hierarchical star topology for horizontal and backbone cabling, the standardization of 100 Ohm balanced twisted-pair cabling in the horizontal with a maximum channel distance of 100 meters, and the standardization of the venerable 8-position modular (RJ45) connector termination in the work area.

A lot of changes since then, but certain principles remain the same. One such principle is the principle of "backwards compatibility" with previous generations of cabling. For example, category 6A cabling, which is designed to support 10GBASE-T (10 Gigabit) applications up to 100 meters is also fully backwards compatible with all lower categories of cabling and will support 1000BASE-T, 100BASE-TX and 10BASE-T applications. Higher category components (cables, cords and connectors) can be substituted for lower category components in a channel and provide full functionality with equal to or better performance.
For example when planning a new data center, it is therefore sensible to install a cabling system that can support at least two generations of Ethernet over a 10 year lifespan.
For copper switch-to-server connections that means category 6A (for now) and for fiber backbone switch-to-switch connections that means either 0M4 laser-optimized multimode or single mode optical fiber. But watch out—category 8 has just started to be used.

2.1.1 Future of Ethernet

Each and every person talk of the need for higher speed switch-to-server connections in today's virtualized data centers, the IEEE 802.3bq 40GBASE-T task force is well on their way to defining the topology, reach objectives, cabling requirements and the power consumption of the next generation physical layer interface (PHY) to transmit at speeds of 40 Gb/s. There is TIA 42.7 subcommittee developing the category 8 cabling standards to support the future 40GBASE-T standard.

The Evolution of Ethernet Standards to Meet Higher Speeds				
Date	**IEEE Std.**	**Name**	**Data Rate**	**Type of Cabling**
1990	802.3i	10BASE-T	10 Mb/s	Category 3 cabling
1995	802.3u	100BASE-TX	100 Mb/s*	Category 5 cabling
1998	802.3z	1000BASE-SX	1 Gb/s	Multimode fiber
	802.3z	1000BASE-LX/EX		Single mode fiber
1999	802.3ab	1000BASE-T	1 Gb/s*	Category 5e or higher Category
2003	802.3ae	10GBASE-SR	10 Gb/s	Laser-Optimized MMF
	802.3ae	10GBASE-LR/ER		Single mode fiber
2006	802.3an	10GBASE-T	10 Gb/s*	Category 6A cabling
2015	802.3bq	40GBASE-T	40 Gb/s*	Category 8 (Class I & II) Cabling
2010	802.3ba	40GBASE-SR4/LR4	40 Gb/s	Laser-Optimized MMF or SMF
	802.3ba	100GBASE-SR10/LR4/ER4	100 Gb/s	Laser-Optimized MMF or SMF
2015	802.3bm	100GBASE-SR4	100 Gb/s	Laser-Optimized MMF
2016	SG	Under development	400 Gb/s	Laser-Optimized MMF or SMF
Note: *with auto negotiation				

Figure 20: Ethernet Standard

It (40GBASE-T) is intended to be used for switch-to-server connections in the data center; the ability to interoperate with legacy slower-speed Ethernet technologies through auto negotiation is a goal that is here to stay. This is facilitated by the use of the 8-pin modular (RJ-45) connector, and TIA is therefore developing the category 8 cabling standard to include this recognized de facto interface. Category 8 cabling is specified to 2 GHz, or four times the bandwidth specified for Category 6A cabling and twice the bandwidth specified for Category 7A cabling.

2.2 Ethernet Some Key Considerations

Category 8 on the horizon might be a reason for data center managers to hold off on a cabling upgrade to a lesser performing category 7 or 7A shielded cabling. One drawback of Category 7 and 7A is that it uses a different IEC (International Electro technical Commission) connector design that is not backwards compatible with the 8-pin modular (RJ-45) connector, and therefore would require an adapter patch cord to interface with legacy equipment. And just to make it more interesting, there is a version of Category 8 cabling called Class II that is based on improved Category 7A components.

To understand the crucial part of networks ahead you have to get to the bottom of this.

Ethernet operate on the data link layer and physical layer specifications. Go through the terms used in the Ethernet technology below;

2.2.1 Collision Domain

A collision domain is, as the name implies, a part of a network where packet collisions can occur. A collision occurs when two devices send a packet at the same time on the shared

network segment. The packets collide and both devices must send the packets again, which reduces network efficiency. Collisions are often in a hub environment, because each port on a hub is in the same collision domain. By contrast, each port on a bridge, a switch or a router is in a separate collision domain. The following example illustrates collision domains.

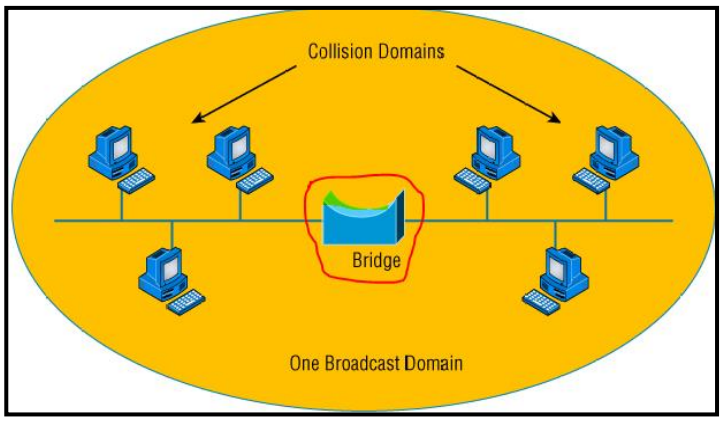


Figure 21: Bridges breaks up collision domain

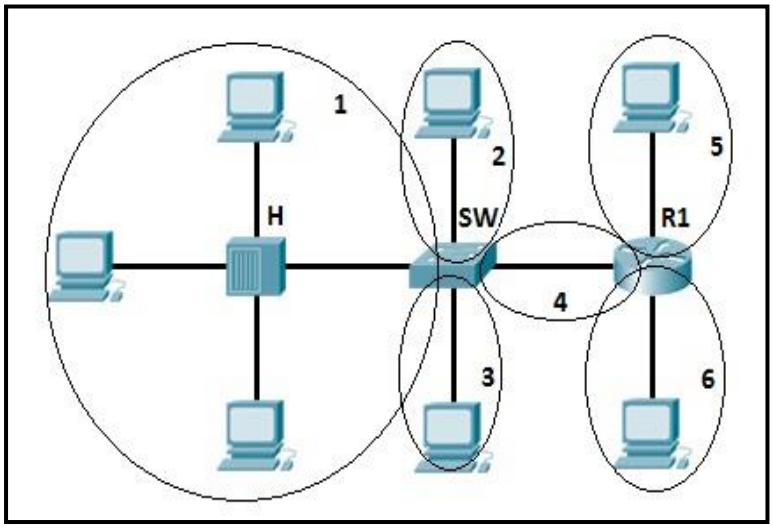


Figure 22: Collision Domain

As you can see, we have 6 collision domains.

Note: *remember, each port on a hub is in the same collision domain. Each port on a bridge, a switch or router is in a separate collision domain.*

2.2.2 Broadcast Domain

A broadcast domain is a domain in which a broadcast is forwarded. A broadcast domain contains all devices that can reach each other at the data link layer (OSI layer 2) by using broadcast. All ports on a hub or a switch are by default in the same broadcast domain. All

ports on a router are in the different broadcast domains and routers don't forward broadcasts from one broadcast domain to another.

The following example clarifies the concept.

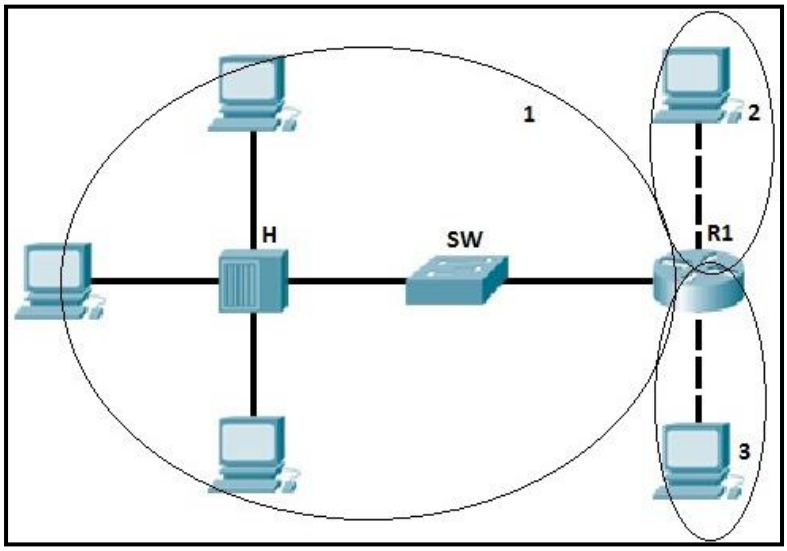


Figure 23: Three Broadcast Domain

From the above diagram we have three broadcast domains, since all ports on a hub or a switch are in the same broadcast domain, and all ports on a router are in a different broadcast domain.

2.2.3 CSMA/CD

CSMA/CD (Carrier sense multiple access with collision detection) acts as decision maker to hosts when to send packets on a shared network segment and to detect collisions. For example, in a hub network, if the two devices can send packets at the same time can cause a collision. CSMA/CD enables devices to "sense" the wire to ensure that no other device is currently transmitting packets. Also, if two devices "sense" that the wire is clear and send packets at the same time, a collision can occur. If the collision occurs, packets have to be resend after a random period of time. This is what will happen if there happen to have a collision.

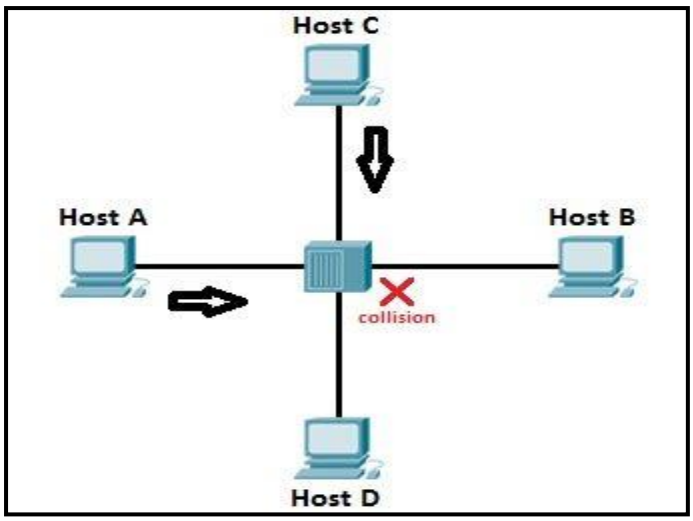


Figure 24: Collision

From the diagram above we have a hub network. Host A is trying to communicate with host B. Host A "senses" the wire and decides to send packets. But, in the same time, host C sends its packets to host D and the collision occurs. The sending devices (host A and host C) detect the collision and resend the packet after a random period of time for 15 tries and time out.

When a collision happens on Ethernet LAN;

- A Jam signal will inform all the devices to stop sending because there is a collision.
- By default the collision invokes a random back off algorithm
- All the devices on that network segment will wait for the timer to lapse and start sending again
- All hosts have the same priority to transmit after the expiry of the timer

These are the effects of having a CSMA/CD network that has sustained a substantial collision;

- Delay
- Low throughput
- Congestion

2.2.4 Broadband /Baseband

We can send signals in two ways for Analog and digital by Broadband and Baseband.

Broadband is commonly used in homes using same medium and for all our analog voice and digital data using same cable.

Baseband is used by LANs. This is where Ethernet uses one digital signal at a time also requiring the available bandwidth. There is a possibility of collisions if all hosts send multiple signals at a time even on wireless, except of those using analog signaling.

- Baseband transmission
    - Digital signals that are carried on a single channel
    - Requires exclusive use of wire's capacity

- Example: Ethernet

- Broadband transmission
  - Multiple transmissions share a single media
  - Transmission sharing the same media rely on multiplexing to manage multiple signals
  - Example: Cable TV and cable Internet

2.2.5 Bit Rates vs Baud Rate

Both are commonly expressed as bits transmitted per second, called bit rate. A ***Baud rate*** was used in 1970 and 1980s measuring the speed of telegraph transmissions and named after a French Engineer Jean-Maurice-Emile-Baudot by ***bps.***

- Throughput
  - Amount of data transmitted during given time period
  - Also called payload rate or effective data rate
  - Expressed as bits transmitted per second
- Bandwidth (strict definition)
  - Difference between highest and lowest frequencies medium can transmit
  - Range of frequencies

Quantity	Prefix	Abbreviation
1 bit per second	n/a	1 bps = 1 bit per second
1000 bits per second	kilo	1 Kbps = 1 kilobit per second
1,000,000 bits per second	mega	1 Mbps = 1 megabit per second
1,000,000,000 bits per second	giga	1 Gbps = 1 gigabit per second
1,000,000,000,000 bits per second	tera	1 Tbps = 1 terabit per second

Figure 25: Throughput and bandwidth measure

So one Baud is one electronic state change per second, 0.2 volts to 3 volts or from binary 0 to 1. However, since a single state change can involve more than a single bit of data, the bps unit of measurement has replacement has substituted it as a more accurate meaning of how much data you're transmitting or receiving.

2.2.6 Wavelength

Have you ever heard of someone saying “You are not in the same wavelength with me” meaning you are not thinking the same. Just like electromagnetic radiation, radio waves, light waves, or infrared (heat) waves have got characteristic patterns as they move around the space. These patterns are not the same in shape and length. Refer to the diagram below,

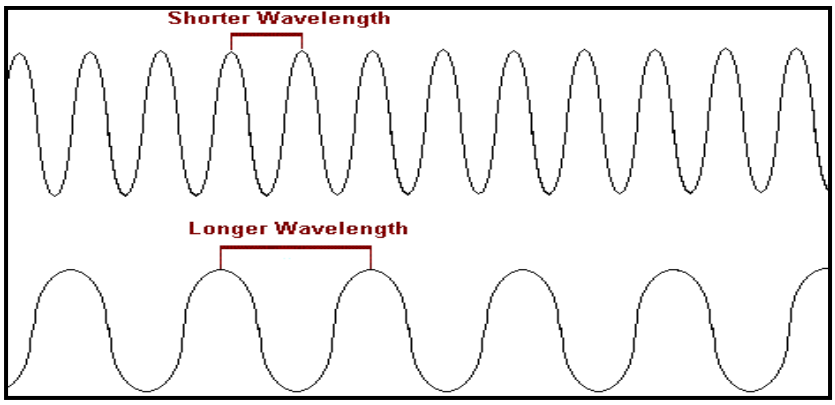


Figure 26: Short & Long Wavelength

The distance between the peaks (high point) is wavelength. So when the two wavelengths are different they are not in the same wavelength and that’s how we can tell the different kind of electromagnetic energy apart. That’s our advantage in electronics to send traffic on different wavelengths at the same simultaneously.

2.3 Ethernet in the Datalink and Physical Layers

2.3.1 Sampling sizes

Computers use 0s or 1s, on and off of switches at very fast speed working on one step at a time. Using succession of discrete amplitude values, computers can convert analog audio signals to digital signals using a discrete step the analog waveform mathematically. The computer captures a series of samples in specified sizes called sampling sizes when converting to analog. Check above on the Bit rate and Baud rate subtitle.

2.3.2 Half and full duplex Ethernet

Ethernet **IEEE 802.3 standard** defines the **half duplex;** Cisco describes, it uses a digital signal on a wire pair flowing in both tracks on the wire. Half Duplex always employs the Carrier Sense Multiple Access with Collision Domain (CSMA/CD), so that it can retransmit the transmission if collision occurs. It is a one way side talk at a time. Once one side has finished transmitting its data, the other side can respond. Only one node can talk at a time. As you can recognize, this method of communication is not very efficient and requires more time to send/receive larger amounts of data.

The older networks used to work with half duplex mode, because of the constraints of the network medium (coax cable) and hardware equipment (hubs). It has a limit of up to 10BaseT, as Cisco describes 10BaseT is not more than 3 to 4Mbps.

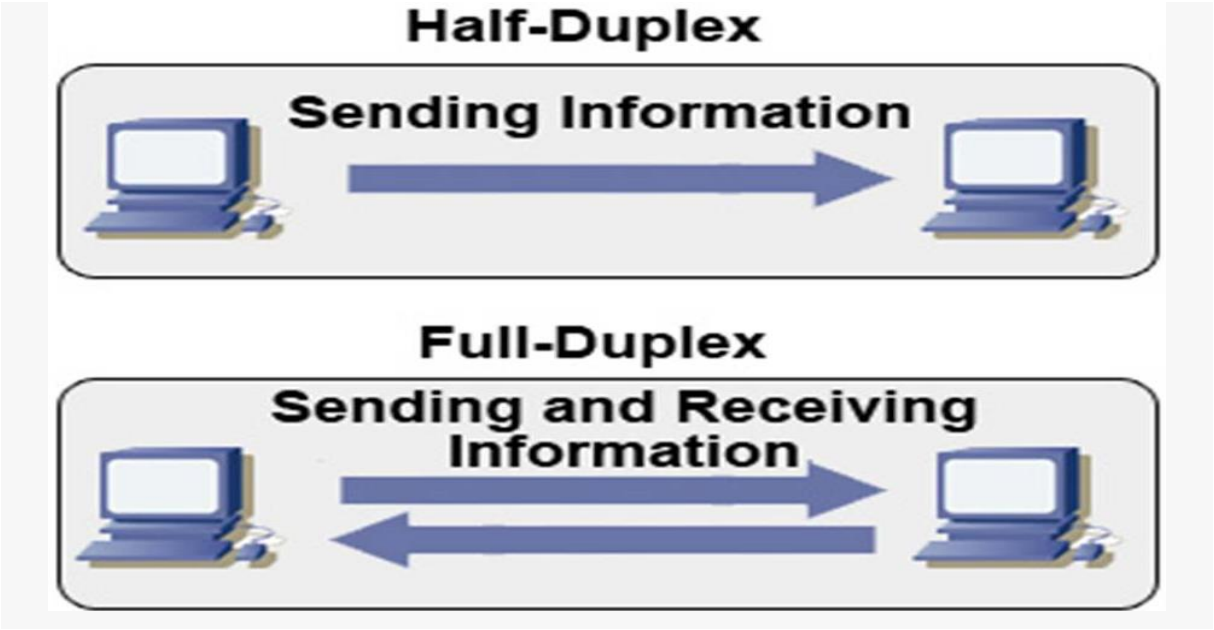


Figure 27: Half Duplex Sending Information and Full Duplex Receiving information

2.3.3 Full Duplex Ethernet Mode

On the other hand, full-duplex is used to describe communication where both sides are able to send and receive data at the same time. In these cases, there is no danger of a collision **"auto detect mechanism"** and therefore the transfer of data is completed much faster (it can run at 10Mpbs or 100Mbps).

These days, all networks make use of switches (rather than hubs) and UTP Ethernet cabling, which allow full-duplex communication between all connected hosts.

Full duplex Ethernet mode can be deployed in the following conditions:

- Switch to host connection
- Switch to a switch connection
- Host to host connection by using crossover cable

A few points to remember for deploying full duplex Ethernet mode:

- No collisions in full duplex.
- A dedicated port is necessary for each full duplex device.
- Host' network identification card and switch port have to able to operate in full duplex mode.

2.4 Ethernet at the Data Link Layer

2.4.1 MAC

How Ethernet is used these days? What are MAC addresses and how are they used in networking. This is going to cover the Datalink layer where the Ethernet protocol is found.

In short a media access control **address** (**MAC address**) or Hardware address of a computer is a unique identifier assigned to network interfaces for communications at the data link layer of a network segment. **MAC addresses** are used as a network address for most IEEE 802 network technologies, including Ethernet and Wi-Fi.

It is used for data sharing, printers and videoconferencing. For example there are two computers (Diana and Vicky) personalized connected to the same hub how they communicate. It is responsible for framing packets received from the network layer and preparing them for transmission on the local network through the Ethernet contention media-access method (CSMA/CD as discussed previously.

The Ethernet MAC addresses are made up of Hexadecimal addresses first things first let's start by discussing binary, decimal, and followed by Hexadecimal addresses and how to convert one another.

2.4.2 Binary to decimal and hexadecimal conversion

In networking it is important to follow steps before we talk about TCP/IP protocol stack and IP addressing we must know the difference between binary, decimal, and hexadecimal numbers and how we convert from one format to the other.

Let's start with binary numbering each digit is limited to either a 1 (one) or a 0 (zero) each digit is called 1 bit (short for binary digit). When counting either 4 or 8 bits together and are called ***nibble*** and a ***byte*** respectively. Interestingly binary numbering is the value represented in a decimal format- the typical decimal format being the base-10 number scheme. Binary numbers are from right moving left with each spot having double the value of the previous spot.

Below is a decimal value of each bit location in a nibble or a byte. *Nibble* is four bits and a *byte* is an eight bits. However in network addressing we refer a byte as an ***octet*** or octal addressing. In Mathematics octal addressing refers to base 8, not the same as base 10.

Nibble values	Byte values
8 4 2 1	128 64 32 16 8 4 2 1

Figure 28: Binary Values

Step by step on binary

2.4.3 Binary to Octal

An easy way to convert from binary to octal is to group binary digits into sets of three, starting with the least significant (rightmost) digits.

Then, look up each group in a table:

Binary:	000	001	010	011	100	101	110	111
Octal:	0	1	2	3	4	5	6	7

Binary = 011 100 101

Octal = 3 4 5 = 345 oct

2.4.4 Binary to Hexadecimal

An equally easy way to convert from binary to hexadecimal is to group binary digits into sets of four, starting with the least significant (rightmost) digits.

Binary: 11100101 = 1110 0101

Then, look up each group in a table:

Binary:	0000	0001	0010	0011	0100	0101	0110	0111
Hexadecimal:	0	1	2	3	4	5	6	7
Binary:	1000	1001	1010	1011	1100	1101	1110	1111
Hexadecimal:	8	9	A	B	C	D	E	F

2.4.5 Binary to Decimal

They say there are only 10 people in this world: those that understand binary and those that don't.

If you don't get that joke, you'll need a method to convert from binary to decimal. One method involves addition and multiplication.

1. Start the decimal result at 0.
2. Remove the most significant binary digit (leftmost) and add it to the result.
3. If all binary digits have been removed, you're done. Stop.
4. Otherwise, multiply the result by 2.
5. Go to step 2.

Here is an example of converting 11100000000 binary to decimal:

Binary Digits	Operation	Decimal Result	Operation	Decimal Result
11100000000	+1	1	× 2	2
1100000000	+1	3	× 2	6
100000000	+1	7	× 2	14
00000000	+0	14	× 2	28
0000000	+0	28	× 2	56
000000	+0	56	× 2	112
00000	+0	112	× 2	224
0000	+0	224	× 2	448
000	+0	448	× 2	896
00	+0	896	× 2	1792
0	+0	1792	Done.	

Figure 29: Binary Conversion

Memories the below binary to decimal chat,

Binary value	Decimal value
10000000	128
11000000	192
11100000	224
11110000	240
11111000	248
11111100	252
11111110	254
11111111	255

0	0000	0
1	0001	1
2	0010	2
3	0011	3
4	0100	4
5	0101	5
6	0110	6
7	0111	7
8	1000	8
9	1001	9
A	1010	10
B	1011	11
C	1100	12
D	1101	13
E	1110	14
F	1111	15

Figure 30: Memorization of Binary to decimal chart

Figure 31: Hex-to-binary-to-decimal chart

Hexadecimal addressing is completely different than binary or decimal by reading nibbles not bytes. If we use nibble we can convert bits to hex. On step 1 know that the hexadecimal addressing scheme uses number 0 through 9. (we can't use 10, 11, 12 and so forth because they are two digits) so letters A, B, C, D, E and F are used to represent 10, 11, 12, 13, 14, and 15 as follows;

Further reading: network+ Study Guide, 3rd Edition by Todd Lammle. (Page 104- 22)

2.5 Ethernet Addressing

What is a MAC address? What is OUI?

A MAC address is a unique identifier for network interfaces. It is a 48-bit number (12 hexadecimal characters). They can either be written in either of these formats:

- MM:MM:MM:SS:SS:SS
- MM-MM-MM-SS-SS-SS

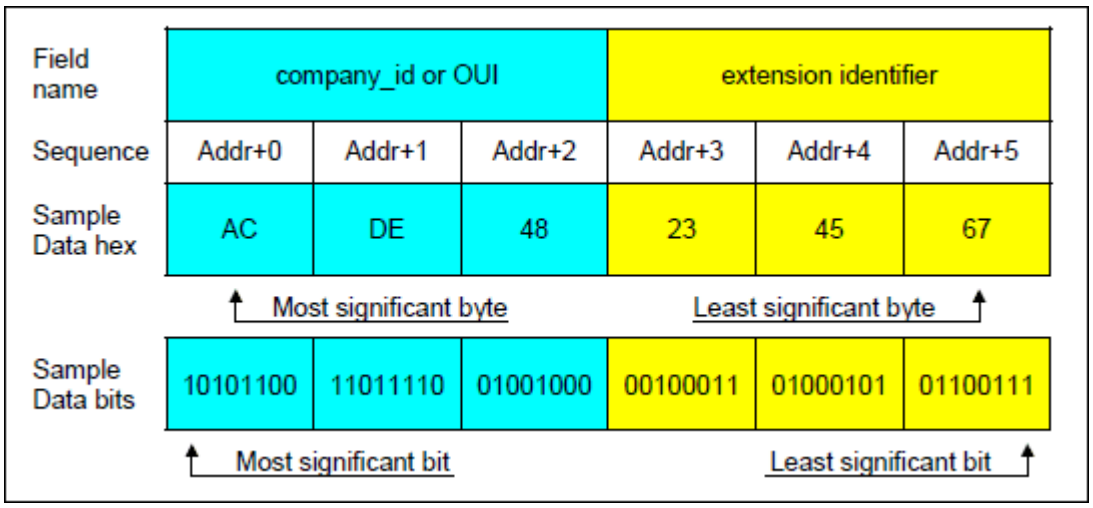


Figure 32: MAC Address and OUI

An OUI {Organizationally Unique Identifier} is a 24-bit number that uniquely identifies a vendor or manufacturer. They are purchased and assigned by the IEEE. The OUI is basically the first three octets of a MAC address. For example, these are examples of OUI:

- 00:00:0A -- this is owned by Omron
- 00-0D-4B -- this is owned by Roku, LLC

How to find the MAC address

2.5.1 How to find the MAC address in Windows

- Go to the DOS prompt
  - Click on Start button, select Run
  - Type cmd and press ENTER
- In the DOS prompt, type ipconfig /all and press ENTER
  You should see the MAC address in this format 00:00:00:00:00:00

2.5.2 How to find the MAC address in UNIX or Linux

- Go to a terminal
- In the terminal, type ifconfig and press ENTER
  You should see the MAC address in this format 00:00:00:00:00:00 following *HWaddr*

2.5.3 How to find the MAC address in Mac OS

- Go to Utilities (Finder > Utilities OR command + shift + u) and start the Terminal app.
- Type this in the Terminal:

```
networksetup -listallhardwareports
```

- You will be a list of all the network interfaces with their MAC addresses (Ethernet Address).
  An example output is this:

```
$ networksetup -listallhardwareports
```

```
•
• Hardware Port: Bluetooth DUN
•    Device: Bluetooth-Modem
• Ethernet Address: N/A
•
• Hardware Port: Ethernet
•    Device: en0
• Ethernet Address: 10:dd:b1:xx:xx:xx
•
• Hardware Port: FireWire
•    Device: fw0
• Ethernet Address: 10:dd:b1:ff:fe:xx:xx:xx
•
• Hardware Port: Wi-Fi
•    Device: en1
• Ethernet Address: 4c:8d:79:xx:xx:xx
•
• Hardware Port: Bluetooth PAN
•    Device: en3
• Ethernet Address: N/A
•
• VLAN Configurations
• ===================
```

2.6 Ethernet Frames

Ethernet Frame DEST MAC SOURCE MAC ET PAYLOAD 46-1500 BYTES Ethernet Frame consists of: 6 Byte: Destination MAC address 6 Byte: Source MAC addresses 2 Byte: Ethertype 46 - 1500 Bytes Payload. There are other Ethernet frame formats but they are the minority.

MAC Addresses MAC address (also known as hardware address or physical address) is a 6 byte address assigned by the IEEE Standards Association and is unique for every Ethernet device ever manufactured. The first three bytes are the OUI (Organizationally Unique Identifier) the second three bytes is a unique identifier assigned by the vendor. Thus OUI Card Specific ID.

MAC Address MAC Address of Ethernet NIC

```
   Connection-specific DNS Suffix  . : HEADOFFICE.local
   Description . . . . . . . . . . . : Realtek PCIe GBE Family Controller
   Physical Address. . . . . . . . . : 40-61-86-F8-A5-07
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes
   Link-local IPv6 Address . . . . . : fe80::4970:4f22:2da1:6832%3(Preferred)
   IPv4 Address. . . . . . . . . . . : 192.168.0.214(Preferred)
   Subnet Mask . . . . . . . . . . . : 255.255.254.0
   Lease Obtained. . . . . . . . . . : Tuesday, November 29, 2016 7:58:30 AM
   Lease Expires . . . . . . . . . . : Wednesday, December 7, 2016 7:58:58 AM
   Default Gateway . . . . . . . . . : 192.168.0.254
   DHCP Server . . . . . . . . . . . : 192.168.0.3
   DHCPv6 IAID . . . . . . . . . . . : 54550918
   DHCPv6 Client DUID. . . . . . . . : 00-01-00-01-1D-D5-25-E7-40-61-86-F8-A5-07
   DNS Servers . . . . . . . . . . . : 192.168.0.3
                                       192.168.0.252
   Primary WINS Server . . . . . . . : 192.168.0.3
   NetBIOS over Tcpip. . . . . . . . : Enabled

Tunnel adapter isatap.HEADOFFICE.local:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . : HEADOFFICE.local
   Description . . . . . . . . . . . : Microsoft ISATAP Adapter
   Physical Address. . . . . . . . . : 00-00-00-00-00-00-00-E0
   DHCP Enabled. . . . . . . . . . . : No
   Autoconfiguration Enabled . . . . : Yes
```

Figure 33: Identifying the MAC address

Class activity: Identify the MAC address and show it to your Lecturer

Source: http://standards-oui.ieee.org/oui.txt

```
OUI/MA-L                                                    Organization
company_id                                                  Organization
                                                            Address

E0-43-DB   (hex)                Shenzhen ViewAt Technology Co.,Ltd.
E043DB     (base 16)            Shenzhen ViewAt Technology Co.,Ltd.
                                9A,Microprofit,6th Gaoxin South Road, High-Tech Industrial Park, Nanshan, Shenzhen, CHINA.
                                shenzhen  guangdong  518057
                                CN

24-05-F5   (hex)                Integrated Device Technology (Malaysia) Sdn. Bhd.
2405F5     (base 16)            Integrated Device Technology (Malaysia) Sdn. Bhd.
                                Phase 3, Bayan Lepas FIZ
                                Bayan Lepas  Penang  11900
                                MY

2C-30-33   (hex)                NETGEAR
2C3033     (base 16)            NETGEAR
                                350 East Plumeria Drive
                                San Jose  null  95134
                                US

3C-D9-2B   (hex)                Hewlett Packard
3CD92B     (base 16)            Hewlett Packard
                                11445 Compaq Center Drive
                                Houston    77070
                                US
```

Figure 34: OUI

The above diagram shows the OUI of a few companies but the above.

Below is a NIC card showing the MAC address on the white strap of paper.

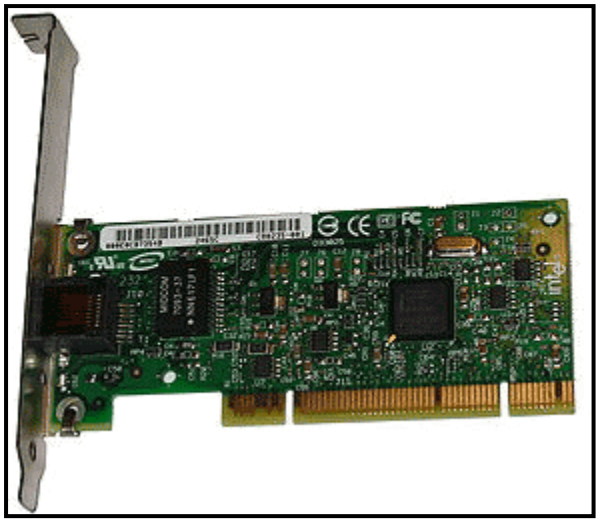


Figure 35: NIC Card

There are other ethernet over other standards like IEEE 1905.1-2013.

Further reading: Network+ Study Guide by Todd Lammle 3rd Edition on, pp. 105- 116.

Special MAC Addresses Broadcast: A MAC with all bits set FF FF FF FF FF FF is a BROADCAST. It is received by all devices on the Ethernet segment Multicast: A MAC address with the least significant bit of the most significant byte set is a MULTICAST address. 01 00 00 00 00 00 00 Note: Ethernet frames are always displayed from most significant to least significant. In actual transmission, each byte is transmitted from least significant bit to most significant bit. Some RFCs reference this as "first bit transmitted".

2.6.1 Ethertype: The two bytes after the source MAC in Ethernet II are the Ethertype Identifies the type of frame: 0800 is IP 0806 is ARP 8137 is Novell IPX 8100 is VLAN 802.3 Ethernet uses these two bytes as a length field How does a device know which field is referred to?

2.6.2 Data (Payload): Following the 14 bytes of Ethernet header will be between 46 and 1500 bytes of payload. This will give a minimum Ethernet frame of 60 bytes and a maximum of 1514 bytes 14 bytes header + 46 bytes payload = 60 14 bytes header + 1500 bytes payload = 1514

2.6.3 PDU Encapsulation

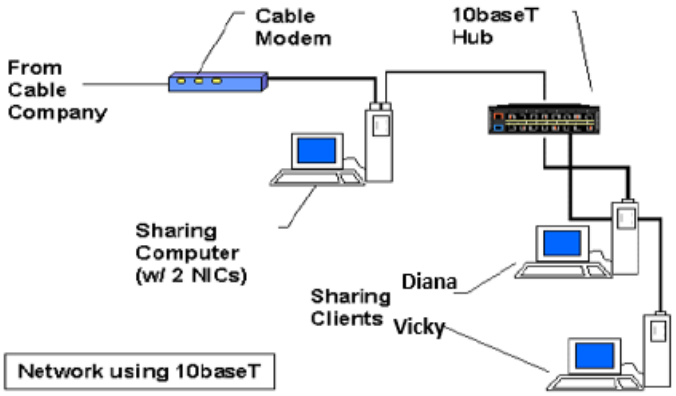
The "payload" portion of the Ethernet frame usually contains the protocol information from higher layer PDUs such as IP and TCP. Details are in Topic two, Semester 1 Learner guide.

Basic Concepts: To address a particular network node you must have the hardware MAC address If the destination MAC isn't right, it doesn't get there All higher level protocols sent over Ethernet are encapsulated in an Ethernet frame.

Summary

This topic was about Ethernet Networking basics on how work stations can communicate or share data and CSMA/CD how Ethernet half-dupllex network. Don't forget the diference between half and full duplex modes. Revisit this topic to grasp it properly.

EXTRA GROUP READING EXERCISE:



These computers are all connected to the Hub (Repeater) but how are they communicating? They use a MAC address also known as the hardware address that is burned in the Network card of all the computers. To talk to each other they use Name Resolution (Hostname to IP resolution) done by **(DNS)** Domain Name service is the task of converting domain names to their corresponding IP address. If the two computers (Hosts) are in the same LAN Vicky can just do **Broadcasting** to Diana without the use of a DNS that's Microsoft Windows.

An example of a Name resolution process for the Vicky and Diana Hosts output are as follows:

Time	*Source*	*Destination*	*Protocol*	*Infor*
53.892794	*192.168.0.1*	*192.168.0.255*	*NBNS*	*Name query NB Vicky<00>*

The way the operating system resolves the domain name is based upon its configuration.

2.7 TROUBLESHOOTING TOOLS

Transmission Control Protocol/ Internet Protocol (TCP/IP) version 4 is mostly used in networks so to test IP connectivity and verify IPv6 networks and TCP/IP utilities for windows.

In the Network+ troubleshooting model, there are eight steps you must follow:
1. Establish symptoms.

2. Identify the affected area.

3. Establish what has changed.

4. Select the most probable cause.

5. Implement a solution.

6. Test the result.

7. Recognize the potential effects of the solution.

8. Document the solution.

Step 1: Establish Symptoms

Obviously, if you can't identify a problem, you can't begin to solve it. Typically, you need to ask some questions to begin to clarify exactly what is happening. In our example, we should ask the user the following:

- Which part of the Internet can't you access?
- A particular website?
- A particular address?
- Any website?
- Can you use your web browser?

The user cannot access the corporate intranet or get to any sites on the Internet. User can use web browser to access the corporate FTP site, which he has bookmarked (by IP address 10.0.0.2). We can, therefore, rule out the web browser as the source of the problem.

Step 2: Identify the Affected Area

Computers and networks are fickle; they can work fine for months, suddenly malfunction horribly, and then continue to work fine for several more months, never again exhibiting that particular problem. And that's why it's important to be able to reproduce the problem and identify the affected area. Identifying the affected area narrows down what you have to troubleshoot. One of your goals is to make problems easier to troubleshoot and, thus, get users working again as soon as possible.

Step 3: Establish What Has Changed

If you can reproduce the problem, your next step is to attempt to determine what has changed. Drawing on your knowledge of networking, you might ask yourself and your user questions such as the following:

Were you ever able to do this?

If not, then maybe this is not an operation the hardware or software is designed to do. You can inform the user that the system won't do the operation (or that they may need additional hardware or software to do it).

If so, when did you become unable to do it?

If the computer was able to do the operation and then suddenly could not, the conditions that surround this change become extremely important. You may be able to discover the cause of the problem if you know what happened immediately before the change. It is likely that the cause of the problem is related to the conditions surrounding the change.

Has anything changed since you were last able to do this?

Most often, the thing that changed before the problem started is the source of the problem. When you ask this question of a user, the answer is typically that nothing has changed, so you might need to rephrase it. For example, you can try asking, "Did anyone add anything to your computer?" or "Are you doing anything that's different from the way you normally precede?"

Were any error messages displayed?

Are other people experiencing this problem?

For further reading: http://www.spiceworks.com/free-network-troubleshooting-tool/

After reading the above please read the below to be fully equipped with the troubleshooting tools

2.7.1 Troubleshooting Tools to consider

Both using GUI and Command line to verify TCP/IP function on windows workstations. Check listed which are the common ones to be used:

.Traceroute (tracert in Microsoft)

.ipconfig/winipcfg (ifconfig in UNIX)

.ping

.arp

.nslookup (dig in UNIX)

.Mtr (pathping)

.route

.nbtstat

.netstat

.ftp

.telnet

For the last two .ftp and .telnet has been explained under the Server Application subtopic.

2.7.2 Protocol analyser

Sometimes called **Network Monitors** or **Sniffers** used to capture packets as they cross the network.

There are commercial Sniffers like Wireshark and OmniPeek that can capture any packet because of its NIC configuration that can capture any packet that it sees.

Protocol Analysers are used to determine the type of traffic in a network and depending on the product and the bells and whistles contained therein, through using the port numbers, protocols etc.

DHCP has its packets as follows;

- DHCP Discover
- DHCP Offer
- DHCP Request
- DHCP ACK

Through executing *ipconfig/release* and *ipconfig/renew* commands when the analyser is on the client four packets below will be seen in the analyser's capture file. See the example of a DHCP process in the below capture;

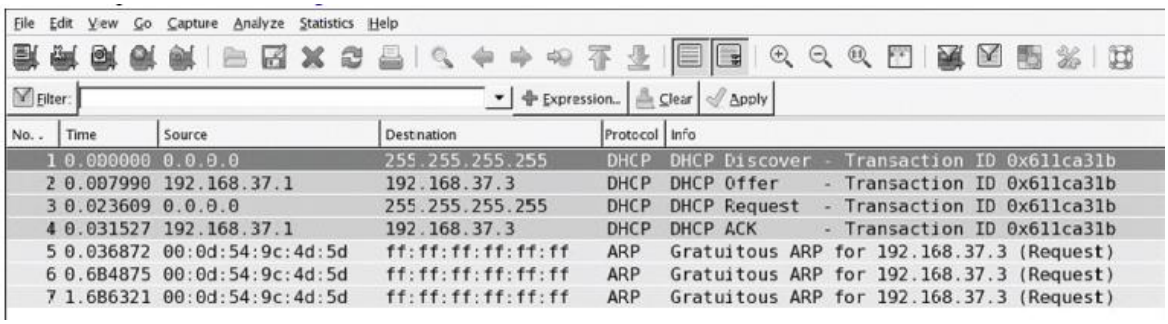


Figure 36: DHCP Capture

Examining clearly the capture were the DHCP DISCOVER packets with no DHCPOFFER packets, it shows the DHCP server is not receiving the DHCPDISCOVER packets On the DHCPOFFER packets shows the DHCP server is out of addresses. The point will be the tool is used to troubleshoot the issue.

2.7.3 Throughput Testers

People complain about the speed of the network and some don't complain but to check whether the slowdown is real as opposed to perceived you need to measure throughput with the tester.

Check the diagram below showing results of a throughput tester:

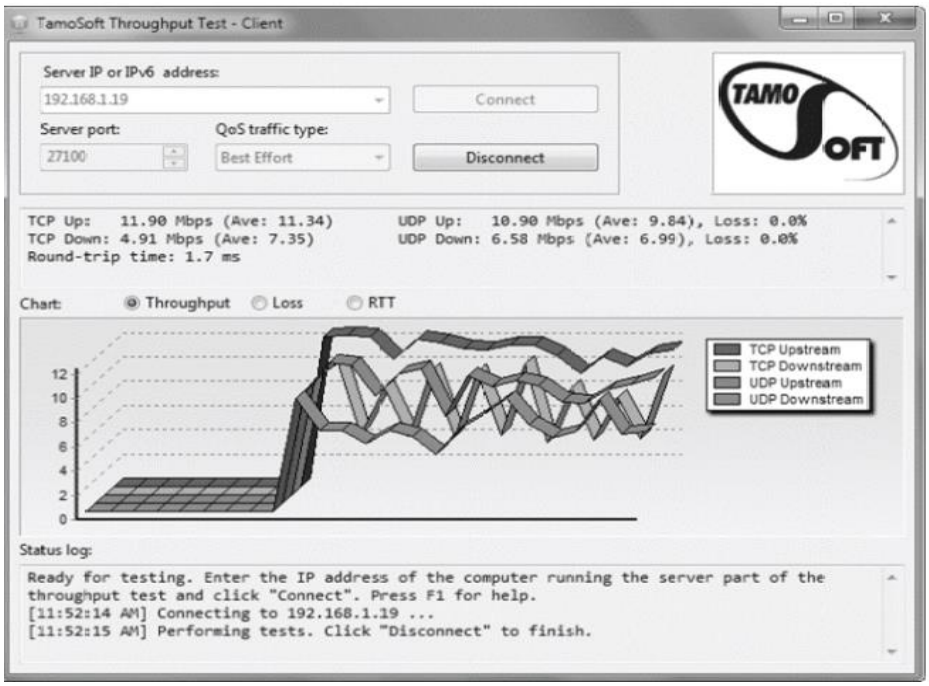


Figure 37: Throughput software by TamoSoft

Remember, When installing it should be installed on both the client and the server to measure the traffic between the client and a server. It shows throughput for traffic in real time and in this short is breaking that traffic up by unicast (TCP) and broadcast (UDP) types and by direction.

2.7.4 Connectivity Software

Remote desktop (Windows Products) can let you access your machine to perform troubleshooting this is an example of connectivity software. There are other commercial tools like LogMeIn.com, GoToMyPC, and TeamViewer, etc. The figure below shows the session window for LogMeIn.com.

There are advantages of this software obviously you can do whatever you want to do when they are doing it wrongly.



Figure 38: LogMeIn.com

Traceroute

Latency is the delay a message takes to travel form the source to destination. It travels faster but it uses time let's see how distance affects the RTT (Round Trip Time).

Open a command prompt and run ***tracert*** on a web site whose server is on a different continent from you. E.g. **tracert** www.mediacollege.com

When running on Ubuntu Platform install ***traceroute*** *utility and use it. Installing use this command:* ***sudo apt-get install traceroute***

Traceroute is a command which can show you the path a packet of information taken from your computer to one you specify. It will list all the routers it passes through until it reaches its destination, or fails to and is discarded. In addition to this, it will tell you how long each 'hop' from router to router takes.

In Windows, select *Start > Programs > Accessories > Command Prompt*. This will give you a window like the one below.

Enter the word *tracert*, followed by a space, then the domain name.

The following is a successful traceroute from your computer in New Zealand to mediacollege.com:

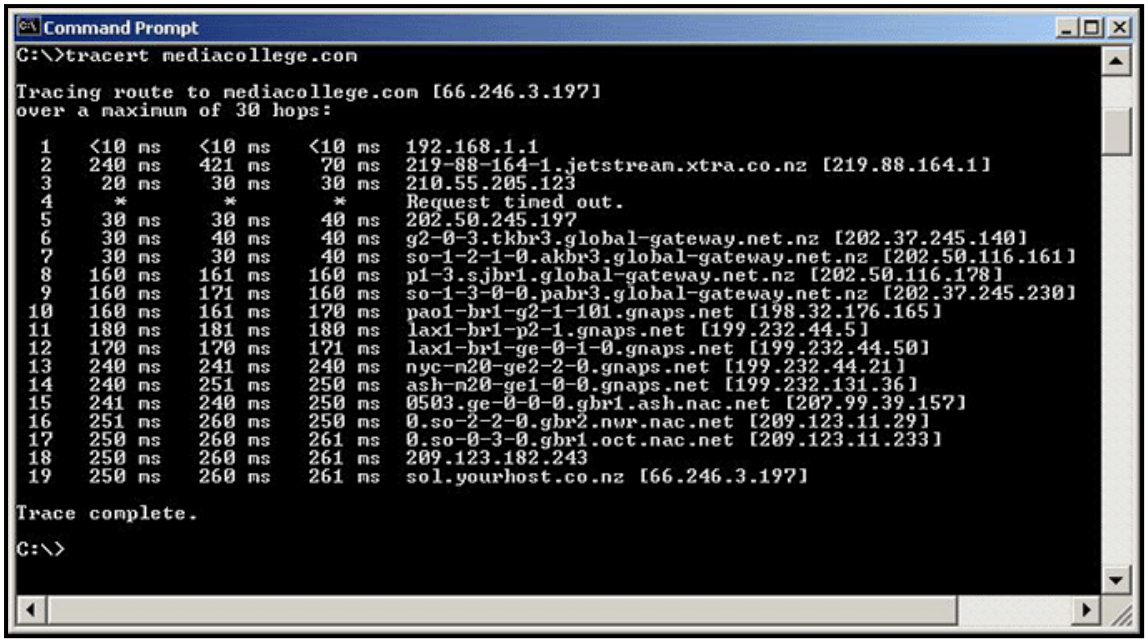


Figure 39: Latency test

Firstly it tells you that it's tracing the route to mediacollege.com, tells you the IP address of that domain, and what the maximum number of hops will be before it times out.
Examine the output and find at what point in the route messages started jumping across the ocean.

Next it gives information about each router it passes through on the way to its destination.

1 is the internet gateway on the network this traceroute was done from (an ADSL modem in this case)
2 is the ISP the origin computer is connected to (xtra.co.nz)
3 is also in the xtra network
4 timed out
5 - 9 are all routers on the global-gateway.net.nz network (the domain that is the internet gateway out of New Zealand)
10 - 14 are all gnaps.net in the USA (a telecom supplier in the USA)
15 - 17 are on the nac network (Net Access Corporation, an ISP in the New York area)
18 is a router on the network mediacollege.com is hosted on
and finally, line

19 is the computer mediacollege.com is hosted on (sol.yourhost.co.nz)

Each of the 3 columns are a response from that router, and how long it took (each hop is tested 3 times). For example, in line 2, the first try took 240ms (240 milliseconds), the second took 421 ms,

and the third took 70ms.
You will notice that line 4 'timed out', that is, there was no response from the router, so another one was tried (202.50.245.197) which was successful.
You will also notice that the time it took quadrupled while passing through the global-gateway network.

This is extremely useful when trying to find out why a website is unreachable, as you will be able to see where the connection fails. If you have a website hosted somewhere, it would be a good idea to do a traceroute to it when it is working, so that when it fails, you can do another traceroute to it (which will probably time out if the website is unreachable) and compare them. Be aware though, that it will probably take a different route each time, but the networks it passes through will generally be very similar.
If the example above had continued to time out after line 9, you could suspect that global-gateway.co.nz was the problem, and not mediacollege.com.
If it timed out after line 1, you would know there was a problem connecting to your ISP (in this case you would not be able to access anything on the internet).

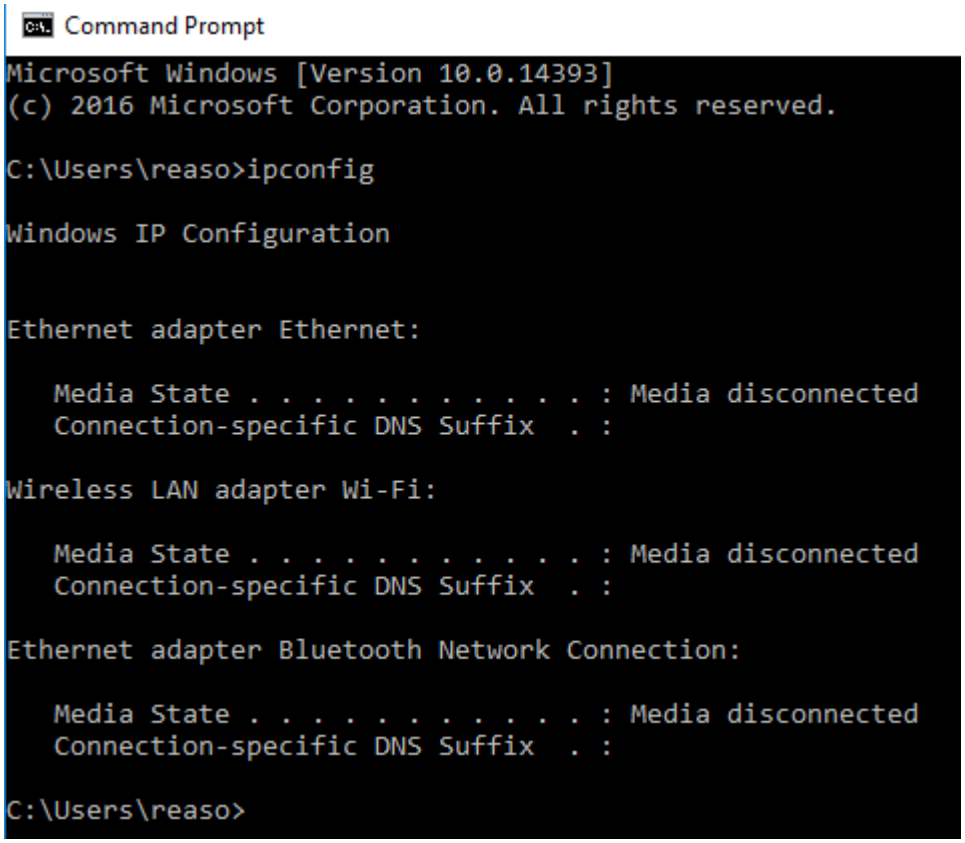
It is generally recommended that if you have a website that is unreachable, you should use both the traceroute and ping commands before you contact your ISP to complain. More often than not, there will be nothing to your ISP or hosting company can do about it.

ipconfig and ifconfig

Windows uses ipconfig and UNIX/LINUX/Mac uses ifconfig to display the current configuration of TCP/IP on a given workstation including the current IP address, DNS configuration, Windows Internet Naming Services (WINS) configuration, and default gateway.

Ipconfig utility

All new operating systems such as Mac, Windows Vista, Windows 7, 8 and 10 and Windows server 2008/ 2012 IPv6 configuration is available by default. The output if the command ipconfig is executed you see something like this because I am not connected to the internet:



But when connected to the internet you will see something like this:

```
C:\Users\tlammle>ipconfig
Windows IP Configuration
Ethernet adapter Local Area Connection:
   Connection-specific DNS Suffix  . : domain.actdsltmp
   Link-local IPv6 Address . . . . . : fe80::2836:c43e:274b:f08c%11
   IPv4 Address. . . . . . . . . . . : 192.168.0.6
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : 192.168.0.1
Wireless LAN adapter Wireless Network Connection:
   Connection-specific DNS Suffix  . : qwest.net
   Link-local IPv6 Address . . . . . : fe80::20e7:7fb8:8a00:832b%10
   IPv4 Address. . . . . . . . . . . : 10.0.1.198
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : fe80::21b:63ff:fef3:3694%10
                                       10.0.1.1
Tunnel adapter Local Area Connection* 6:
   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
Tunnel adapter Local Area Connection* 7:
```

Note: Your lecturer will demonstrate ipconfig for you or try it yourself.

Using ifconfig

Normally for Mac, Linux and UNIX uses this utility and gives you the similar information to the *ipconfig.* ***Interface configuration****(ifconfig)* there are major differences *ipconfig* is used to view TCPP/IP configuration for a computer where as ifconfig do the same with extra as configuring protocol or a particular network interface. The command is as follows:

ifconfig interface [address [parameters]]

the interface parameter euqals the unix name of the interface, such as eth0. If the optional address parameter is specified, the ifconfig command sets the IP address for the interface to the address you've specified. When the ifconfig command is used by itself with no parameters, all configured interfaces will be reported on. But if only the interface name is specified, you will get output that looks like this:

```
# ifconfig eth0
eth0  Link encap 10Mbps Ethernet  HWaddr 00:00:C0:90:B3:42
inetaddr 172.16.0.2 Bcast 172.16.0.255 Mask 255.255.255.0 UP
BROADCAST RUNNING  MTU 1500  Metric 0
   RX packets 3135 errors 217 dropped 7 overrun 26
   TX packets 1752 errors 25 dropped 0 overrun 0
```

The eth0 interface is a 10Mbps Ethernet interface. The MAC and IP address information is displayed in this output as well. The result above didn't show the output for DNS information but this ifconfig for the tool can.

Ping Utility

The most commonly used network tool is the ping utility. This utility is used to provide a basic connectivity test between the requesting host and a destination host. This is done by using the Internet Control Message Protocol (ICMP) which has the ability to send an echo packet to a destination host and a mechanism to listen for a response from this host. Simply stated, if the requesting host receives a response from the destination host, this host is reachable. This utility is commonly used to provide a basic picture of where a specific networking problem may exist. For example, if an Internet connection is down at an office, the ping utility can be used to figure out whether the problem exists within the office or within the network of the Internet provider. Below shows an example of the ping utility to select;

```
Options:
    -t             Ping the specified host until stopped.
                   To see statistics and continue - type Control-Break;
                   To stop - type Control-C.
    -a             Resolve addresses to hostnames.
    -n count       Number of echo requests to send.
    -l size        Send buffer size.
    -f             Set Don't Fragment flag in packet (IPv4-only).
    -i TTL         Time To Live.
    -v TOS         Type Of Service (IPv4-only. This setting has been deprecated
                   and has no effect on the type of service field in the IP
                   Header).
    -r count       Record route for count hops (IPv4-only).
    -s count       Timestamp for count hops (IPv4-only).
    -j host-list   Loose source route along host-list (IPv4-only).
    -k host-list   Strict source route along host-list (IPv4-only).
    -w timeout     Timeout in milliseconds to wait for each reply.
    -R             Use routing header to test reverse route also (IPv6-only).
                   Per RFC 5095 the use of this routing header has been
                   deprecated. Some systems may drop echo requests if
                   this header is used.
    -S srcaddr     Source address to use.
    -c compartment Routing compartment identifier.
    -p             Ping a Hyper-V Network Virtualization provider address.
    -4             Force using IPv4.
    -6             Force using IPv6.
```

Below shows an example of the ping utility is used to obtain the reachability status of the locally connected router.

```
Command Prompt
Microsoft Windows [Version 6.1.7600]
Copyright (c) 2009 Microsoft Corporation.  All rights reserved.

C:\Users\Sean Wilkins>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:
Reply from 192.168.1.1: bytes=32 time=1ms TTL=64
Reply from 192.168.1.1: bytes=32 time<1ms TTL=64
Reply from 192.168.1.1: bytes=32 time=2ms TTL=64
Reply from 192.168.1.1: bytes=32 time=2ms TTL=64

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 2ms, Average = 1ms
```

Figure 40:Pimg Utility

Using ARP Utility

Windows 2000 TCP/IP allows an application to communicate over a network with another computer by using either an IP address, a host name, or a NetBIOS name. However, regardless of which naming convention is used, the destination must ultimately be resolved to a hardware address (media access control (MAC) address) for shared access media such as Ethernet and Token Ring. The *Address Resolution Protocol (ARP)* allows a host to find the MAC address of a node with an IP address on the same physical network, when given the node's IP address. To make ARP efficient, each computer caches IP-to-MAC address mappings to eliminate repetitive ARP broadcast requests. The Arp tool allows a user to view and modify ARP table entries on the local computer.

The **arp** command is useful for viewing the ARP cache and resolving address resolution problems.

A static entry can be added to an ARP file by issuing the **arp -s** < *IP address* > < *MAC address* > command. However, adding such static ARP cache entries must be used with caution as it is easy to enter the wrong MAC address for an IP address.

Detecting Duplicate IP Addresses Using ARP

When starting up, Windows performs a gratuitous ARP to detect any duplication with its own IP address. While this detects most cases of duplicate IP addresses, in a few situations two TCP/IP hosts (either Microsoft or non-Microsoft) on the same network can be configured for the same IP address. The MAC and IP address mapping is done by the ARP module, which uses the first ARP response it receives. Therefore, the impostor computer's reply sometimes comes back before the intended computer's reply.

These problems are difficult to isolate and track down. Use the **arp -a** command to display the mappings in the ARP cache. If you know the Ethernet address for the remote computer you wish to use, you can easily determine whether the two match. If not, use the **arp -d** command to delete the entry, then use Ping with the same address (forcing an ARP), and check the Ethernet address in the cache again by using **arp -a**.

If both computers are on the same network, you will eventually get a response from the imposter computer. If not, you might have to capture the traffic from the impostor host with Network Monitor to determine the owner or location of the system. For more information about Network Monitor, see "Monitoring Network Performance" in the *Server Operations Guide*.

Using nslookup utility

Some of the most common networking issues revolve around issues with Dynamic Name System (DNS) address resolution issues. DNS is used by everyone using the Internet to resolve commonly known domain names (i.e. google.com) to commonly unknown IP addresses (i.e. 74.125.115.147). When this system does not work, most of the functionality that people are used to goes away, as there is no way to resolve this information. The nslookup utility can be used to lookup the specific IP address(es) associated with a domain name. If this utility is unable to resolve this information, there is a DNS issue. Along with simple lookup, the nslookup utility is able to query specific DNS servers to determine an issue with the default DNS servers configured on a host. Figure below shows an example of how the nslookup utility can be used to query the associated IP address information.

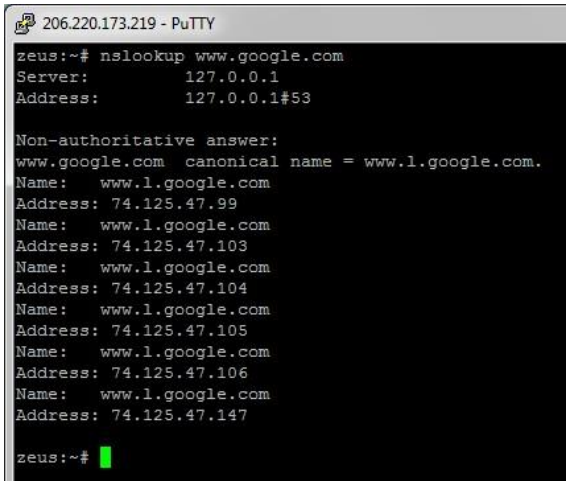


Figure 41: nslookup utility

Resolving names with the host table

A *host name* is an alias assigned to an IP node to identify it as a TCP/IP host. The host name can be up to 255 characters long and can contain alphabetic and numeric characters and the "-" and "." characters. Multiple host names can be assigned to the same host. For Windows 2000–based computers, the host name does not have to match the Windows 2000 computer name.

Windows Sockets applications, such as Microsoft® Internet Explorer and the FTP utility, can use one of two values for the destination to be connected: the IP address or a host name. When the IP address is specified, name resolution is not needed. When a host name is specified, the host name must be resolved to an IP address before IP-based communication with the desired resource can begin.

Host names can take various forms. The two most common forms are a nickname and a domain name. A nickname is an alias to an IP address that individual people can assign and use. A *domain name* is a structured name that follows Internet conventions.

2.7.5 Domain Names

To facilitate different organizations and their desires to have scalable, customizable naming scheme in which to operate, the InterNIC has created and maintains a hierarchical namespace called the *Domain Name System* (DNS). DNS is a naming scheme that looks similar to the directory structure for files on a disk. However, instead of tracing a file from the root directory through subdirectories to its final location and its file name, a host name is traced from its final location through its parent domains back up to the root. The unique name of the host, representing its position in the hierarchy, is called its *Fully Qualified Domain Name* (FQDN). The top-level domain namespace is shown in Figure 1.11 with example second-level and subdomains.

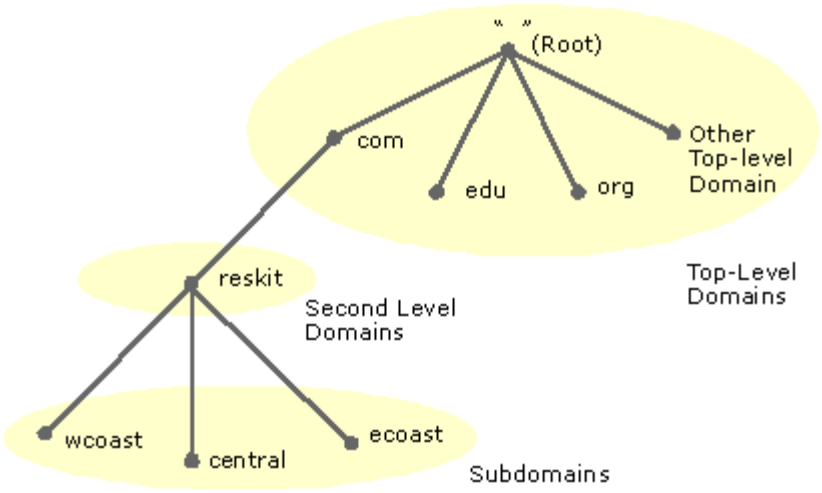


Figure 42: Domain Name System

The domain namespace consists of:

- The *root domain,* representing the root of the namespace and indicated with a "" (null).
- *Top-level domains*, those directly below the root, indicating a type of organization. On the Internet, the InterNIC is responsible for the maintenance of top-level domain names. Table below has a partial list of the Internet's top-level domain names.

Domain Name	Meaning
COM	Commercial organization
EDU	Educational institution
GOV	Government institution
MIL	Military group
NET	Major network support center
ORG	Organization other than those above
INT	International organization
< *country/ region code* >	Each country/region (geographic scheme)

Figure 43: Internet Top-Level Domain Names

- *Second-level domains*, below the top level domains, identifying a specific organization within its top-level domain. On the Internet, the InterNIC is responsible for the maintenance of second-level domain names and ensuring their uniqueness.

- *Subdomains* of the organization, below the second-level domain. The individual organization is responsible for the creation and maintenance of subdomains.

For example, for the FQDN **ftpsrv.wcoast.reskit.com**:

- The trailing period (**.**) denotes that this is an FQDN with the name relative to the root of the domain namespace. The trailing period is usually not required for FQDNs and if it is missing it is assumed to be present.
- **com** is the top-level domain, indicating a commercial organization.
- **reskit** is the second-level domain, indicating the Windows 2000 Resource Kit organization.
- **wcoast** is a subdomain of **reskit.com** indicating the West Coast division of the Windows 2000 Resource Kit organization.
- **ftpsrv** is the name of the FTP server in the West Coast division.

Domain names are not case sensitive.

Organizations not connected to the Internet can implement whatever top and second-level domain names they want. However, typical implementations do adhere to the InterNIC specification so that eventual participation in the Internet will not require a renaming process.

Using the Mtr command (pathping)

In an effort to take advantage of the benefits of both the ping and tracert/traceroute commands, the pathping and mtr utilities were developed. Both of these tools take the functionality and information that can be obtained from these types of tools and provide a more detailed single picture of the path characteristics from a specific host to a specific destination. Figure 11 and 12 below show examples of these two tools and what information they provide.

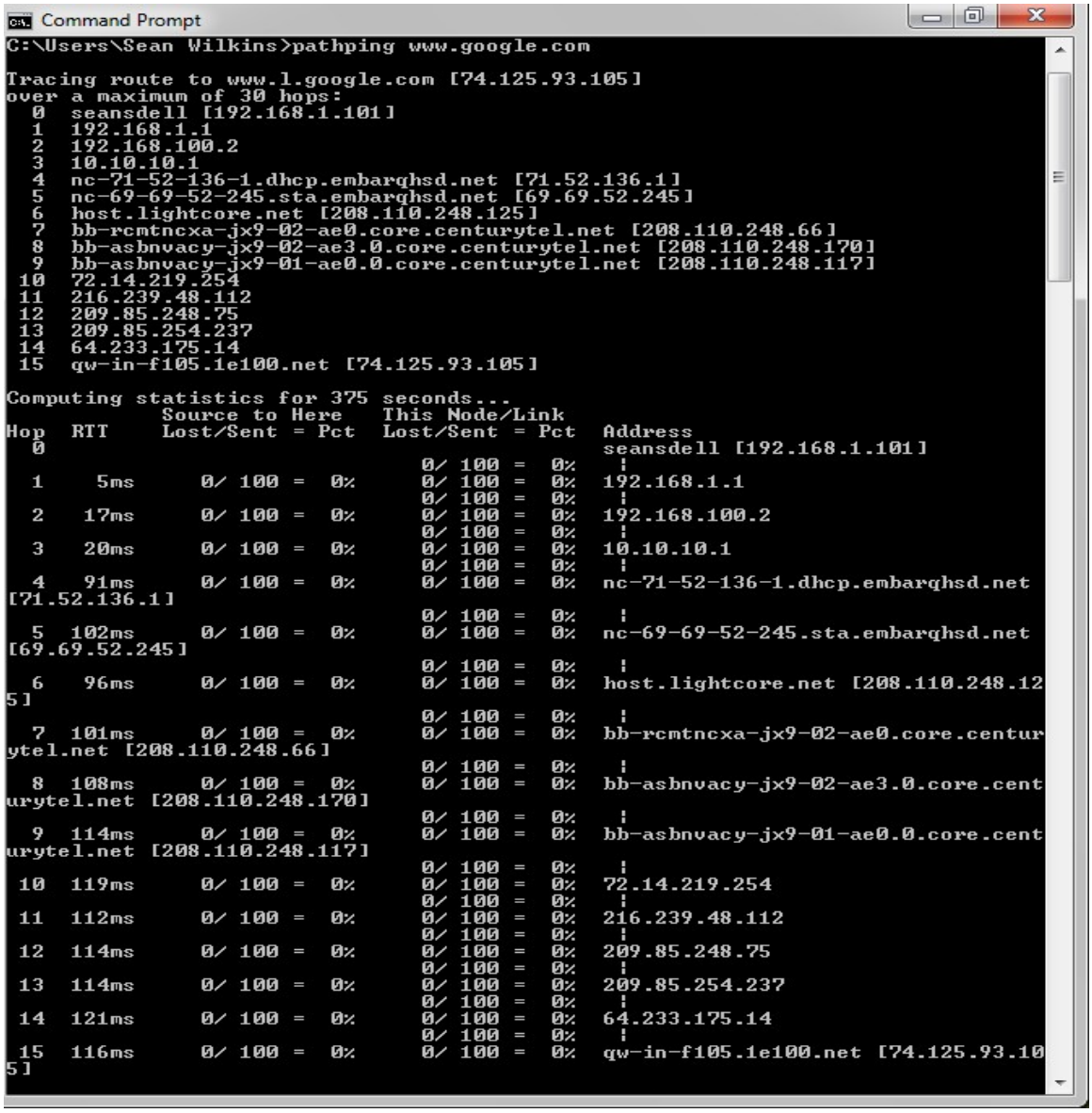


Figure 44: Path Ping

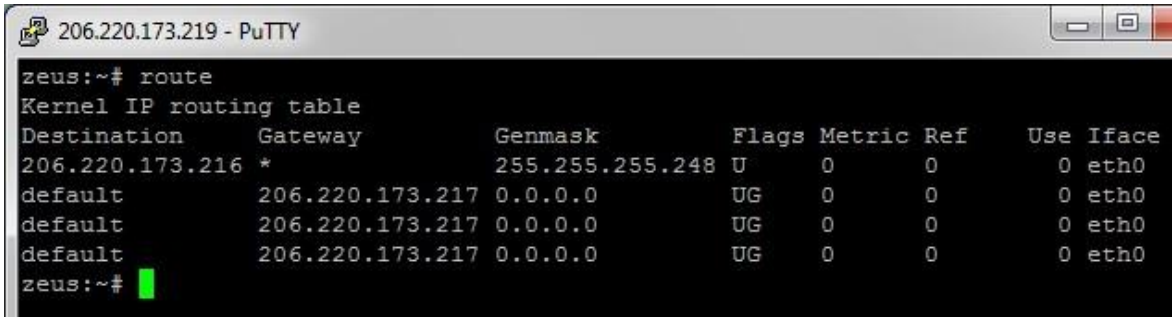


Figure 45: Mtr

Using the route command

The last of the tools covered in this article is the route utility. This utility is used to display the current status of the routing table on a host. While the use of the route utility is limited in common situations where the host only has a single IP address with a single gateway, it is vital in other situations where

multiple IP address and multiple gateways are available. Figure below shows an example of the route utility being used on a Windows machine.



Using the nbtstat command

Displays NetBIOS over TCP/IP (NetBT) configuration and allows management of the NetBIOS name cache.

Nbtstat is designed to help troubleshoot NetBIOS name resolution problems. When a network is functioning normally, NetBIOS over TCP/IP (NetBT) resolves NetBIOS names to IP addresses. It does this through several options for NetBIOS name resolution, including local cache lookup, WINS server query, broadcast, LMHOSTS lookup, Hosts lookup, and DNS server query.
The **nbtstat** command removes and corrects preloaded entries using a number of case-sensitive switches. The **nbtstat - a <** *name* **>** command performs a NetBIOS adapter status command on the computer name specified by < *name*> . The adapter status command returns the local NetBIOS name table for that computer as well as the MAC address of the adapter card. The **nbtstat -A** < *IP address* > command performs the same function using a target IP address rather than a name.
The **nbtstat - c** option shows the contents of the NetBIOS name cache, which contains NetBIOS name-to-IP address mappings.
nbtstat -n displays the names that have been registered locally on the system by NetBIOS applications such as the server and redirector.
The **nbtstat -r** command displays the count of all NetBIOS names resolved by broadcast and by querying a WINS server. The **nbtstat -R** command purges the name cache and reloads all #PRE entries from the LMHOSTS file. #PRE entries are the LMHOSTS name entries that are preloaded into the cache.
Nbtstat -RR sends name release packets to the WINS server and starts a refresh, thus re-registering all names with the name server without having to reboot. This is a new option in Windows NT 4.0 with Service Pack 4 as well as in Windows 2000.
You can use **nbtstat -S** to list the current NetBIOS sessions and their status, including statistics. Sample output looks like this:
C:\>nbtstat -S
Local Area Connection:
Node IpAddress: [172.16.0.142] Scope Id: []
NetBIOS Connection Table
Local Name State In/Out Remote Host Input Output

TESTPC1 <00> Connected Out 172.16.210.25 6MB 5MB
TESTPC1 <00> Connected Out 172.16.3.1 108KB 116KB
TESTPC1 <00> Connected Out 172.16.3.20 299KB 19KB
TESTPC1 <00> Connected Out 172.16.3.4 324KB 19KB
TESTPC1 <03> Listening

Finally, **nbtstat -s** provides a similar set of session listings, but provides the remote computer names, rather than their IP addresses.

Note

The options for the **Nbtstat** command are case sensitive.

The Nbtstat switches are listed in Table below;

Nbtstat Switches

Switch	Name	Function
-a < *name* >	adapter status	Returns the NetBIOS name table and MAC address of the address card for the computer name specified.
-A < *IP address* >	Adapter status	Lists the same information as -a when given the target's IP address.
-c	cache	Lists the contents of the NetBIOS name cache.
[*Number*]	Interval	Typing a numerical value tells Nbtstat to redisplay selected statistics each interval seconds, pausing between each display. Press Ctrl+C to stop redisplaying statistics.
-n	names	Displays the names registered locally by NetBIOS applications such as the server and redirector.
-r	resolved	Displays a count of all names resolved by broadcast or WINS server.
-R	Reload	Purges the name cache and reloads all #PRE entries from LMHOSTS.
-RR	ReleaseRefresh	Releases and reregisters all names with the name server.
-s	sessions	Lists the NetBIOS sessions table converting destination IP addresses to computer NetBIOS names.
-S	Sessions	Lists the current NetBIOS sessions and their status, with the IP address.
/?	Help	Displays this list.

Using the netstat utility

Often, one of the things that are required to be figured out is the current state of the active network connections on a host. This is very important information to find for a variety of reasons. For example, when verifying the status of a listening port on a host or to check and see what remote hosts are connected to a local host on a specific port? It is also possible to use the netstat utility to

determine which services on a host that is associated with specific active ports. Figure below shows an example of the netstat utility being used to display the currently active ports on a Linux machine.

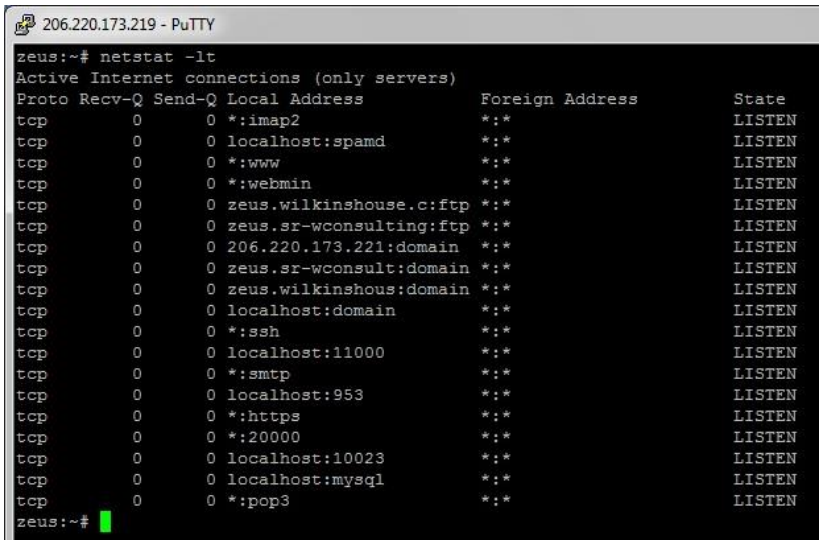


Figure 46: netsat utility

Summary

A lot has been talked about troubleshooting TCP/IP tools they include GUI tools like protocol analysers, throughput testers , and connectivity software like Remote Desktop, and they include command-line utilities like tracert, ping, arp, netstat, nbtstat, ipconfig, ifconfig and nslookup.

Also these utilities how they work and various options and switches and how they all affect the use of the utilities. Lastly how the utilities work within the TCP/IP suite.

2.8 MODULATION TECHNIQUES

- Data relies on digital transmission
- Network connection may handle only analog signals
- Modem



RESOURCE

Further reading: Network+ study guide, 3rd Edition by Todd Lammle. Chapter 17. pages 485-517.
Web link: https://technet.microsoft.com/en-us/library
Videos: https://www.youtube.com/watch?v=BVBm-mO6cyQ

- Accomplishes this translation

- Modulator/demodulator

- Data modulation
  - Technology modifying analog signals into digital signals and vice versa
  - Make analog signals suitable for carrying data over a communication path

2.8.1 A Modem

When in the communications environment a modem is a combination of two networking devices: a modulator and a demodulator (modem for short). These devices perform modulation and demodulation algorithms concurrently, to convert a signal from analog-to-digital and digital-to-analog, enabling data transmission to and from various computing resources. Below is a modem picture.



Figure 47: Modem

2.8.2 Modulation/Demodulation

Modulation is a carrier signal that varies in accordance with the message signal. The technique used to change the signal characteristics. Basically, the modulation is of following two types: Digital and Analog modulation.

Using the telephone example on how typical data flow using the illustration below as a reference. When making a call we speak into a microphone, sound waves from our voice are transmitted by the telephone in the form of radio waves thus analog signals, which usually vary in signal strength or frequency. The analog signals travel from the telephone into the modem. It is the modem's job to demodulate, which is transforming analog signals into digital signals for computer processing, and then eventually modulate the signal, backsliding the digital signals back into analog signals.

A **transmitter** is a group of electronic circuits designed to convert the information into a signal for transmission over a given communication medium.

A **receiver** is a group of electronic circuits designed to convert the signal back to the original information.

The communication channel is the medium which is designed to transmit the electronic signal from one place to another.

Classification of Modulation Techniques: There are high-frequency sinusoidal waveforms used as carrier signal. For this purpose ,if the variation in the parameter of the carrier is continuous in accordance to the input analog signal the modulation technique is termed as analog modulation scheme if the variation is discrete then it is termed as Digital Modulation Technique.

Type of Modulation Techniques

2.8.3 Analog Modulation Techniques

(i) Amplitude Modulation
(ii) Frequency Modulation
(iii) Phase Modulation A.M. F.M. P.M. 02

Analog

One important characteristic of data transmission is the type of signalling involved. On a data network, information can be transmitted via one of two signalling methods: Analog or Digital. Computers generate and interpret digital signals as electrical current, the pressure of which is measured in volts. The strength of an electrical signal is directly proportional to its voltage. Thus, when network engineers talk about the strength of a signal, they often refer to the signal's voltage. After being generated, signals travel over copper cabling as electrical current. Over fiber-optic cable, they travel as light pulses. And through the atmosphere, they travel as electromagnetic waves.

Analog data signals are also generated as voltage. However, in analog signals, voltage varies continuously and appears as a wavy line when graphed over time, as shown in Figure 2-1.

An analog signal, like other waveforms, is characterized by four fundamental properties: amplitude, frequency, wavelength, and phase. A wave's amplitude is a measure of its strength at any given point in time. On a wave graph, the amplitude is the height of the wave at any point in time. In Figure below, for example, the wave has amplitude of 5 volts at .25 seconds, amplitude of 0 volts at .5 seconds, and amplitude of −5 volts at .75 seconds.

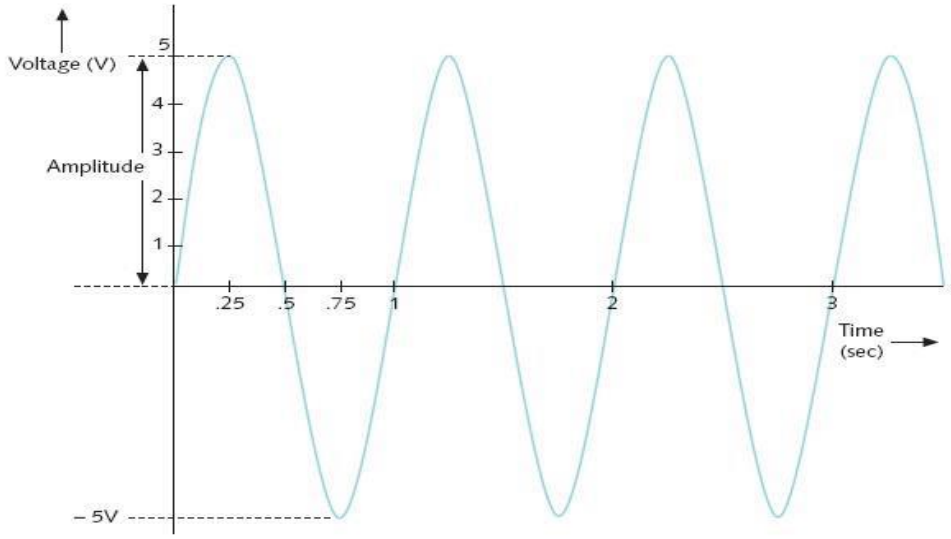


Figure 48: Analog

Whereas amplitude indicates an analog wave's strength, frequency is the number of times that a wave's amplitude cycles from its starting point, through its highest amplitude and its lowest amplitude, and back to its starting point over a fixed period of time. Frequency is expressed in cycles per second, or hertz (Hz), named after German physicist Heinrich Hertz, who experimented with electromagnetic waves in the late nineteenth century. For example, in Figure 48 above the wave cycles to its highest then lowest amplitude and returns to its starting point once in 1 second. Thus, the frequency of that wave would be 1 cycle per second, or 1 Hz—which, as it turns out, is an extremely low frequency.

Frequencies used to convey speech over telephone wires fall in the 300 to 3300 Hz range. Humans can hear frequencies between 20 and 20,000 Hz. An FM radio station may use a frequency between 850,000 Hz (or 850 kHz) and 108,000,000 Hz (or 108 MHz) to transmit its signal through the air.

The distance between corresponding points on a wave's cycle—for example, between one peak and the next—is called its wavelength. Wavelengths can be expressed in meters or feet. A wave's wavelength is inversely proportional to its frequency. In other words the higher the frequency, the shorter the wavelength. For example, a radio wave with a frequency of 1,000,000 cycles per second (1 MHz) has a wavelength of 300 meters, while a wave with a frequency of 2,000,000 Hz (2 MHz) has a wavelength of 150 meters.

The term phase refers to the progress of a wave over time in relationship to a fixed point. Suppose two separate waves have identical amplitudes and frequencies. If one wave starts at its lowest amplitude at the same time the second wave starts at its highest amplitude, these waves

will have different phases. More precisely, they will be 180 degrees out of phase (using the standard assignment of 360 degrees to one complete wave). Had the second wave also started at its lowest amplitude, the two waves would be in phase. Figure 2-2 illustrates waves with identical amplitudes and frequencies whose phases are 90 degrees apart.

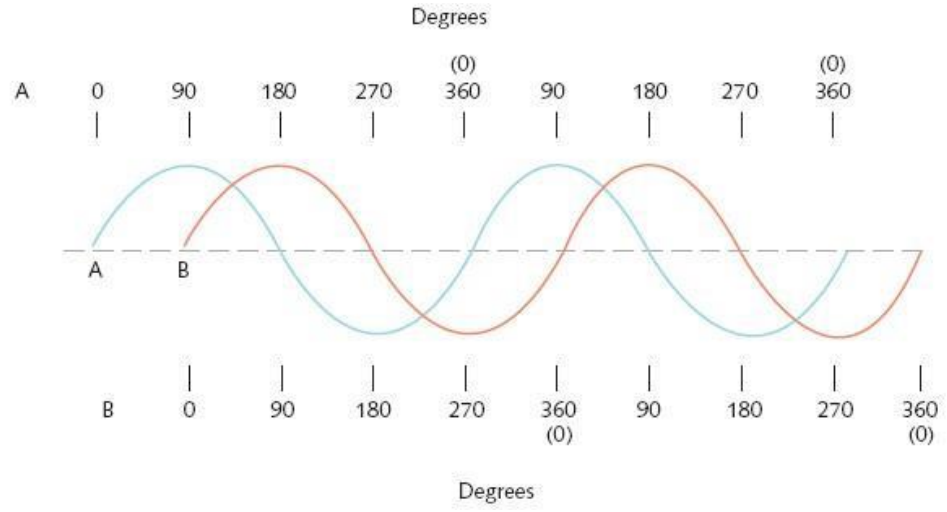


Figure 49: Waves with a 90-degree phase difference

One benefit to analog signals is that, because they are more variable than digital signals, they can convey greater subtleties with less energy. For example, think of the difference between your voice and a digital voice, such as the automated service that some libraries use to notify you when a book you have requested is available. The digital voice has a poorer quality than your own voice—that is, it sounds like a machine. It can't convey the subtle changes in inflection that you expect in a human voice. Only very high-quality digital signals—for example, those used to record music on compact discs—can achieve such accuracy.

One drawback to analog signals is that their voltage is varied and imprecise. Thus, analog transmission is more susceptible to transmission flaws such as noise, or any type of interference that may degrade a signal, than digital signals. If you have tried to listen to AM radio on a stormy night, you have probably heard the crackle and static of noise affecting the signal. Now contrast the analog signals pictured in Figures 48 and 49 to a digital signal, as shown in Figure 50.

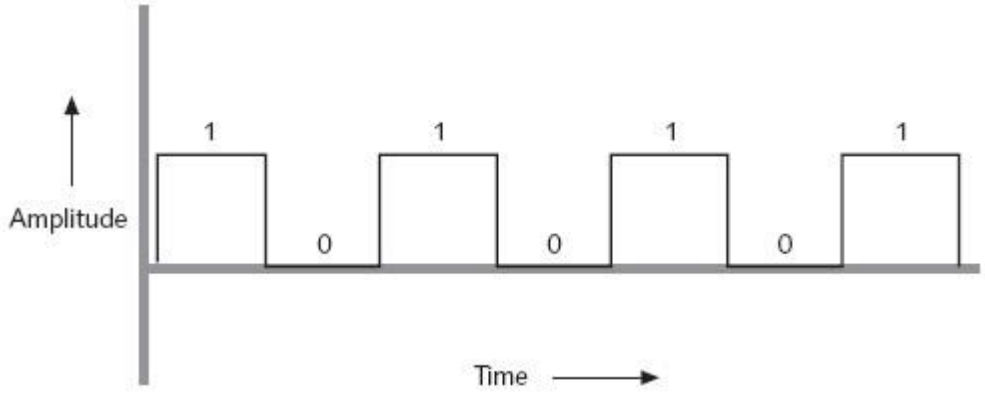


Figure 50: Digital signal

2.8.4 Digital Modulation Techniques

(i) Amplitude Shift Keying
(ii) Frequency Shift Keying
(iii) Phase Shift Keying A.S.K. F.S.K. P.S.K.

Digital signals

Are composed of pulses of precise, positive voltages and zero voltages. A pulse of positive voltage represents a 1. A pulse of zero voltage (in other words, the lack of any voltage) represents a 0. The use of 1s and 0s to represent information is characteristic of a binary system. Every pulse in the digital signal is called a binary digit, or bit.

Figure 50 an example of a digital signal

A bit can have only one of two possible values: 1 or 0. Eight bits together form a byte. In broad terms, one byte carries one piece of information. For example, the byte 01111001 means 121 on a digital network.

Computers read and write information—for example, program instructions, routing information, and network addresses—in bits and bytes. When a number is represented in binary form (for example, 01111001), each bit position, or placeholder, in the number represents a specific multiple of 2. Because a byte contains eight bits, it has eight placeholders. When counting placeholders in a byte, you move from right to left. The placeholder farthest to the right is known as the zero position; the one to its left is in the first position, and so on. The placeholder farthest to the left is in the seventh position, as shown in Figure 51.

Bit position:	7	6	5	4	3	2	1	0
Binary exponential:	2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0
Value if bit = 1:	128	64	32	16	8	4	2	1

Figure 51: seventh position

To find the decimal value of a bit, you multiply the 1 or 0 (whichever the bit is set to) by 2^x, where *x* equals the bit's position. For example, the 1 or 0 in the zero position must be multiplied by 2 to the 0 power, or 2_0, to determine its value. Any number (other than zero) raised to the power of 0 has a value of 1. Thus, if the zero-position bit is 1, it represents a value of $1 \times 2_0$, or 1×1, which equals 1. If a 0 is in the zero position, its value equals $0 \times 2_0$, or 0×1, which equals 0. In every position, if a bit is 0, that position represents a decimal number of 0.

To convert a byte to a decimal number, determine the value represented by each bit, and then add those values together. If a bit in the byte is 1 (in other words, if it's "on"), the bit's numerical equivalent in the coding scheme is added to the total. If a bit is 0, which position has no value and nothing is added to the total. For example, the byte 11111111 equals: $1 \times 2_7 + 1 \times 2_6 + 1 \times 2_5 + 1 \times 2_4 + 1 \times 2_3 + 1 \times 2_2 + 1 \times 2_1 + 1 \times 2_0$, or 128 + 64 + 32 + 16 + 8 + 4 + 2 + 1. Its decimal equivalent, then, is 255. In another example, the byte 00100100 equals: $0 \times 2_7 + 0 \times 2_6 + 1 \times 2_5 + 0 \times 2_4 + 0 \times 2_3 + 1 \times 2_2 + 0 \times 2_1 + 0 \times 2_0$, or 0 + 0 + 32 + 0 + 0 + 4 + 0 + 0. Its decimal equivalent, then, is 36.

Figure 51 illustrates placeholders in a byte, the exponential multiplier for each position, and the different decimal that are represented by a 1 in each position.

To convert a decimal number to a byte, you reverse this process. For example, the decimal number 8 equals 2_3, which mean a single "on" bit would be indicated in the fourth bit position as follows: 00001000. In another example, the decimal number 9 equals 8 + 1, or $2_3 + 2_0$, and would be represented by the binary number 00001001.

The binary numbering scheme may be used with more than eight positions. However, in the digital world, bytes form the building blocks for messages, and bytes always include eight positions. In a data signal, multiple bytes are combined to form a message. If you were to peek at the 1s and 0s used to transmit an entire email message, for example, you might see millions of zeros and ones passing by. A JPEG, that expresses letters, numbers, and pictures.

THINK POINT

Discussion Question

Research Study and further reading:

Statistical Multiplexing, Wavelength Division Multiplexing, Dense Wavelength Division Multiplexing and CWDM (Coarse Wave length Division Multiplexing)

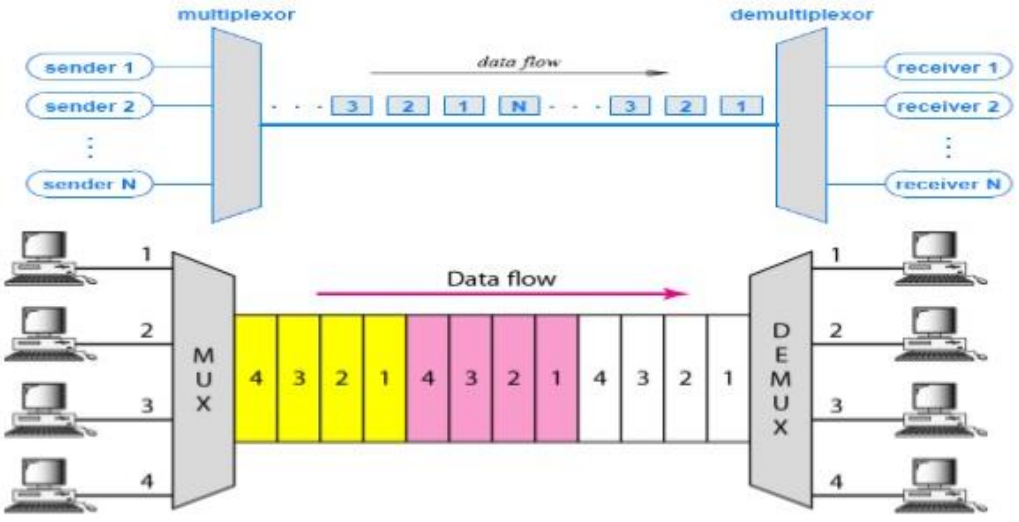


Figure 52: TDM Concept

Source: http://www.slideshare.net/nimay1/multiplexing-fdmatdmacdma

2.8.5 Frequency Division Multiplexing is a networking method in which multiple data signals are joint for concurrent transmission via a shared communication medium. **FDM** uses a carrier signal at a discrete frequency for each data stream and then combines many modulated signals. Below **blue waves:** A circuit switch put together the Multiplexes multiple voice calls on a high- bandwidth link.

Each circuit receives a fixed bandwidth the frequency of each call is shifted, to avoid interference.

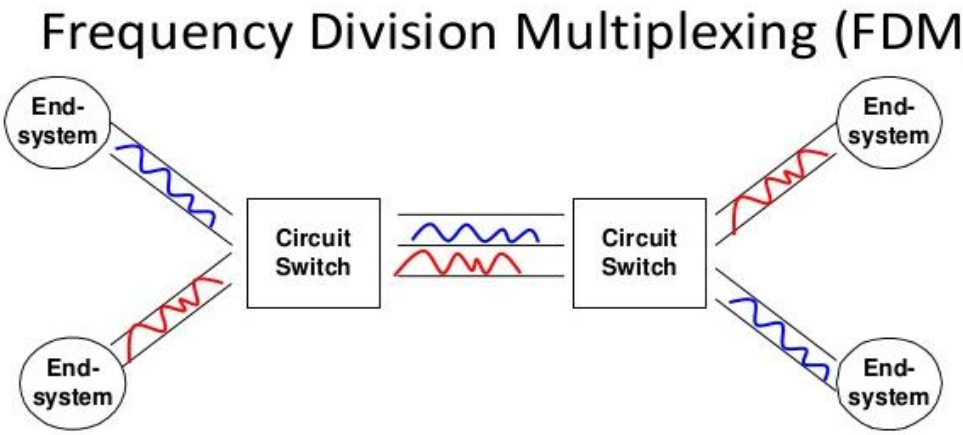


Figure 53: FDM

Source: http://www.slideshare.net/AmarPanchal1/complete-computer-network

There are other Multiplexing techniques that were left out of this guide for the purpose of research listed below;

Note: For further reading use recommended text book: Networking Network+ guide to Networks 7th Edition (Tamara Dean) pp 211-218.

1. https://www.edgefx.in/types-of-modulation-techniques-with-applications/
2. https://www.edgefx.in/types-of-modulation-techniques-with-applications/

Recommended books

CompTIA Network + Study guide by Todd Lammle (page 27- 53)
CompTIA Network + Guide to Networks by Tamara Dean (pages 18-25)
WEB:

	Review Questions

Answers are available on Network+ study guide, 3rd Edition by Todd Lammle. Chapter 17. pages 636 Appendix A

1. What command can you type from a command prompt to see the hopes a packet takes to get to a destination host?
2. What tool would you use to verify a complaint about a slow network?
3. You need your IP address, subnet mask, default gateway and DNS information. What command will you type from a windows command prompt?
4. You need to login as a dumb terminal to a server or UNIX host and run programs. What application will you use?
5. What command will you use to add a route on a windows server routing table?
6. What application will you use to log in to a server and transfer files?
7. You are asked to check your name resolution information on your host. What command will you type from the command prompt?
8. You want to use netstat, but you want to see only the IP address, not the names of the hosts. Which modifier will you use?
9. You want the IP configuration on a UNIX host. What command will you type at the command prompt?
10. Which windows command will show you the routing table of your host or server?

Assessment questions

Answers are available on Network+ study guide, 3rd Edition by Todd Lammle. Chapter 17. pages 637 Appendix B

1. Which TCP/IP utility is most often used to test whether an IP host is up and functional?
    a. ftp
    b. telnet
    c. ping
    d. netstat
2. Which TCP/IP utility will produce the following result?

```
Interface: 199.102.30.152
Internet Address    Physical Address    Type
199.102.30.152      A0-ee-00-5b-0e-ac    dynamic
```

a. Arp

b. netstat

c. tracert

d. nbtstat

3. Which windows utility can you use to connect to a machine 50 miles away to troubleshoot?

a. Remote Desktop

b. netstat

c. arp

d. Wireshark

4. Which TCP/IP utility might produce the following output

```
Reply from 204.153.163.2: bytes=32 time=1ms TTL=128
Reply from 204.153.163.2: bytes=32 time=1ms TTL=128
Reply from 204.153.163.2: bytes=32 time=1ms TTL=128
Reply from 204.153.163.2: bytes=32 time<10ms TTL=128
```

a. tracert

b. ping

c. wins

d. ipconfig

5. Which utility can you use to find the MAC and TCP/IP addresses of your Windows workstation?

a. ping

b. ipconfig

c. ipconfig /all

d. tracert

e. telnet

6. which ping commands will verify that your local TCP/IP interface is working (choose all that apply)

a. ping 204.153.163.2

b. ping 127.0.0.1

c. ping localif

d. ping localhost

e. ping impost

7. Which switch for the windows nbtstat utility will display all NetBIOS name resolution statics?

a. –r

b. /r

c. –R

d. –R

8. You need to find a NICs specific MAC address. Which command-line tool can you use to find this information without physically going to the computer?

    a. ping

    b. nbtstat

    c. netstat

    d. ftp

9. Which nbtstat utility switch will purge and reload the remote NetBIOS name table cache?

    a. –r

    b. –R

    c. /r

    d. /R

10. Wireshark is an example of a_____?

    a. Throughput

    b. Protocol analyser

    c. Remote connection tool

11. Which utility products output similar to the following?

```
1  110 ms  96 ms  107 ms fgo1.corpcomm.net [209.74.93.10]
2  96 ms  126 ms  95 ms someone.corpcomm.net [209.74.93.1]
3  113 ms  119 ms  112 ms Serial5-1-1.GW2.MSP1.alter.net     [157.130.100.185]
4  133 ms  123 ms  126 ms 152.ATM3-0.XR2.CHI6.ALTER.NET      [146.188.209.126]
5  176 ms  133 ms  129 ms 290.ATM2-0.TR2.CHI4.ALTER.NET   [146.188.209.10]
6  196 ms  184 ms  218 ms 106.ATM7-0.TR2.SCL1.ALTER.NET   [146.188.136.162]
7  182 ms  187 ms  187 ms 298.ATM7-0.XR2.SJC1.ALTER.NET   [146.188.146.61]
8  204 ms  176 ms  186 ms 192.ATM3-0-0.SAN-JOSE9- GW.ALTER.NET [146.188.144.133]
9  202 ms  198 ms  212 ms atm3-0-622M.cr1.sjc.globalcenter.net [206.57.16.17]
10 209 ms  202 ms  195 ms pos3-1-155M.br4.SJC.globalcenter.net [206.132.150.98]
11 190 ms   *   191 ms pos0-0-0-155M.hr3.SNV.globalcenter.net [206.251.5.93]
12 195 ms  188 ms  188 ms pos4-1-0-   155M.hr2.SNV.globalcenter.net
[206.132.150.206]

13 198 ms   202 ms   197 ms www10.yahoo.com [204.71.200.75]
```

    a. Arp

    b. Tracert

    c. Nbtstat

    d. Netstat

12. You are the network administrator. A user calls you, complaining that the performance of the intranet web server is slugging. When you try to ping the server, it takes several seconds for the server to respond. You suspect that the problem is realistic to a router that is seriously overloaded. Which workstation utility could you use to find out which router is causing this problem?

    a. Netstat
    b. Nbtstat
    c. Tracert
    d. Ping
    e. Arp

13. Which ipconfig switch will display the most complete listing of IP configuration information for a station?

    a. /all
    b. /renew
    c. /release
    d. /?

14. Which utility will display a list of all the routers that a packet passes though on the way to an IP destination?

    a. netstat
    b. nbtstat
    c. tracert
    d. ping
    e. arp

15. Which Windows TCP/IP utility could you use to find out whether a server is responding on TCP port 21?

    a. tcp
    b. Port
    c. Ping
    d. Nbtstat
    e. telnet

16. Which arp command can you use to display the currently cached ARP entries?

    a. arp

b. arp-all

c. arp/a

d. ipconfig/arp

e. arp/ipconfig

17. Which command-line tool would best be used to verify DN5 functionality?

a. netstat

b. nbtstat

c. dig

d. icmp

e. arp

18. Which of the following arp utility switches perform the same function?(choose all that apply)

a. –g

b. –A

c. –d

d. –a

19. Which command would you use at a workstation prompt to see the DNS servers that are configured to use?(choose three)

a. arp

b. nslookup

c. netstat

d. nbtstat

e. ipconfig/all

f. ifconfig

20. Which nbtstat switch displays a list of all the NetBIOS sessions currently active on the local workstation?

a. –a

b. –r

c. –s

d. -i

CHAPTER 3: NETWORK CONNECTIVITY DEVICES

Learning outcomes

- **Understand switches, hubs, routers and bridges**
- **Know how to configure network connectivity devices**
- **Understand and implement and configure a firewall**
- **Understand a DHCP**
- **Understand a Proxy and reverse proxy how they work and types**
- **Be able to understand VLANS**

3.1 A Switch

- Typically connects individual computers
  - A switch is essentially the same as a bridge
  - though typically used to connect hosts, not LANs
  - Like bridges, support concurrent communication
- Host A can talk to C, while B talks to D

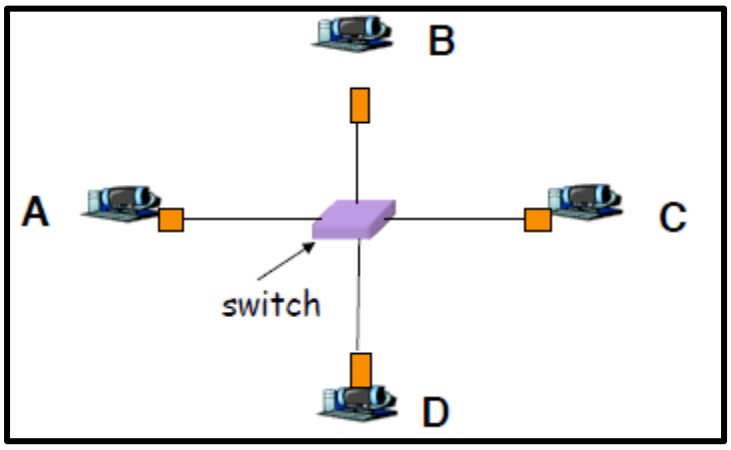


Figure 54: Switch

3.1.1 Dedicated Access and Full Duplex

• Dedicated access

– Host has direct connection to the switch rather than a shared LAN connection

• Full duplex

– Each connection can send in both directions

– Host sending to switch, and host receiving from switch

– E.g., in 10BaseT and 100Base T

• Completely supports concurrent transmissions

– Each connection is a bidirectional point-to-point link

3.1.2 Switches: Traffic Isolation

• Switch breaks subnet into LAN segments
• Switch filters packets
– Frame only forwarded to the necessary segments
– Segments can support separate transmissions

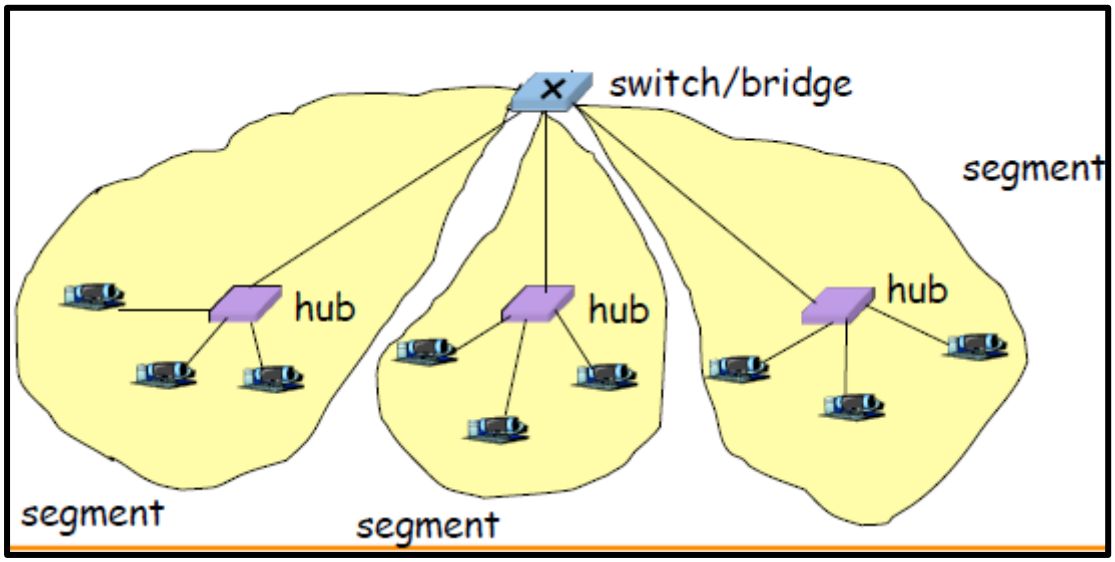


Figure 55: Switch and Bridge

3.2 Multilayer switch

Traditionally, switches are the network devices that forward data packets based on the Layer 2 information like media access control (MAC) addresses. Routers forward packets based on IP addresses. The router strips off the old Layer 2 header, slaps on the new one and queues the packet for transmission.

As multilayer switching technology developed, higher level functions were also added such as the ability to look deep inside packets for information that could aid in the packet-forwarding process. Thus, multilayer switches became devices that examine Layer 2 through Layer 7.

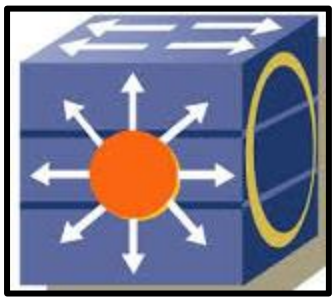


Figure 56: Multi-layer switch

3.4 Router



Figure 57: Cisco Router

Routers are small electronic devices that join multiple computer networks together via either wired or wireless connections.

3.4.1 How Routers Work

In technical terms, a router is a Layer 3 network gateway device, meaning that it connects two or more networks and that the router operates at the network layer of the OSI model.

Routers contain a processor (CPU), several kinds of digital memory, and input-output (I/O) interfaces.

They function as special-purpose computers, one that does not require a keyboard or display.

The router's memory stores an embedded operating system (O/S). Compared to general-purpose OS products like Microsoft Windows or Apple Mac OS, router operating systems limit what kind of applications can be run on them and also need much smaller amounts of storage space. Examples of popular router operating systems include *Cisco Internetwork Operating System (IOS)* and DD-WRT. These operating systems are manufactured into a binary firmware image and are commonly called *router firmware*. By maintaining configuration information in a part of memory called the routing table, routers also can filter both incoming and outgoing traffic based on the addresses of senders and receivers.

3.4.2 Routers for Business Networks and the Internet

Before home networking became popular, routers could be found only the closets of businesses and schools. Each cost thousands of dollars and required special technical training to set up and manage. The largest and most powerful network routers form the Internet backbone. These routers must manage many terabits of data flowing through and between Internet Service Provider (ISP) networks

3.4.3 Home Broadband Routers

Routers became mainstream consumer devices when households began to accumulate multiple computers and wanted to share the home Internet connection

Home networks use Internet Protocol (IP) routers to connect computers to each other and to the Internet. Early generations of home routers supported wired networking with Ethernet cables while newer wireless routers supported Wi-Fi together with Ethernet. The term broadband router applies to any home wired or wireless router being used for sharing a broadband Internet connection.

Home routers often cost USD $100 or less. They are manufactured to be much more affordable than business routers in part because they offer fewer features. Still, home routers provide many essential home networking functions:

- sharing of home Internet connections for dozens of devices
- basic home network firewall and other security support
- ability to change router configuration settings from a Web browsers

3.4.4 Switch versus a Router

Switches create a network. Routers connect networks. A router links computers to the Internet, so users can share the connection. A router acts as a dispatcher, choosing the best path for information to travel so it's received quickly.

3.5 Firewall

A gateway that selectively blocks or filters traffic between networks. As with any other type of gateway, firewalls may be devices optimized for performing their tasks or computers installed with software necessary to accomplish those tasks.

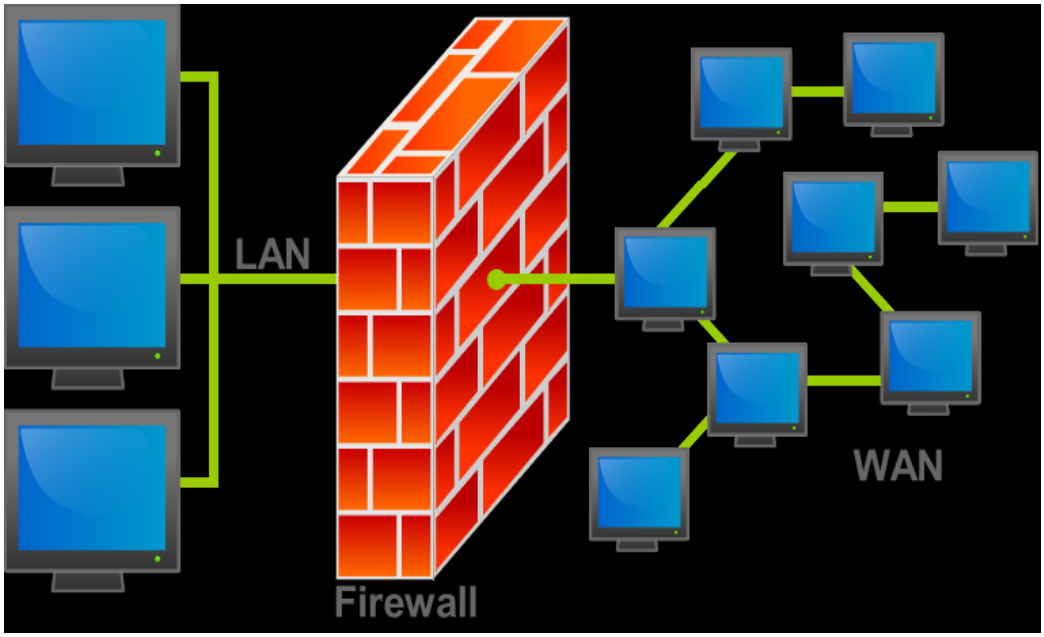


Figure 58: Firewall

Types firewall

3.4.1 Proxy firewall

An early type of firewall device, a proxy firewall serves as the gateway from one network to another for a specific application. Proxy servers can provide additional functionality such as content caching and security by preventing direct connections from outside the network. However, this also may impact throughput capabilities and the applications they can support.

3.4.2 Stateful inspection firewall

Now thought of as a "traditional" firewall, a stateful inspection firewall allows or blocks traffic based on state, port, and protocol. It monitors all activity from the opening of a connection until it is closed. Filtering decisions are made based on both administrator-defined rules as well as context, which refer to using information from previous connections and packets belonging to the same connection.

3.4.3 Unified threat management (UTM) firewall

A UTM device typically combines, in a loosely coupled way, the functions of a stateful inspection firewall with intrusion prevention and antivirus. It may also include additional services and often cloud management. UTMs focus on simplicity and ease of use.

3.4.4 Next-generation firewall (NGFW)

Firewalls have evolved beyond simple packet filtering and stateful inspection. Most companies are deploying next-generation firewalls to block modern threats such as advanced malware and application-layer attacks.

According to Gartner, Inc.'s definition, a next-generation firewall must include:

- Standard firewall capabilities like stateful inspection
- Integrated intrusion prevention
- Application awareness and control to see and block risky apps
- Upgrade paths to include future information feeds
- Techniques to address evolving security threats

While these capabilities are increasingly becoming the standard for most companies, NGFWs can do more.

3.4.5 Threat-focused NGFW

These firewalls include all the capabilities of a traditional NGFW and also provide advanced threat detection and remediation. With a threat-focused NGFW you can:

- Know which assets are most at risk with complete context awareness
- Quickly react to attacks with intelligent security automation that sets policies and hardens your defences dynamically
- Better detect evasive or suspicious activity with network and endpoint event correlation
- Greatly decrease the time from detection to clean-up with retrospective security that continuously monitors for suspicious activity and behaviour even after initial inspection
- Ease administration and reduce complexity with unified policies that protect across the entire attack continuum

Further reading: http://www.cisco.com/c/en/us/products/security/firewalls/what-is-a-firewall.html

Video: http://www.cisco.com/c/en/us/products/security/firewalls/what-is-a-firewall.html

3.4.6 HIDS

Host based intrusion detection (HIDS) refers to intrusion detection that takes place on a single host system. Currently, HIDS involves installing an agent on the local host that monitors and reports on the system configuration and application activity. Some common abilities of HIDS systems include log analysis, event correlation, integrity checking, policy enforcement, rootkit detection, and alerting. They often also have the ability to baseline a host system to detect variations in system configuration. In specific vendor implementations these HIDS agents also allow connectivity to other security systems. For example, Cisco CSA has the ability to send host data upstream to Cisco network IPS devices, Checkpoint Integrity can be integrated with Checkpoint Secure Client (Client VPN), and IBM Proventia Desktop is Cisco NAC certified.

HIDS Intrusion Prevention

Most HIDS packages now have the ability to actively prevent malicious or anomalous activity on the host system. Due to the potential impact this can have on the end user, HIDS is frequently deployed in "monitor only" mode initially. This enables the administrator to create a baseline of the system configuration and activity. Active blocking of applications, system changes, and network activity is limited to only the most egregious activities. Administrators can then tune the system policy based on what is considered "normal activity".

3.4.7 IDS/IPS

- **Intrusion Detection System** - A device or application that analyzes whole packets, both header and payload, looking for known events. When a known event is detected a log message is generated detailing the event.
- **Intrusion Prevention System** - A device or application that analyzes whole packets, both header and payload, looking for known events. When a known event is detected the packet is rejected.

The functional difference between an IDS and an IPS is a fairly subtle one and is often nothing more than a configuration setting change. For example, in a Juniper IDP module, changing from Detection to Prevention is as easy as changing a drop-down selection from LOG to LOG/DROP. At a technical level it can sometimes require redesign of your monitoring architecture.

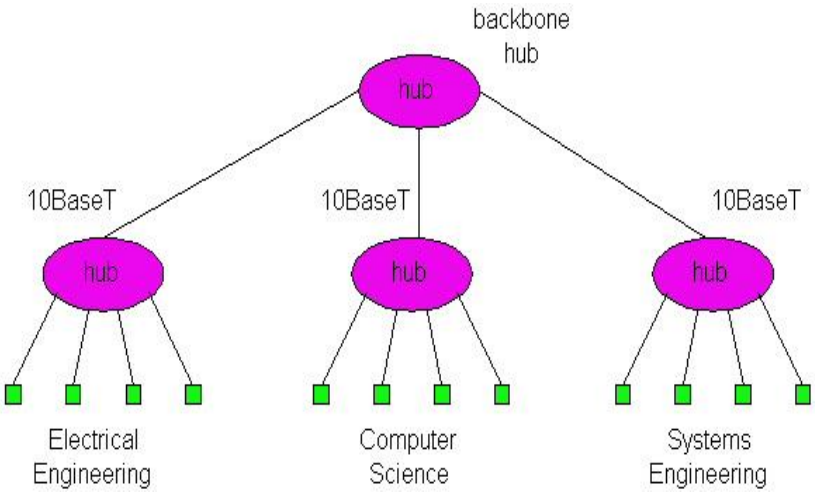
Given the similarity between all three systems there has been some convergence over time. The Juniper IDP module mentioned above, for example, is effectively an add-on component to a firewall. From a network flow and administrative perspective the firewall and IDP are functionally indistinguishable even if they are technically two separate devices.

There is also much market discussion of something called a Next Generation Firewall (NGFW). The concept is still new enough that each vendor has their own definition as to what constitutes a NGFW but for the most part all agree that it is a device that enforces policy unilaterally across more than just network packet header information. This can make a single device act as both a traditional Firewall and IPS. Occasionally additional information is gathered, such as from which user the traffic originated, allowing even more comprehensive policy enforcement.

3.5 Hubs and Bridges

- Physical Layer devices: essentially repeaters operating at bit levels: repeat received bits on one interface to all other interfaces
- Hubs can be arranged in a hierarchy (or multi-tier design), with backbone hub at its top

Example of a Hub below:



Each connected LAN referred to as LAN **segment**. Hubs do not isolate collision domains: node may collide with any node residing at any segment in LAN

Hub Advantages:

- simple, inexpensive device
- Multi-tier provides graceful degradation: portions of the LAN continue to operate if one hub malfunctions
- extends maximum distance between node pairs (100m per Hub)

Hub limitations

- single collision domain results in no increase in max throughput
  - multi-tier throughput same as single segment throughput
- individual LAN restrictions pose limits on number of nodes in same collision domain and on total allowed geographical coverage
- Cannot connect different Ethernet types (e.g., 10BaseT and 100baseT) why?

3.6 DHCP

What Is DHCP?

3.6.1 Applies To: Windows Server 2008

Dynamic Host Configuration Protocol (DHCP) is a client/server protocol that automatically provides an Internet Protocol (IP) host with its IP address and other related configuration information such as the subnet mask and default gateway. RFCs 2131 and 2132 define DHCP as an Internet Engineering

Task Force (IETF) standard based on Bootstrap Protocol (BOOTP), a protocol with which DHCP shares many implementation details. DHCP allows hosts to obtain required TCP/IP configuration information from a DHCP server.

Windows Server® 2008 includes the DHCP Server service, which is an optional networking component. All Windows-based clients include the DHCP client as part of TCP/IP, including Windows Vista®, the Windows Server®°2003 operating system, the Windows® XP Professional operating system, Microsoft Windows®°2000 Professional operating system, Microsoft Windows°NT® Workstation°4.0 operating system, Microsoft Windows® Millennium Edition operating system, and the Microsoft Windows®°98 operating system.

3.6.2 Why use DHCP?

Every device on a TCP/IP-based network must have a unique unicast IP address to access the network and its resources. Without DHCP, IP addresses for new computers or computers that are moved from one subnet to another must be configured manually; IP addresses for computers that are removed from the network must be manually reclaimed.

With DHCP, this entire process is automated and managed centrally. The DHCP server maintains a pool of IP addresses and leases an address to any DHCP-enabled client when it starts up on the network. Because the IP addresses are dynamic (leased) rather than static (permanently assigned), addresses no longer in use are automatically returned to the pool for reallocation.

The network administrator establishes DHCP servers that maintain TCP/IP configuration information and provide address configuration to DHCP-enabled clients in the form of a lease offer. The DHCP server stores the configuration information in a database that includes:

- Valid TCP/IP configuration parameters for all clients on the network.
- Valid IP addresses, maintained in a pool for assignment to clients, as well as excluded addresses.
- Reserved IP addresses associated with particular DHCP clients. This allows consistent assignment of a single IP address to a single DHCP client.
- The lease duration, or the length of time for which the IP address can be used before a lease renewal is required.

A DHCP-enabled client, upon accepting a lease offer, receives:

- A valid IP address for the subnet to which it is connecting.

- Requested DHCP options, which are additional parameters that a DHCP server is configured to assign to clients. Some examples of DHCP options are Router (default gateway), DNS Servers, and DNS Domain Name. For a full list of DHCP options, see DHCP Tools and Options.

3.6.3 Benefits of DHCP

In Windows Server 2008, the DHCP Server service provides the following benefits:

- **Reliable IP address configuration**. DHCP minimizes configuration errors caused by manual IP address configuration, such as typographical errors, or address conflicts caused by the assignment of an IP address to more than one computer at the same time.

- **Reduced network administration**. DHCP includes the following features to reduce network administration:
  - Centralized and automated TCP/IP configuration.
  - The ability to define TCP/IP configurations from a central location.
  - The ability to assign a full range of additional TCP/IP configuration values by means of DHCP options.
  - The efficient handling of IP address changes for clients that must be updated frequently, such as those for portable computers that move to different locations on a wireless network.
  - The forwarding of initial DHCP messages by using a DHCP relay agent, which eliminates the need for a DHCP server on every subnet.

3.7 Proxy Reverse Proxy

3.7.1 Web acceleration refers to speeding up the transfer of content between web servers and client browsers by using a variety of techniques such as caching and compression.

High-traffic websites must support hundreds of thousands, if not millions, of users in a fast, reliable manner. To scale to meet such high volumes, modern computing best practice usually calls for adding more server hardware, which can become expensive. Web acceleration instead employs other methods to speed delivery of both static and dynamic content, enabling your web servers to handle more client requests without the need for more hardware.

3.7.2 HTTP Optimization

One of the primary ways to accelerate web traffic is by using a load balancer or reverse proxy server to optimize HTTP traffic as it flows between clients and backend servers. Users access the web from a variety of devices – laptops, mobile phones, smart TVs, and cameras – over connections that vary widely in bandwidth. Content acceleration is important because the web server can have

its resources tied up as the client is accepting and processing the data it has just received. This can create inefficiencies in server utilization and lead to poor performance for other users as well. Through HTTP optimization techniques, the load balancer sits in between clients and servers, forwarding requests for content to backend servers in a streamlined and efficient manner that maximizes speed and server utilization.

3.7.3 Caching and Prefetching

Web acceleration servers can cache, or locally store, commonly requested information rather than fetching it over and over from backend servers, which wastes server resources and slows down content delivery. The web accelerator can refresh cached content at a specified interval to avoid serving content that is out of date.
The web accelerator can also pre-fetch and cache content that the user is likely to ask for, such as the next page of a document, making it ready for delivery as soon as the user requests it.

Compression

A web accelerator can compress large files, such as image or video files, to reduce transfer times.

3.7.4 SSL/TLS Processing

Some advanced web accelerators can offload computationally intensive processing from backend servers, freeing them to serve content faster. A common example is encryption and decryption of documents during transmissions secured with Secure Sockets Layer (SSL) or Transport Layer Security (TLS).

3.8 VLANS

A VLAN is a group of devices on one or more LANs that are configured to communicate as if they were attached to the same wire, when in fact they are located on a number of different LAN segments. Because VLANs are based on logical instead of physical connections, they are extremely flexible.

VLANs define broadcast domains in a Layer 2 network. A broadcast domain is the set of all devices that will receive broadcast frames originating from any device within the set. Broadcast domains are typically bounded by routers because routers do not forward broadcast frames. Layer 2 switches create broadcast domains based on the configuration of the switch. Switches are multiport bridges that allow you to create multiple broadcast domains. Each broadcast domain is like a distinct virtual bridge within a switch.

You can define one or many virtual bridges within a switch. Each virtual bridge you create in the switch defines a new broadcast domain (VLAN). Traffic cannot pass directly to another VLAN (between broadcast domains) within the switch or between two switches. To interconnect two different VLANs, you must use routers or Layer 3 switches. See the "Overview of Layer 3 Interfaces" section for information on inter-VLAN routing on Catalyst 4500 series switches. Figure below shows an example of three VLANs that create logically defined networks.

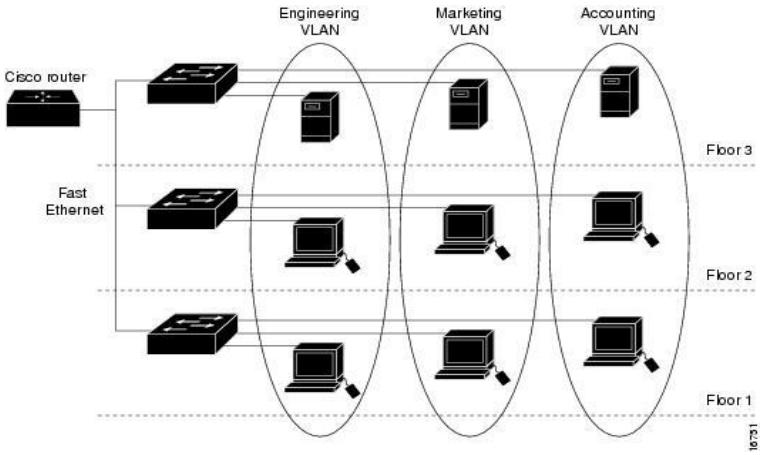


Figure 59: Sample VLANs

VLANs are often associated with IP subnetworks. For example, all of the end stations in a particular IP subnet belong to the same VLAN. Traffic between VLANs must be routed. You must assign LAN interface VLAN membership on an interface-by-interface basis (this is known as interface-based or static VLAN membership).

You can set the following parameters when you create a VLAN in the management domain:

- VLAN number
- VLAN name
- VLAN type
- VLAN state (active or suspended)
- Maximum transmission unit (MTU) for the VLAN
- Security Association Identifier (SAID)
- VLAN number to use when translating from one VLAN type to another

3.9 PoE (Power over Ethernet) and STP (Spanning Tree Protocol)

Introduction

This document presents a list of recommendations that help to implement a safe network with regard to bridging for Cisco Catalyst switches that run Catalyst OS (CatOS) and Cisco IOS® Software. This document discusses some of the common reasons that Spanning Tree Protocol (STP) can fail and the information for which to look to identify the source of the problem. The document also shows the kind of design that minimizes spanning tree-related issues and is easy to troubleshoot.

Prerequisites

There are no specific requirements for this document.

Components Used

This document is not restricted to specific software and hardware versions.

Background Information

This document does not discuss the basic operation of STP. To learn how STP works, refer to this document:

3.9.1 Understanding and Configuring Spanning Tree Protocol (STP) on Catalyst Switches

This document does not discuss Rapid STP (RSTP), defined in IEEE 802.1w. Also, this document does not discuss Multiple Spanning Tree (MST) protocol, defined in IEEE 802.1s. For more information on RSTP and MST, refer to these documents:

Understanding Multiple Spanning Tree Protocol (802.1s)

Understanding Rapid Spanning Tree Protocol (802.1w)

For a more specific STP troubleshooting document for Catalyst switches that run Cisco IOS Software, refer to the document Troubleshooting STP on Catalyst Switch Running Cisco Integrated IOS (Native Mode).

3.9.2 Spanning Tree Protocol Failure

The primary function of the spanning-tree algorithm (STA) is to cut loops that redundant links create in bridge networks. The STP operates at Layer 2 of the Open System Interconnection (OSI) model. By means of bridge protocol data units (BPDUs) that exchange between bridges, the STP elects the ports that eventually forward or block traffic. This protocol can fail in some specific cases, and troubleshooting the resulting situation can be very difficult, which depends on the design of the

network. In this particular area, you perform the most important part of the troubleshooting before the problem occurs.

A failure in the STA generally leads to a bridging loop. Most customers that call Cisco Technical Support for spanning tree problems suspect a bug, but a bug is seldom the cause. Even if the software is the problem, a bridging loop in an STP environment still comes from a port that should block, but instead forwards traffic.

3.9.3 Spanning Tree Convergence

Refer to the Spanning Tree Flash animation to see an example that explains how the Spanning Tree initially converges. The example also explains why a blocked port goes into the forwarding mode because of an excessive loss of BPDUs, resulting in STA failure.

The rest of this document lists the different situations that can cause the STA to fail. Most of these failures relate to a massive loss of BPDUs. The loss causes blocked ports to transition to forwarding mode.

3.9.4 Duplex Mismatch

Duplex mismatch on a point-to-point link is a very common configuration error. If you manually set the duplex mode to full on one side of the link and leave the other side in auto negotiation mode, the link ends up in half-duplex. (A port with duplex mode set to full no longer negotiates.)

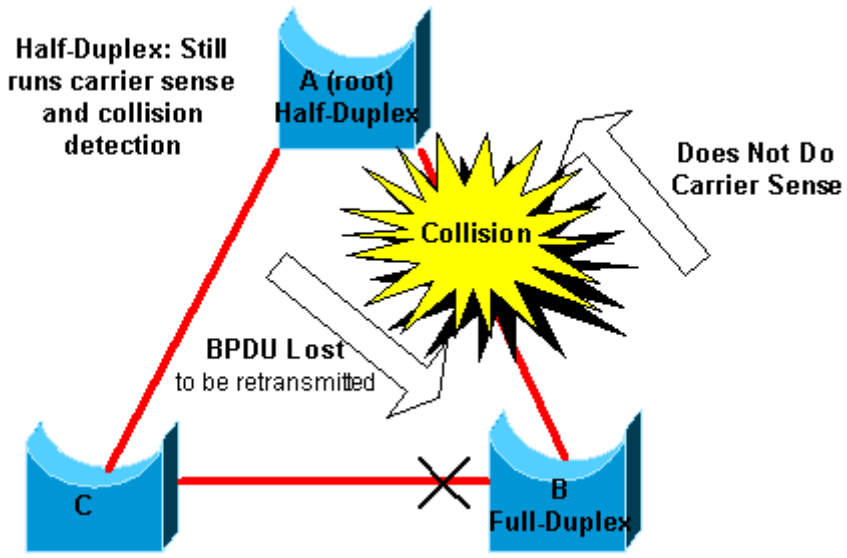


Figure 60: carrier sense and collision

The worst-case scenario is when a bridge that sends BPDUs has the duplex mode set to half-duplex on a port, but the peer port on other end of link has the duplex mode set to full-duplex. In the above example, the duplex mismatch on the link between bridge A and B can easily lead to a bridging loop. Because bridge B has configuration for full-duplex, it does not perform carrier sense before link access. Bridge B starts to send frames even if bridge A is already using the link. This situation is a problem for A; bridge A detects a collision and runs the back off algorithm before the bridge attempts another transmission of the frame. If there is enough traffic from B to A, every packet that A sends, which includes the BPDUs, undergoes deferment or collision and eventually gets dropped. From an STP point of view, because bridge B does not receive BPDUs from A anymore, bridge B has lost the root bridge. This leads B to unblock the port connected to bridge C, which creates the loop.

Further reading: Chapter 5, Network+, Study guide, 3rd Edition, page 126-43, by Todd Lammle.

Weblink: http://www.cisco.com/c/en/us/support/docs/lan-switching/spanning-tree-protocol/10556-16.html

Video: https://www.youtube.com/watch?v=u4E_mG0pe00

https://www.youtube.com/watch?v=w6TTDEyRI1o

Review questions

What is the purpose of Spanning Tree Protocol in a switched LAN?

A. To provide a mechanism for network monitoring in switched environments

B. To prevent routing loops in networks with redundant paths

C. To prevent switching loops in networks with redundant switched paths

D. To manage the VLAN database across multiple switches

Answer: Option C

Which statement describes a spanning-tree network that has converged?

A. All switch and bridge ports are in the forwarding state.

B. All switch and bridge ports are assigned as either root or designated ports.

C. All switch and bridge ports are in either the forwarding or blocking state.

D. All switch and bridge ports are either blocking or looping.

Answer: Option C

What does a switch do when a frame is received on an interface and the destination hardware address is unknown or not in the filter table?

A. Forwards the switch to the first available link

B. Drops the frame

C. Floods the network with the frame looking for the device

D. Sends back a message to the originating station asking for a name resolution

Answer: Option C

In which circumstance are multiple copies of the same unicast frame likely to be transmitted in a switched LAN?

A. During high-traffic periods

B. After broken links are re-established

C. When upper-layer protocols require high reliability

D. In an improperly implemented redundant topology

Answer: Option D

If you want to disable STP on a port connected to a server, which command would you use?

A. disable spanning-tree

B. spanning-tree off

C. spanning-tree security

D. spanning-tree portfast

Answer: Option D

Which of the following statement is true?

A. A switch creates a single collision domain and a single broadcast domain. A router creates a single collision domain.
B. A switch creates separate collision domains but one broadcast domain. A router provides a separate broadcast domain.
C. A switch creates a single collision domain and separate broadcast domains. A router provides a separate broadcast domain as well.
D. A switch creates separate collision domains and separate broadcast domains. A router provides separate collision domains.

Answer: Option B

You have two switches connected together with two crossover cables for redundancy, and STP is disabled. Which of the following will happen between the switches?

A. The routing tables on the switches will not update.
B. The MAC forward/filter table will not update on the switch.
C. Broadcast storms will occur on the switched network.
D. The switches will automatically load-balance between the two links.

Answer: Option C

Layer 2 switching provides which of the following?
Hardware-based bridging (ASIC)
Wire speed
Low latency

Low cost
A. 1 and 3
B. 2 and 4
C. 1, 2 and 4
D. All of the above
Answer: Option D

Your switch has a port status LED that is alternating between green and amber. What could this indicate?
A. The port is experiencing errors.
B. The port is shut down.
C. The port is in STP blocking mode.
D. Nothing; this is normal.
Answer: Option A

You want to run the new 802.1w on your switches. Which of the following would enable this protocol?
A. Switch(config)# spanning-tree mode rapid-pvst
B. Switch# spanning-tree mode rapid-pvst
C. Switch(config)# spanning-tree mode 802.1w
D. Switch# spanning-tree mode 802.1w
Answer: Option A

CHAPTER 4: NETWORK DESIGN

Learning outcome

- Network Structure and equipment in commercial buildings and work areas
- Identify significant components of an industrial control system or SCADA system
- Inventory and manage network assets and identify significant business documents
- Create and follow appropriate change management procedures for major and minor network changes
- Identify significant physical security controls to limit or monitor access to secure areas
- Describe the components of a reliable disaster recovery plan and a defensible incident response plan

4.1 Networks Cable type mastering

INTRODUCTION

In this topic we are discovering the basic essentials required to build a network. It is not an easy job to know where to place walls, doors where to install electrical and plumbing systems management of traffic patterns through rooms to make the building liveable. As a network architect you must consider hardware and software factors when designing a network.

We are going to discuss the hardware used in the network architecture; location of the devices their environment and the equipment limitations and its attentions. Interestingly you will be able to see how Ethernet make it easier to move data across the physical network and how to trace the failure of the devices to play its role. As you continue with your studies you will learn deeper on various layers of network components cable details and wireless networking (802.1) will be explained in due course.

4.2 Implementing a Network Infrastructure

The setup and cabling of most of the big organisations are the same due to the TIA (Telecommunication Industrial Association) standards and its former mother company EAI (Electronic Industries Alliance).

The architecture for commercial buildings is called **structured Cabling** from the TIA/EIA-568 Commercial Building Wiring Standard it doesn't care where the cable is from. This structured principle applies from the cabling media, transmission technology or networking speed. It starts from the Demarc and ends at the user's work station. **Demarc (Short for Demarcation)** is the device at the point where a telecommunications service providers network ends (public switched telephone network) and the organisational network starts.

It is only describing the good way of the network cable installation maximising performance and minimising maintenance.

However the structured cabling is based on a hierarchical design and assumes a network is based on the star topology.

Below is the type of a Demarc;

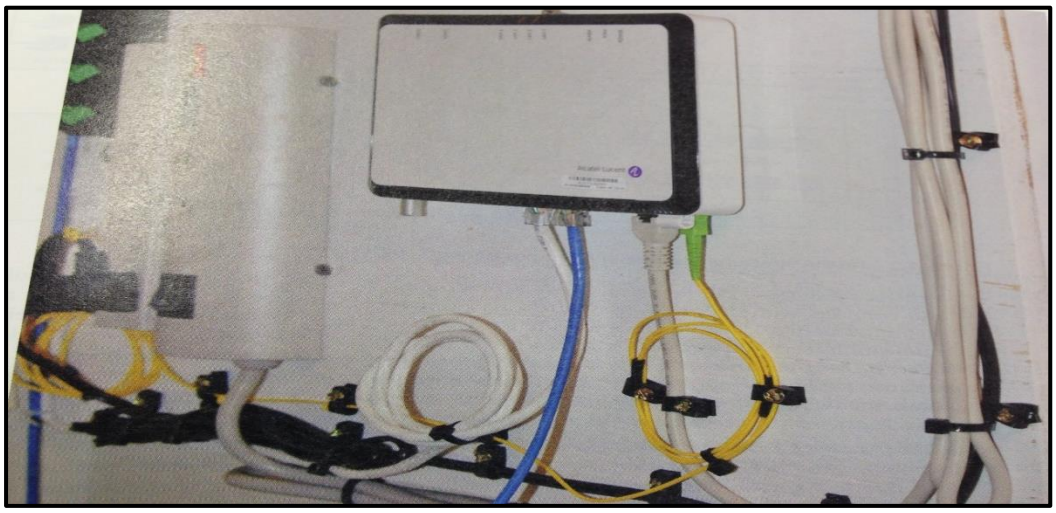


Figure 61: example of a Demarc

Source: Networking: Guide to Networks 7th Edition by Tamara Dean *et al (pp. 158-9)*

4.3 Devices for Structured Cabling

The benefits of these standards include:

- Consistency of design and installation;
- Conformance to physical and transmission line requirements;
- A basis for examining a proposed system expansion and other changes; and
- Uniform documentation.

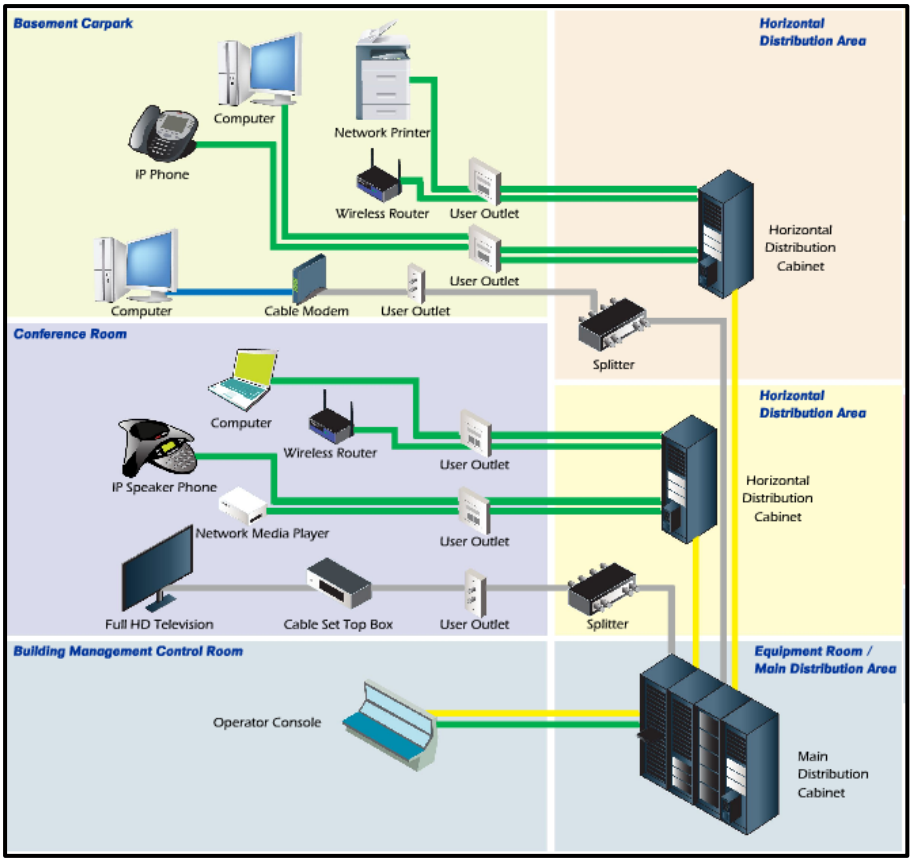


Figure 62: TIA/EIA Structured Cabling in a Multistory Building

Source: http://www.elvtek.com/Solution.aspx?catelist=0,5,36,19&cateid=19&parentid=36&itemid=7

Structured cabling design and installation is governed by TIA/EIA as said above, that specify wiring data centers, offices, and apartments building for data or voice communication using various kind of cable, most commonly used category 5e (CAT5e), (CAT6) or (CAT7) and fibre optic cabling and modular connector.

There are components like;

1. Entrance Facility (EF): This is the main incoming entrance for the network interface connecting to the building backbone cabling. It is composed of cables, network demarcation point(s), connecting hardware, protection devices and other equipment that connect to the access provider (AP) or private network cabling. It includes connections between outside plant and inside building cabling.

2. Equipment Room (ER)

   The environmentally well-ordered centralized space for telecommunications equipment is usually more complex than a telecommunications room (TR) or telecommunications enclosure (TE). This is where you find the main cross-connect (MC) {Distributor C} and may also contain the intermediate cross-connects (ICs) {Distributor B}, horizontal cross-connects (HCs) {Distributor A}, or both.

3. Backbone Cabling

The backbone cabling provides interconnection between telecommunications rooms, equipment rooms, access provider (AP) spaces and entrance facilities. There are two subsystems defined for backbone cabling:

- **Cabling Subsystem 2** – Backbone cabling between the horizontal cross-connect (HC) {Distributor A (DA)} and the intermediate cross-connect (IC) {Distributor B (DB)}
- **Cabling Subsystem 3** – Backbone cabling between an intermediate cross-connect (IC) {Distributor B (DB)} and the main cross-connect (MC) {Distributor C (DC)}

Recognized cabling:

- **100-ohm twisted-pair cabling**: Category 3, Category 5e, Category 6 or Category 6A and Category 7.
- **Multimode optical fiber cabling**: 850 nm laser-optimized 50/125 µm is recommended.
- 50/125 µm refer to the diameters of the glass or plastic core, the part of the fiber that carries the light which encodes your data. See below diagram depicting the structure,
- **Single-mode optical fiber cabling**

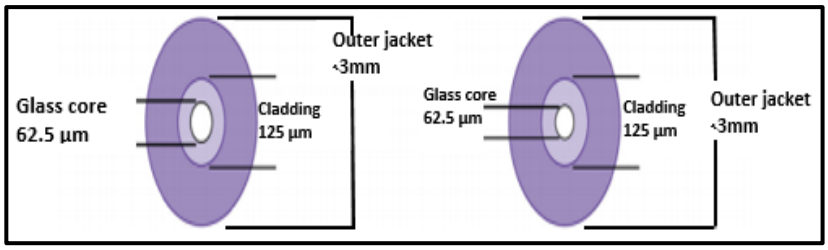


Figure 63: Cross section of Multimode fiber optic cables

4. Telecommunications Room (TR) or Telecommunications Enclosure (TE)

A TR or TE houses the terminations of horizontal and backbone cables to connecting hardware including any **jumpers** or **patch cords** (is a length of fiber cabling fitted with LC, SC, MTRJ or ST connectors at each end. It may also contain the IC (**Intermediate Crossconnect**) or MC (**Main Crossconnect**) for different portions of the backbone cabling system.

5. Horizontal Cabling (Cabling Subsystem 1)

The horizontal cabling system extends from the work area's telecommunications information outlet to the telecommunications room (TR)

or telecommunications enclosure (TE). It includes horizontal cable, mechanical terminations, jumpers and patch cords located in the TR or TE and may incorporate **Multi-User Telecommunications Outlet Assemblies** (MUTOAs) and consolidation points (CPs). The maximum horizontal cable length shall be not more than 90 m (295 ft.), independent of media type. If a MUTOA is deployed, the maximum horizontal balanced twisted-pair copper cable length shall be reduced.

Recognized and recommended cabling *(to be discussed in the next Chapters)***:**

- 4-pair 100-ohm unshielded or shielded twisted-pair cabling:
  Category 5e, Category 6, Category 6A or Category 7
- Multimode optical fiber cabling, 2-fiber (or higher fiber count)
- Single-mode optical fiber cabling, 2-fiber (or higher fiber count)

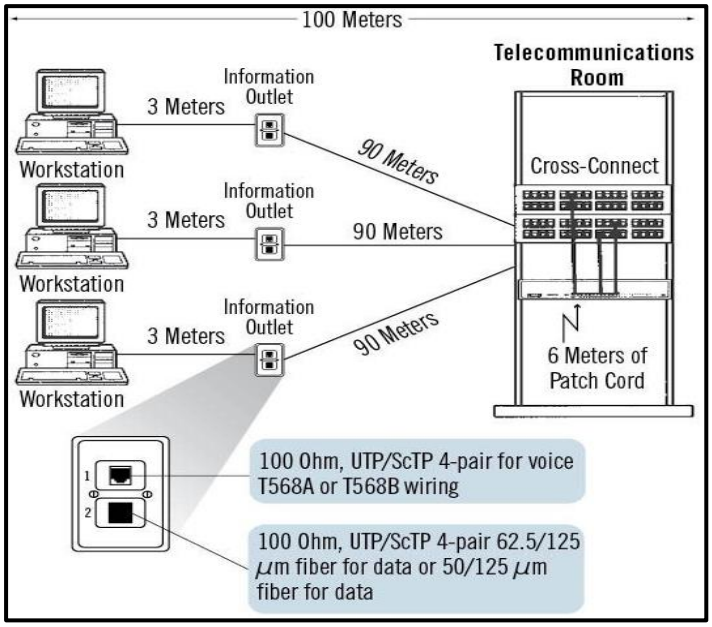


Figure 64; MUTOAs

6. Work Area

Work area (WA) components extend from the telecommunications outlet/connector end of the horizontal cabling system to the WA equipment.

A minimum of two telecommunications outlets (permanent links) should be provided for each work area. Multi-user telecommunications outlet assemblies (MUTOAs), if used, are part of the WA.

Further reading:

Networking: Guide to Networks 7th Edition by Tamara Dean *et al (pp. 160-2)*

Network+ study Guide, 3th Edition by Todd Lammle (*pp. 116-122)*

4.6 Enterprise Networks and SOHO (Small Office/Home Office)

- Identify significant components of an industrial control system or SCADA system
- Inventory and manage network assets and identify significant business documents
- Create and follow appropriate change management procedures for major and minor network changes
- Identify significant physical security controls to limit or monitor access to secure areas
- Describe the components of a reliable disaster recovery plan and a defensible incident response plan

4.7 Components of an Industrial Control System and SCADA Network

- Industrial control system (ICS)
  - Group of network computers used to manage a physical system of industrial processes
- Basic components specific to an ICS:
  - Supervisory control and data acquisition (SCADA)
  - Remote terminal units (RTU)
  - Programmable logic controller (PLC)
  - Communications channels
  - Human-machine interfaces (HMIs)
- Basic components specific to an ICS (cont'd):
  - Software and ICS servers
    - Acquisitions server (I/O server)
    - Control server, MTU (master terminal unit), or SCADA server
    - Historian

4.8 Industrial Control System

- Industrial system
  - A system of machines, such as an assembly line
  - Computers interact with machinery and physical components that are not digital or technical in nature

- ❖ Could be spread over a wide geographical area
    - Such as a public transportation system or a gas pipeline

- Internet of Things (IOT)
    - ❖ Consider by some to be the next generation of the Internet
    - ❖ Connects objects that are not used as computers

4.9 Components of an Industrial Control System and SCADA Network

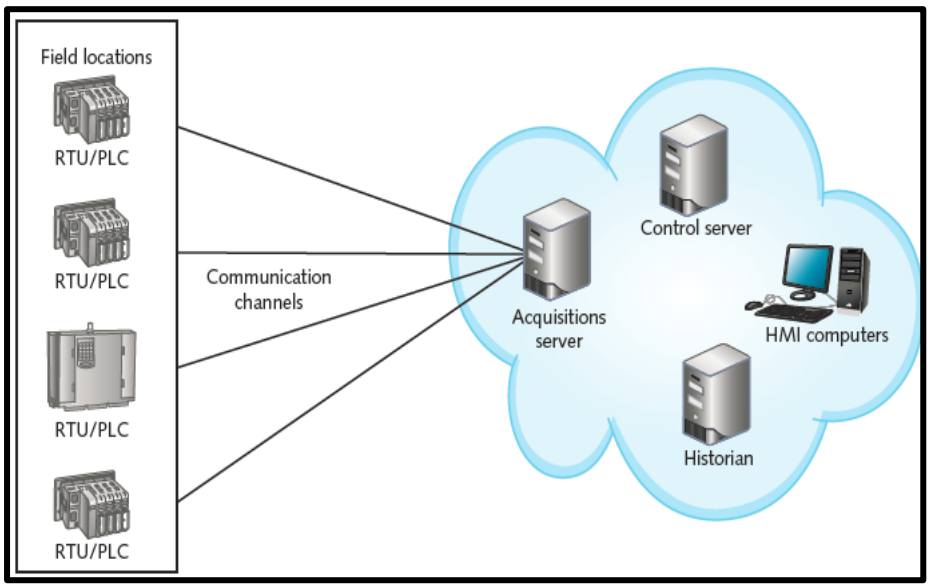


Figure 65: Basic ICS and SCADA Network



Figure 66: Programmable logic controller is programmed to affect a physical system

4.10 Components of an Industrial Control System and SCADA Network

- Two methods that an ICS might use to control the physical system:
  - Open loop system
  - Makes decisions based on predetermined expectations, events, and past history
- Closed loop system
  - Makes decisions based on real-time data
  - Requires field devices distributed throughout they physical system to monitor aspects of the system (called a distributed control system or DCS)

4.10.1 Securing an ICS/SCADA Network

- Recommended best practices:
  - Inventory all connections to your ICS/SCADA network
  - Segment your ICS/SCADA network from the corporate network
  - Isolate your ICS/SCADA network by deploying a DMZ between the corporate network and the ICS network
  - Completely disconnect the ICS/SCADA network from the Internet
  - Secure or harden the ICS/SCADA network by implementing strict firewall rules, IDS, and physical security controls
  - For fault tolerance, deploy redundancy as appropriate
  - Harden the ICS/SCADA network by strictly controlling access to the network with encrypted authentication
  - Protect the historian
  - Make sure vendors responsible for supporting hardware and software on your network fully disclose any backdoor entrance into your network
  - If the ICS network provides W-Fi, consider installing Faraday cages around the Wi-Fi hot spots
  - Keep current all documentation needed for configuration management
  - Keep well-documented and well-maintained backups of the system and its data
  - Clearly define risk management practices and establish risk management teams
  - Implement role-based access control (RBAC) to the system
  - Asset Management and Business Documents
- Asset management documentation
  - Important when managing large numbers of devices

- Essential in an enterprise environment

4.10.2 Asset Management

- Refers to monitoring and maintaining all assets that make up a network
- First step is to inventory all network components:
- Nodes or hardware devices
- Software
- Organization needs determine appropriate asset management tool
- Benefits
- Simplifies maintaining and upgrading network
- Provides information about hardware and software costs and benefits

4.10.3 Business Documents

- Standard business documents you may encounter:
  - RFP (request for proposal)
- Request to vendors to submit a proposal for a product or service your company wants to purchase
- MOU (memorandum of understanding)
- Documents the intentions of two or more parties to enter into a binding agreement, or contract
- SOW (statement of work)
- Documents in detail the work that must be completed for a particular project
- Standard business documents you may encounter (cont'd):
- SLA (service-level agreement)
- A legally binding contract or part of a contract that defines the aspects of a service provided to a customer
- Example: the service provided by an ISP
- MLA (master license agreement)
- Grants a license from a creator, developer, or producer to a third party for the purposes of marketing or sublicensing, or distributing the product to consumers

4.10.4 Change Management

- Managing change while maintaining network's efficiency and availability:
  - Requires good planning
- Common software and hardware changes
  - Range from installing patches to replacing network backbone
- Several ways to approach changes

4.10.5 Software and Hardware Changes

- Three types of changes to existing software:
  - Patch
    - A correction, improvement, or enhancement
  - Upgrade
    - A major change to a software package that enhances the functionality and features of the software
  - Rollback
    - Also called back leveling or downgrading
    - Process of reverting to a previous version after attempting to patch or upgrade it
- General steps to change software or hardware
  - Don't allow patches to be automatically installed
  - Determine whether patch or upgrade is necessary
  - Research change purpose, compatibility, and effects
  - Test the patch or upgrade in a testing lab to make sure it acts as expected
  - Determine whether changes should apply to all users, network segments, or devices
  - Schedule change for completion during off-hours
    - Called the maintenance window
  - Notify appropriate personnel of intent to change
  - Back up current system
  - Prevent users from accessing system during change
  - Keep instructions handy as you install revision
  - Implement the change and test system fully
  - Re-enable access to the system
    - Or revert to previous version
  - Inform personnel that change is complete
  - Record change in the change management system
- Reversing a software upgrade
  - Software change may create unexpected problems
  - Be prepared to reverse an upgrade
- Backleveling
  - Reverting to previous version of software after attempting upgrade
  - No hard-and-fast rules for Backleveling
  - Always refer to software vendor's documentation to reverse an upgrade
  - For NOS: consult other professionals as well

Type of upgrade	Options for reversing
Operating system patch	Use the patch's automatic uninstall utility.
Client software upgrade	Use the upgrade's automatic uninstall utility, or reinstall the previous version of the client on top of the upgrade.
Shared application upgrade	Use the application's automatic uninstall utility, or maintain a complete copy of the previous installation of the application and reinstall it over the upgrade.
Operating system upgrade	Prior to the upgrade, make a complete backup of the system; to backlevel, restore the entire system from the backup; uninstall an operating system upgrade only as a last resort.

Figure 67: Reversing a software upgrade

4.10.6 Change Management Documentation

Generally, the larger an organization, the more documentation required when making changes.

Required process will vary but expect the following:

- Submit a change request document
- Understand and follow the approval process
- The change is project managed (change coordinator)
- Provide additional documentation
- Close the change

4.10.7 Physical Security Controls

- Restrict physical access to critical components
  - Only trusted networking staff should have access
  - A security policy should define who has access
- Sophisticated door access controls:
  - Keypad or cipher locks
    - Cipher locks are physical or electronic locks requiring a code to open the door
  - Access badges
    - Proximity cards, passive cards, and active cards



Figure 68: cipher lock can document who enters an area and when he was out

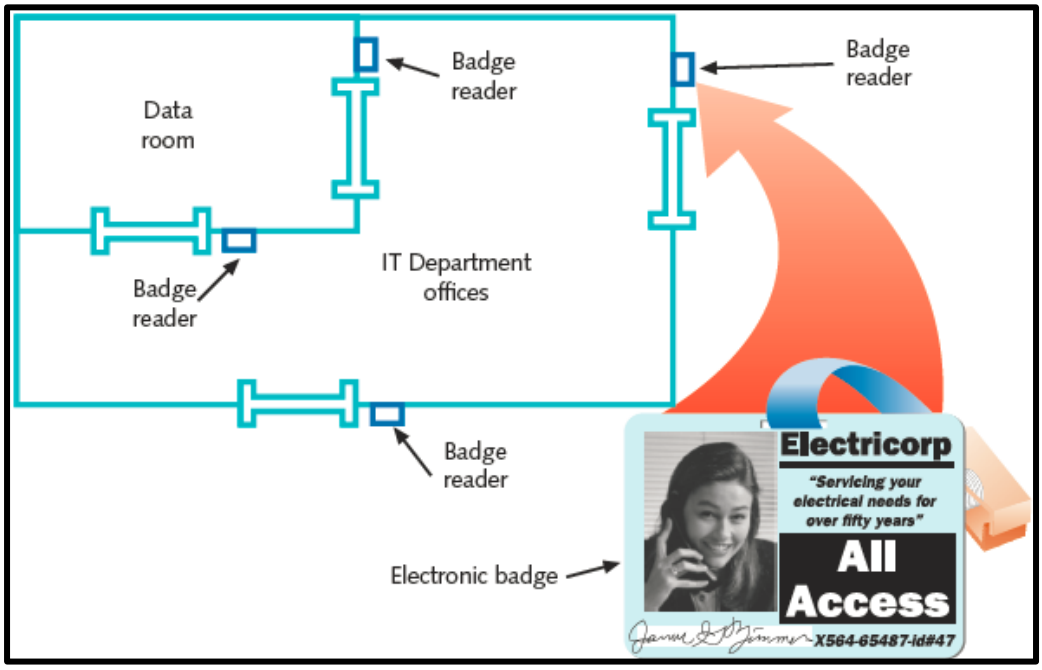


Figure 69: Badges access security system

4.10.8 Physical Security Controls

- Sophisticated door access controls (cont'd):
  - Biometrics
    - Scans an individual's unique physical characteristics
  - Mantraps
    - Consists of two doors on either end of a small entryway
    - First door must close before the second door can open
- AIT (advanced imaging technology) machines

- Use millimeter-wave scanners to indicate on cartoonlike images any areas of concern to security personnel



Figure 70: finger Print Scanner

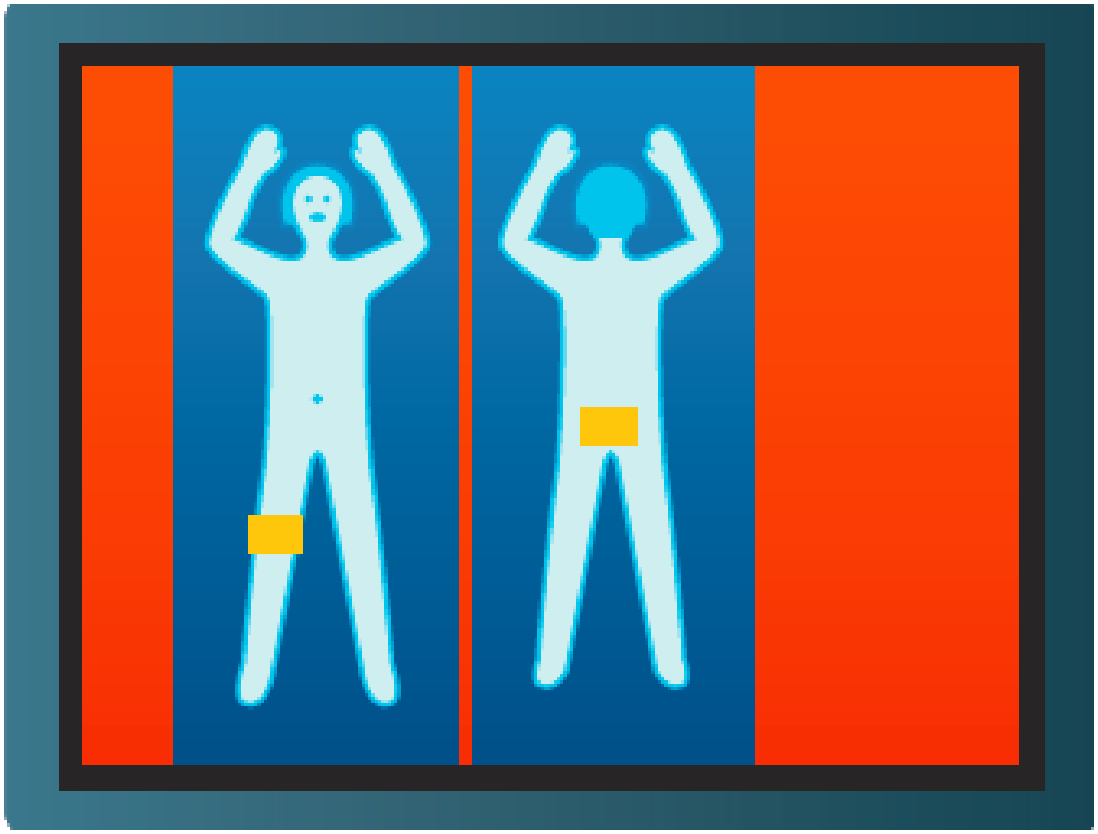


Figure 71: Result of an AIT Scan

- Many IT departments use video surveillance systems (closed-circuit TV) to monitor activity in secured rooms
- IP cameras can be placed in data centers
- A central security office might display several camera views at once
  - Or it might switch from camera to camera
  - Important questions to ask when planning for physical security:
  - Which rooms contain critical systems or data and must be secured?
  - Through what means might intruders gain access to the facility, computer room, data room, network closet, or data storage areas?
  - How and to what extent authorized personnel are granted entry?
  - Are employees instructed to ensure security after entering or leaving secured areas?
- Important questions to ask when planning for physical security (cont'd):
  - Are authentication methods difficult to forge or circumvent?
  - Do supervisors or security personnel make periodic physical security checks?
  - Are all combinations, codes, or other access means to computer facilities protected at all times, and are those combinations changed frequently?
  - Do you have a plan for documenting and responding to physical security breaches?
  - To guard against information being stolen from a decommissioned hard drive
    - Run a specialized drive sanitizer program to make file recovery impossible
  - Degausser
    - A magnetic hard drive eraser

4.10.9 Troubleshooting and Response Policies

- Disasters and security breaches to happen
- Training and preparation can make all the difference in your company's ability to respond and adapt to these situations

4.11 Disaster recovery

- Restoring critical functionality, data
  - After enterprise-wide outage
  - Affecting more than single system, limited group
- Consider possible extremes

- Enterprise-wide outage
- Failures
- Security breaches
- Data corruption

4.11.1 Disaster Recovery Planning

- Account for worst-case scenarios
- Identify disaster recovery team (red team)
- Provide contingency plans for restoring and replacing:
  - Computer systems
  - Power
  - Telephony systems
  - Paper-based files
- Goal is to ensure business continuity
  - Ability to continue to do business

- Sections of the plan related to computer systems should include the following:
  - Contact information for emergency coordinators
  - Details on which data and servers are being backed up, how frequently backups occur, where backups are kept, and how backed-up data can be recovered
  - Details on network topology, redundancy, and agreements with national service carriers
  - Regular strategies for testing the disaster recovery plan
  - A plan for managing the crisis, including regular communications with employees and customers
  - Having a comprehensive disaster recovery plan
  - Lessens the risk of losing critical data
  - Makes potential customers and insurance providers look more favorably on your organization

4.11.2 Disaster Recovery Contingencies

i. Cold site

- Components necessary to rebuild network exist
- Not appropriately configured, updated, or connected

ii. Warm site

- Components necessary to rebuild network exist

- Some appropriately configured, updated, and connected

iii. Hot site

- Components exist and match network's current state
- All appropriately configured, updated, and connected

4.11.3 Forensics

- Every security policy should include a response policy
  - Defines the characteristics of an event that qualifies as a formal incident and steps that should be followed
- Qualifying incidents might include a:
  - Break-in
  - Fire
  - Weather-related emergency
  - Hacking attack or malware outbreak
  - Discovery of illegal content or activity
- Data collected might be presented in a court of law
  - Data must be carefully collected so that it will stand up to the scrutiny of the court
- First responders may take charge
  - People with training and/or certifications that prepare them to handle evidence
- Every IT technician should know how to safeguard sensitive information, logged data, and other legal evidence until first responder can take over
- A response policy should detail the following steps:
  - Determine if escalation is necessary
  - Secure the area
  - Document the scene
  - Monitor evidence and data collection
  - Protect the chain of custody
  - Monitor transport of data and equipment
  - Create a report
- A response policy should identify members of a response team:
  - Dispatcher
    - The person to call who first notices the problem
  - Manager
    - Coordinates the resources necessary to solve the problem

- ❖ Technical support specialist
  - ➢ Focuses on solving the problem quickly
- ❖ Public relations specialist
  - ➢ Acts as official spokesperson for the organization

4.11.4 Summary

- An industrial system is a system of machines, such as an assembly line at a tire manufacturing plant
- An industrial control system (ICS) is a group of networked computers used to manage a physical system of industrial processes
- Isolate an ICS/SCADA network by deploying a DMZ between the corporate network and the ICS network
- First step in managing assets is to inventory all the components on the network
- Be familiar with the following business documents: RFP, MOU, SOW, SLA, and MLA
- Managing change while maintaining your network's efficiency and availability requires good planning
- Three types of changes to existing software include patches, upgrades or updates, and rollbacks
- The complexity of a change approval process is usually determined by the cost and time involved in making the change, the number of users affected, potential risk to work productivity, and difficulty of rolling back the change
- A security policy defines who has access to the computer room
- A disaster recovery plan should identify a disaster recovery team, sometimes called the red team, with an appointed coordinator
- Every contingency plan necessitates a site other than the building where the network's main components normally reside
- Some forensic data available for analysis can be damaged or destroyed if improperly handled

Further Reading:

i. Networks + Guide to Networks, 7th Edition by Tamara Dean, Chapter 12, page 603-628.
ii. Networks Study guide, 3rd edition, by Todd Lammle, Chapter 20 page 755-793

Review Questions

1. You are a network administrator managing a network backbone upgrade. Your supervisor has scheduled a meeting to discuss the project's status with you. What is the advantage of taking a Gantt chart to the meeting?

a. A Gantt chart will help the supervisor better understand the project's costs.

b. A Gantt chart will determine the maximum possible amount of each employee's time to be spent on each task.

c. A Gantt chart will allow the supervisor to see timelines of each task in addition to the project as a whole.

d. A Gantt chart will demonstrate why some tasks have taken longer to complete than first anticipated.

2. The task that must be completed before another task is begun is called a(n) ___ .

a. Successor

b. Predecessor

c. Antecedent

d. Dependent

3. In a significant network upgrade project, which of the following tasks takes place first?

a. Identify which tasks are dependent on other tasks.

b. Complete a needs assessment survey.

c. Test the proposed solution on a pilot network.

d. Assign tasks to the most qualified or appropriate people on the project team.

4. Which of the following best describes contingency planning?

a. Obtaining support from high-level project sponsors before committing resources to the project

b. Installing identical software and hardware, on a smaller scale, as the project's proposed solution will require, testing the feasibility of the solution

c. Identifying a team and assigning roles to that team in case of disaster

d. Identifying steps that will minimize the impact of unforeseen obstacles

5. True or False? One way of predicting how long a task might take is by examining the time taken to complete previous similar tasks.

Which of the following tools might you use to assess the success of a project whose purpose is to upgrade an entire network from 100 Mbps to 1000 Mbps?

a. Network Monitor

b. Sniffer or packet analyzer

c. Microsoft SQL Server

d. System Monitor

6. What is the purpose of identifying milestones in a project plan?

a. They indicate when project staff changes must occur.

b. They mark significant events of the project's progress.

c. They offer a quick assessment of how successfully the project is staying within budget.

d. They help predict the result of a project.

7. Which of the following are examples of resources related to a project plan that proposes to upgrade the network cards inside each workstation on a network from 100 Mbps to 1000 Mbps? (Choose all that apply.)

a. NIC device drivers

b. A team member's time

c. IP addresses

d. NICs

e. Switches

8. Which of the following are examples of stakeholders of a project whose purpose is to upgrade an entire network from 100 Mbps to 1000 Mbps? (Choose all that apply.)

a. Network users

b. Software vendors of applications running on the client machines

c. High-level managers who approved the project

d. IT staff who helped implements the change

e. Network cabling vendors

9. Which of the following obstacles could halt or seriously impair the progress of a project whose purpose is to upgrade the NICs in all workstations on a network from 100 Mbps to 1000 Mbps?

a. The use of two different NIC models

b. The use of different installation personnel on different shifts

c. Management's requirement that the cost of each NIC remain under $75

d. A large group of defective NICs

10. In a very large company (for example, one with over 10,000 employees), which of the following staff is most likely to decide whether a project such as an entire network upgrade will be funded?

a. Network administrator

b. Personnel director

c. Chief information officer

d. Accountant

11. Which of the following best describes project management?

a. Recording and analyzing the time and resources required for each task in a project

b. Assessing network statistics before and after a project is completed

c. Monitoring the needs of users prior to the beginning of a project, then later assessing how the project's completion met their needs

d. Planning for and handling the steps required to accomplish a goal in a systematic way

12. Which of the following projects is most likely to be driven by a company's security needs?

a. Doubling the RAM in a key file server

b. Installing a firewall on a connection to the Internet

c. Upgrading the version of client software on each workstation

d. Changing from the use of static IP addressing to DHCP on an entire network

13. What is one good technique for assessing the feasibility of a suggested project deadline before the project begins?

a. Begin calculating task timelines from the deadline, working back to the start of a project.

b. Issue a survey to key staff asking their opinion of the suggested deadline.

c. Use the Web to research similar projects completed by other companies.

d. Calculate the ratio of the number of project milestones to the proposed project duration, in months, to check that it does not exceed 2:1.

14. Which of the following situations might necessitate changing all the IP addresses on a company's networked workstations?

a. The company has divided its network into several smaller subnets.

b. The company has hired 50 new employees.

c. The company has decided to use Network Address Translation (NAT) for all connections to public networks.

d. The company has decided to establish a Web server with e-commerce capability.

15. In Microsoft terminology, what is a hot fix?

a. A patch that replaces all or a portion of the NOS

b. A patch that requires that the server be connected to the Microsoft Web site as it is installed

c. A patch that updates a specific type of software, often the operating system

d. A patch that can be installed while users are logged on without causing adverse effects

16. In which of the following situations would it be wise to backlevel?

a. You have just performed a complete backup of your server's data directories, and you cannot confirm that the backup was successful.

b. You have just applied a fix to your network operating system (NOS) and have discovered that the fix resulted in a lack of network access for half of your users.

c. You have just installed a database program on one of your servers and have discovered that you neglected to install an optional component that your users will need.

d. You have just installed Windows Server 2012 R2 on a new computer and you cannot get the operating system to recognize the NIC.

17. Which two of the following can typically be accomplished by applying a patch to an NOS?

a. Replacing all the NOS's program files

b. Modifying an existing feature

c. Removing an old feature

d. Fixing a known bug

18. Before installing a major NOS patch, you should _____ .

a. Remove all protocols installed on the server.

b. Prevent users from logging on.

c. Disable Internet services.

d. Disable network connectivity.

19. What is the primary difference between a software upgrade and a patch?

a. The software manufacturer issues a patch, whereas an upgrade may be issued by any organization that has the software's source code.

b. A patch fixes a specific part of a software program, whereas an upgrade typically replaces much or all of a software program.

c. A patch typically does not require that the network administrator test its changes before applying it to a server; an upgrade does.

d. A patch typically is not supported by the software manufacturer; an upgrade is.

20. You work for the Best Roast Coffee Company, which has 100 employees in nine retail stores. You are looking at case studies for a product you are researching. Which of the following case studies is most likely to be helpful?

a. A case study for an automobile manufacturer with hundreds of thousands of employees

b. A candy maker with 200 employees that sells candy in 13 malls across the country

c. An insurance company with thousands of employees in three locations, plus hundreds of agents all over the country

d. A small school district with 300 employees and 10,000 students

21. You are trying to decide whether to purchase a software package from a vendor. Who among the following is most likely to provide objective information about the software package?

a. The vendor's sales staff

b. A business partner of the vendor

c. A current customer of the vendor

d. A vendor who makes a competing product

22. You are researching a new Internet access router for your company's network. You have also noticed a trend toward greater use of your company's network resources from home. Given this, which of the following features would you expect to be most valuable in a new Internet access router?

a. Support for additional routing protocols

b. Support for virtual private networks

c. Support for additional security features

d. Support for multiple protocols

23. When implementing a network solution, why is it important to anticipate future trends as much as possible?

a. So that the solutions you implement meet the future needs of your users

b. So that the solutions you implement meet the current needs of your users

c. So that the solutions you implement are more scalable

d. So that you can justify the purchase of the solution to management

24. When investigating a network solution, which two of the following should you do?

a. Get information about possible solutions from only one source.

b. Get information about possible solutions from multiple sources.

c. Look at the product lines from only one vendor.

d. Talk to someone who has implemented a similar solution.

25. The hardware devices attached to a network such as computers and printers are also called _____.

a. Communication endpoints

b. Nodes

c. Connection points

d. Contacts

26. Asset management records should be available to _____.

a. All personnel involved in maintaining the network

b. Only the system administrator

c. Everyone in the organization

d. Anyone

27. Who is protected by documenting hardware and software changes?

a. Users

b. Managers

c. The organization

d. All of the above

28. When should changes be made to the asset management database?

a. Monthly

b. Weekly

c. Daily

d. As changes to hardware and software occur

29. True or False? Network asset management can include device locations, serial numbers, and technical support contact information.

Key terms used

802.2: The IEEE standard for error and flow control in data frames.

802.3: The IEEE standard for Ethernet networking devices and data handling (using the CSMA/CD access method).

802.5: The IEEE standard for token ring networking devices and data handling.

802.11: The IEEE standard for wireless networking.

802.16: The IEEE standard for broadband wireless metropolitan area networking (also known as WiMAX).

ACK (acknowledgment): A response generated at the Transport layer of the OSI model that confirms to a sender that its frame was received. The ACK packet is the third of three in the three-step process of establishing a connection.

Acknowledgment: See ACK.

American National Standards Institute: See ANSI.

ANSI (American National Standards Institute): An organization composed of more than 1000 representatives from industry and government who together determine standards for the electronics industry in addition to other fields, such as chemical and nuclear engineering, health and safety, and construction.

API (Application Program Interface): A set of routines that make up part of a software application.

Application Layer: The seventh layer of the OSI model. Application Layer protocols enable software programs to negotiate formatting, procedural, security, synchronization, and other requirements with the network.

Application Programming Interface: See API.

Block ID: The first set of six characters that make up the MAC address and that are unique to a particular manufacturer.

Checksum: A method of error checking that determines if the contents of an arriving data unit match the contents of the data unit sent by the source. **Connection Oriented:** A type of Transport layer protocol that requires the establishment of a connection between communicating nodes before it will transmit data.

Connectionless: A type of Transport layer protocol that services a request without requiring a verified session and without guaranteeing delivery of data. **CRC (cyclic redundancy check):** An algorithm (or mathematical routine) used to verify the accuracy of data contained in a data frame.

Cyclic Redundancy Check: See CRC.

Data Link Layer: The second layer in the OSI model. The Data Link Layer bridges the networking media with the Network Layer. Its primary function is to divide the data it receives from the Network Layer into frames that can then be transmitted by the Physical Layer.

Data Link Layer address: See MAC address. **Device ID:** The second set of six characters that make up a network device's MAC address. The device ID, which is added at the factory, is based on the device's model and manufacture date.

EIA (Electronic Industries Alliance): A trade organization composed of representatives from electronics manufacturing firms across the United States that sets standards for electronic equipment and lobbies for legislation favourable to the growth of the computer and electronics industries.

Encapsulate: The process of wrapping one layer's PDU with protocol information so that it can be interpreted by a lower layer. For example, Data Link Layer protocols encapsulate Network Layer packets in frames.

Ethernet: A networking technology originally developed at Xerox in the 1970s and improved by Digital Equipment Corporation, Intel, and Xerox. Ethernet, which is the most common form of network transmission technology, follows the IEEE 802.3 standard.

FCS (frame check sequence): The field in a frame responsible for ensuring that data carried by the frame arrives intact. It uses an algorithm, such as CRC, to accomplish this verification.

Flow Control: A method of gauging the appropriate rate of data transmission based on how fast the recipient can accept data.

Fragmentation: A Network Layer service that subdivides segments it receives from the Transport layer into smaller packets.

Frame: A package for data that includes not only the raw data, or "payload," but also the sender's and recipient's addressing and control information. Frames are generated at the Data Link Layer of the OSI model and are issued to the network at the Physical Layer. **Frame Check Sequence:** See FCS. **Hardware Address:** See MAC address.

HTTP (Hypertext Transfer Protocol): An Application Layer protocol that formulates and interprets requests between Web clients and servers.

Hypertext Transfer Protocol: See HTTP.

IAB (Internet Architecture Board): A technical advisory group of researchers and technical professionals responsible for Internet growth and management strategy, resolution of technical disputes, and standards oversight.

IANA (Internet Assigned Numbers Authority): A non-profit, United States government funded group that was established at the University of Southern California and charged with managing IP address allocation and the domain name system. The oversight for many of IANA's functions was given to ICANN in 1998; however, IANA continues to perform Internet addressing and domain name system administration.

ICANN (Internet Corporation for Assigned Names and Numbers): The non-profit corporation currently designated by the United States government to maintain and assign IP addresses.

IEEE (Institute of Electrical and Electronics Engineers): An international society composed of engineering professionals. Its goals are to promote development and education in the electrical engineering and computer science fields. **IETF (Internet Engineering Task Force):** An organization that sets standards for how systems communicate over the Internet (for example, how protocols operate and interact).

Institute of Electrical and Electronics Engineers: See IEEE.

International Organization for Standardization: See ISO.

International Telecommunication Union: See ITU.

Internet Architecture Board: See IAB.

Internet Assigned Numbers Authority: See IANA.

Internet Corporation for Assigned Names and Numbers: See ICANN.

Internet Engineering Task Force: See IETF. **Internet Protocol:** See IP.

Internet Protocol address: See IP address.

Internet service provider: See ISP.

Internet Society: See ISOC.

IP (Internet Protocol): A core protocol in the TCP/IP suite that operates in the Network Layer of the OSI model and provides information about how and where data should be delivered. IP is the sub-protocol that enables TCP/IP to internetwork.

IP address (Internet Protocol address): The Network Layer address assigned to nodes to uniquely identify them on a TCP/IP network. IP addresses consist of 32 bits divided into four octets, or bytes.

ISO (International Organization for Standardization): A collection of standards organizations representing 157 countries with headquarters located in Geneva, Switzerland. Its goal is to establish international technological standards to facilitate the global exchange of information and barrier-free trade.

ISOC (Internet Society): A professional organization with members from 90 sections around the world that helps to establish technical standards for the Internet.

ISP (Internet service provider): A business that provides organizations and individuals with Internet access and often, other services, such as e-mail and Web hosting.

ITU (International Telecommunication Union): A United Nations agency that regulates international telecommunications and provides developing countries with technical expertise and equipment to advance their technological bases.

LLC (Logical Link Control) sub-layer: The upper sub-layer in the Data Link Layer. The LLC provides a common interface and supplies reliability and flow control services.

Logical Address: See network address.

Logical Link Control sub-layer: See LLC (Logical Link Control) sub-layer. **MAC address:** A 12-character string that uniquely identifies a network node. The manufacturer hard codes the MAC address into the NIC. This address is composed of the block ID and device ID.

MAC (Media Access Control) sub-layer: The lower sub-layer of the Data Link Layer. The MAC appends the physical address of the destination computer onto the frame. **Maximum Transmission Unit:** See MTU.

Media Access Control sub-layer: See MAC (Media Access Control) sub-layer.

MTU (maximum transmission unit): The largest data unit a network (for example, Ethernet or token ring) will accept for transmission.

Network Address: A unique identifying number for a network node that follows a hierarchical addressing scheme and can be assigned through operating system software. Network addresses are added to data packets and interpreted by protocols at the Network Layer of the OSI model.

Network Layer: The third layer in the OSI model. Protocols in the Network Layer translate network addresses into their physical counterparts and decide how to route data from the sender to the receiver.

Network Layer address: See network address.

Open Systems Interconnection model: See OSI (Open Systems Interconnection) Model.

OSI (Open Systems Interconnection) model: A model for understanding and developing computer-to computer communication developed in the 1980s by ISO. It divides networking functions among seven layers: Physical, Data Link, Network, Transport, Session, Presentation, and Application.

PDU (Protocol Data Unit): A unit of data at any layer of the OSI model.

Physical Address: See MAC address.

Physical Layer: The lowest, or first, layer of the OSI model. Protocols in the Physical Layer generate and detect signals so as to transmit and receive data over a network medium. These protocols also set the data transmission rate and monitor data error rates, but do not provide error correction.

Presentation Layer: The sixth layer of the OSI model. Protocols in the Presentation Layer translate between the application and the network. Here, data are formatted in a schema that the network can understand, with the format varying according to the type of network used. The Presentation Layer also manages data encryption and decryption, such as the scrambling of system passwords.

Protocol Data Unit: See PDU.

Reassembly: The process of reconstructing data units that have been segmented.

Regional Internet Registry: See RIR.

RIR (Regional Internet Registry): A not-for-profit agency that manages the distribution of IP addresses to private and public entities. ARIN is the RIR for North, Central, and South America and sub-Saharan Africa. APNIC is the RIR for Asia and the Pacific region. RIPE is the RIR for Europe and North Africa. **Route:** To intelligently direct data between networks based on addressing, patterns of usage, and availability of network segments.

Router: A device that connects network segments and directs data based on information contained in the data packet.

Segment: A unit of data that results from subdividing a larger protocol data unit.

Segmentation: The process of decreasing the size of data units when moving data from a network that can handle larger data units to a network that can handle only smaller data units.

Sequencing: The process of assigning a placeholder to each piece of a data block to allow the receiving node's Transport layer to reassemble the data in the correct order.

Session: A connection for data exchange between two parties. The term session may be used in the context of Web, remote access, or terminal and mainframe communications, for example.

Session Layer: The fifth layer in the OSI model. The Session Layer establishes and maintains communication between two nodes on the network. It can be considered the "traffic cop" for network communications.

Standard: A documented agreement containing technical specifications or other precise criteria that are used as guidelines to ensure that materials, products, processes, and services suit their intended purpose.

SYN (synchronization): The packet one node sends to request a connection with another node on the network. The SYN packet is the first of three in the three-step process of establishing a connection.

SYN-ACK (synchronization-acknowledgment): The packet a node sends to acknowledge to another node that it has received a SYN request for connection. The SYN-ACK packet is the second of three in the three-step process of establishing a connection.

Synchronization: See SYN.

Synchronization-acknowledgment: See SYN-ACK.

Telecommunications Industry Association: See TIA.

Terminal: A device with little (if any) of its own processing or disk capacity that depends on a host to supply it with applications and data-processing services.

TIA (Telecommunications Industry Association): A subgroup of the EIA that focuses on standards for information technology, wireless, satellite, fiber optics, and telephone equipment. Probably the best known standards to come from the TIA/EIA alliance are its guidelines for how network cable should be installed in commercial buildings, known as the "TIA/EIA 568-B Series."

Token: A special control frame that indicates to the rest of the network that a particular node has the right to transmit data.

Token ring: A networking technology developed by IBM in the 1980s. It relies upon direct links between nodes and a ring topology, using tokens to allow nodes to transmit data.

Transport Layer: The fourth layer of the OSI model. In the Transport layer protocols ensure that data are transferred from point A to point B reliably and without errors. Transport layer services include flow control, acknowledgment, error correction, segmentation, reassembly, and sequencing. Virtual Address: See network address.